

2D Defects

- ❑ Surfaces
- ❑ Grain Boundary
- ❑ Twin Boundary
- ❑ Stacking Fault
- ❑ Anti Phase Boundary

Part of

**MATERIALS SCIENCE
& *A Learner's Guide*
ENGINEERING**

AN INTRODUCTORY E-BOOK

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<http://home.iitk.ac.in/~anandh/E-book.htm>

Further reading

Crystallography and Crystal Defects

A. Kelly, G.W. Groves and P. Kidd

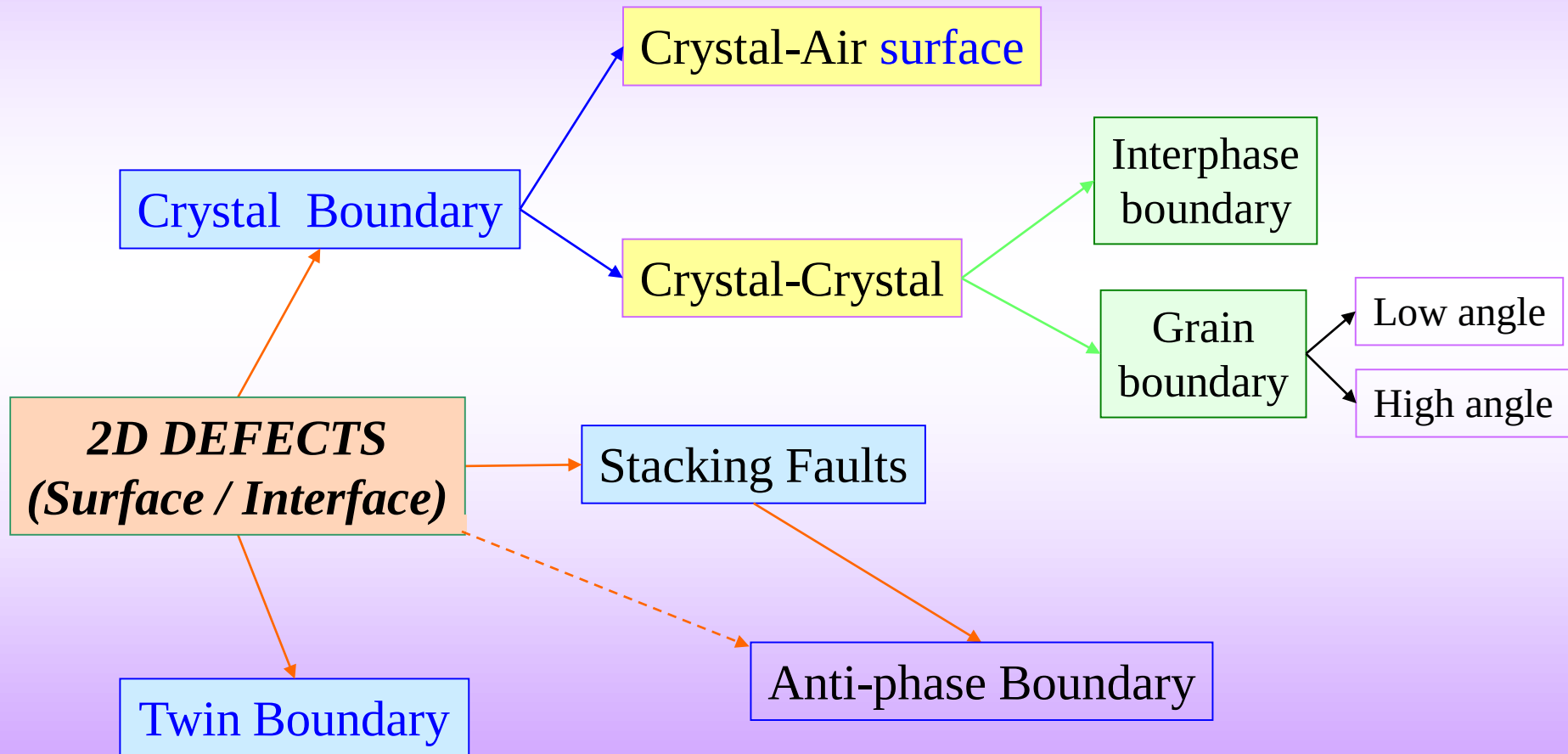
John Wiley & Sons, Chichester (2000)

Caution Note: In any chapter, amongst the first few pages (say 5 pages) there will be some 'big picture' overview information. This may lead to 'overloading' and readers who find this 'uncomfortable' may skip particular slides in the first reading and come back to them later.

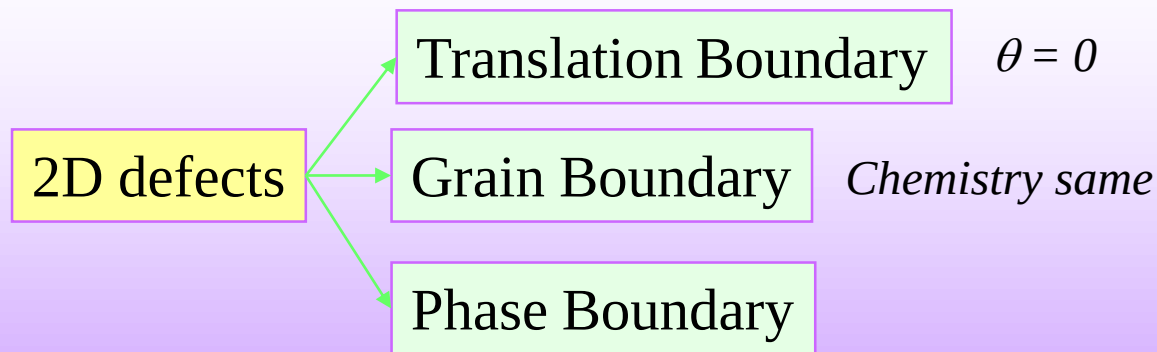
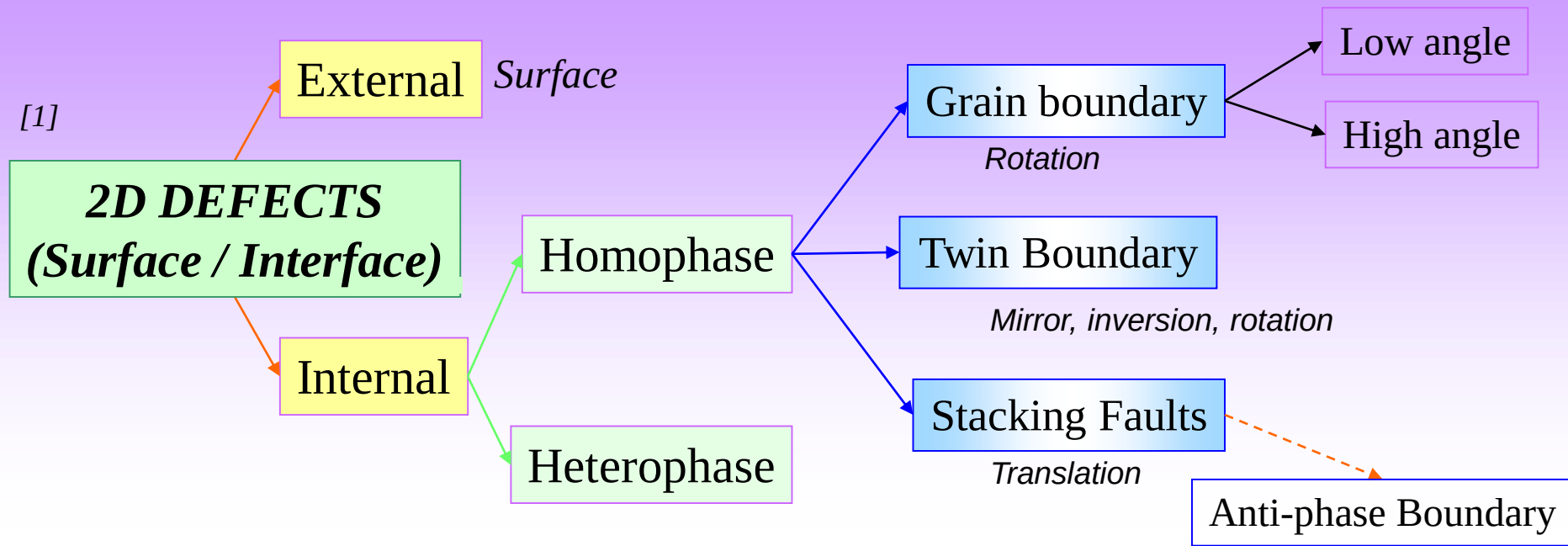
What will you learn in this chapter?

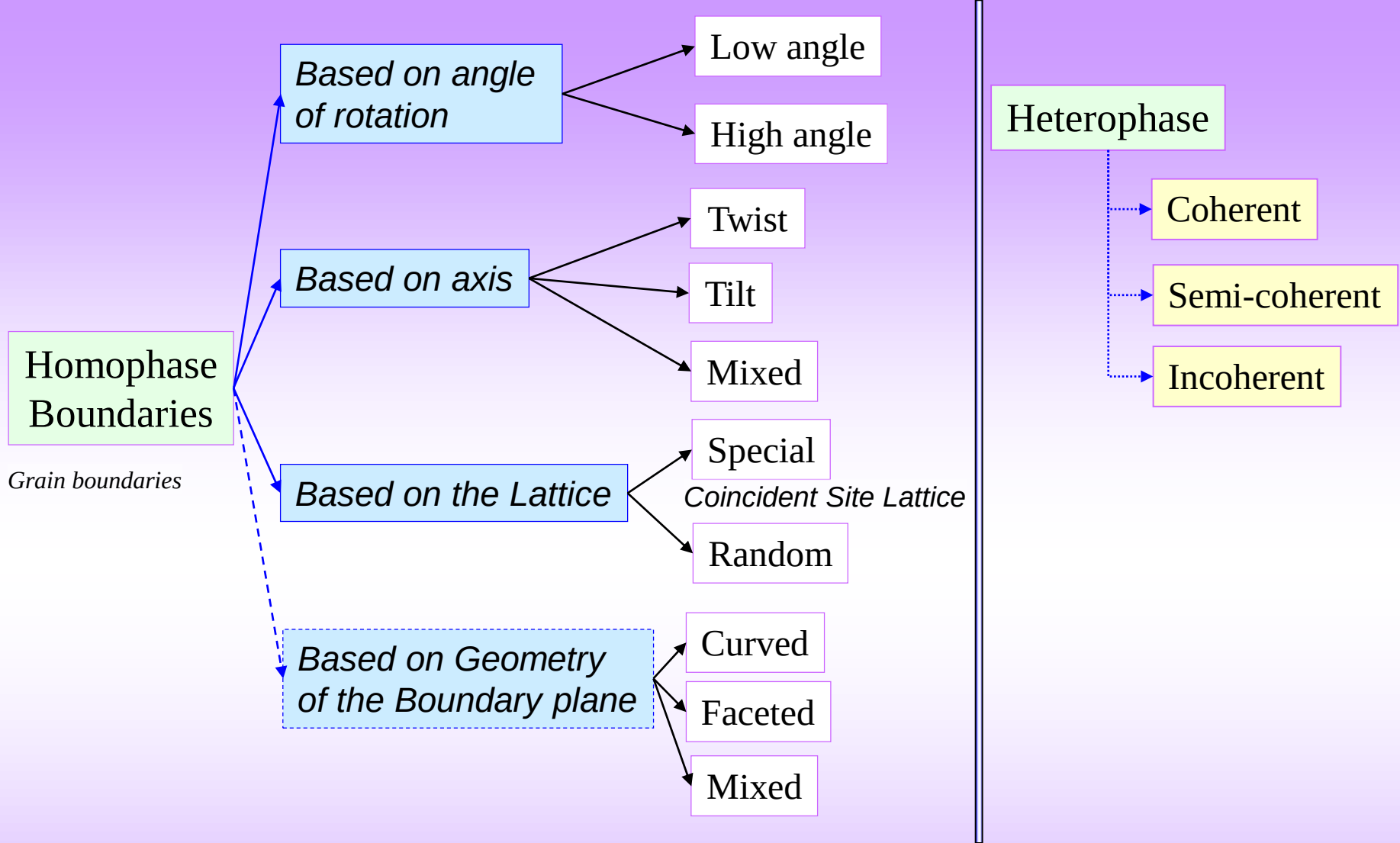
- How to classify 2D defects?
 - How to understand their energies and their structure?
 - How do these defects affect material properties?
-
- Types of interfaces which are possible (in equilibrium).
 - Solid-solid, solid-liquid, solid-gas (solid-vapour)
 - Liquid-liquid, liquid-gas (liquid-vapour)
 - Gases mix with each other and hence gas-gas interface is not possible.

- ❑ We can classify 2D defects in more than one way. These alternate perspectives can give us a broad overview of the defects possible in 2D.
- ❑ Please spend some time looking at these trees before proceeding further. Many of the terms introduced will become clear only later.



Alternate views





■ Variables defining a planar defect (*leading to symmetry breaking*)

➤ Translation [$\mathbf{R}(\mathbf{r})$] + Rotation [Axis ($\mathbf{n}$), Angle (θ)] + Chemistry (*concept of colour symmetry*).

Let us start with a toy model to look at some defects in crystals

(some of these are already familiar to you, while others will be introduced in the current chapter)



Video: Models to understand the formation of defects in crystals

- ☐ *Vacancies*
- ☐ *Grain boundaries*
- ☐ *Surface [(100), (111), (110)]*
- ☐ *Disclination*
- ☐ *Edge dislocation*

SURFACES

- ❑ A surface is 2D in a mathematical sense.
- ❑ The atoms on the surface are 'different' from the 'bulk' atoms in many ways:
 - (i) they are in a state of higher energy due to incomplete bonding,
 - (ii) the lattice parameter of the surface is different from the bulk,
 - (iii) the atoms on the surface are displaced from the 'expected position' (i.e. which is expected from the bulk structure),
 - (iii) the surface atoms experience surface stress.
- ❑ In reality the region of 'distortion' (i.e. the region where atoms which are in a different state from the bulk atoms) is $\sim$ few atomic diameters in thickness.
- ❑ In this chapter we will start with the concept of surface energy.
- ❑ Though we are interested in solid (crystalline) surfaces, we shall compare solid and liquid surfaces to understand the differences
(it will be seen that terms like surface tension are more appropriate for liquids and that the solid surface will be characterized by more parameters than the liquid surface)
- ❑ Often the surface is NOT merely a termination of the bulk (i.e. the surface structure is not like the bulk structure and surfaces can undergo relaxation, reconstruction etc.).

[Click here to get further details on surface energy, surface tension & other topics related to surfaces](#)

→ The 'real surface' is a few atomic layers deep!

Surface Energy

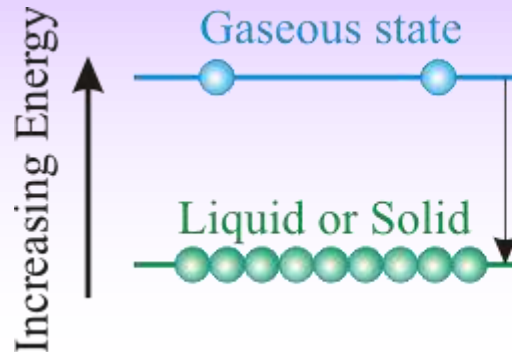
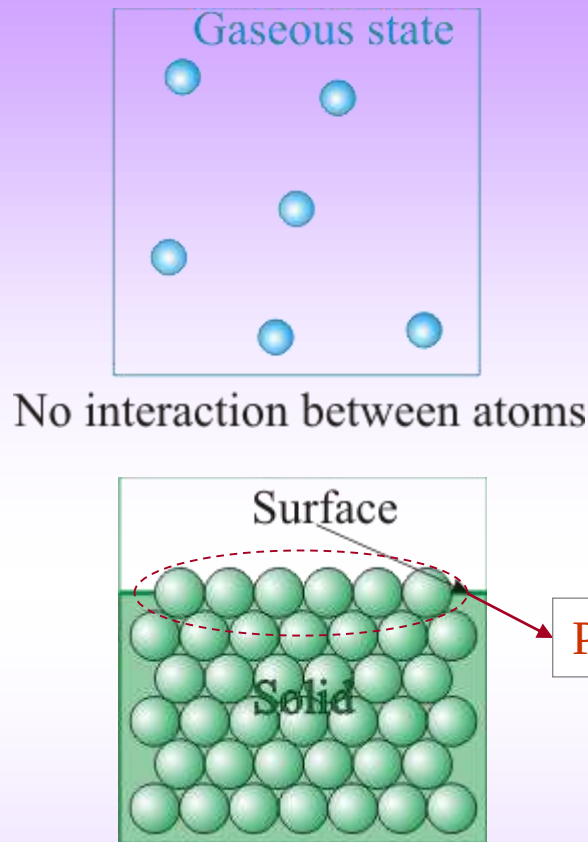
- ❑ Surface is in a state of higher energy as compared to the bulk → surface atoms are “less bonded” as compared to the bulk atoms.
- ❑ In a simple model the ‘missing bonds’ are referred to as broken bonds.
- ❑ Surface Energy and surface tension are concepts associated with liquids and solids (*i.e. not with gases*). As we shall see soon, in the case of solids, the term surface stress is better suited, as compared to surface tension.
- ❑ Under certain thermodynamic parameters, if the Gibbs Free Energy (G) of the solid or liquid is lower than a given gaseous state (wherein the atoms are far apart without any interatomic forces), then the gas will condense (and form a solid or liquid).
- ❑ The lowering in the Gibbs Free Energy is due to the cohesive forces in the liquid or the bonding forces in the solid.
- ❑ The lowering in energy is calculated for an atom (or entity) fully bonded.
- ❑ The atoms on the surface have a higher energy than the bulk atoms (in the regime where the solid or the liquid have a lower energy than the gaseous state).
- ❑ Hence the reference state for the surface is the bulk and not the gaseous state (*which is assumed to be the state where the atoms are non-interacting*).

Is the Surface Energy positive or negative?

We will try to understand as to why we consider the surface energy to be positive.

- Why is the surface energy positive?
- What is the reference state?

[Please click this link to understand this slide better](#)



Atoms lower their energy by forming a liquid or solid
(Assuming the condensed phase is the stable one)

Partly bonded surface atoms

- When we calculate the lowering of the energy of the system on the formation of the condensed state, we counted all atoms to be bulk (interior) atoms → i.e. an *over-counting* was done (as some of the atoms are sitting on the surface → the surface atoms are not as 'well-bonded' as the atoms in the bulk).
- The 'higher energy' of the surface is with respect to the bulk and **not** with respect to the gaseous (non-interacting) state.
- Hence the reference state for the surface is the bulk and not the gaseous state.

- ❑ Hence, it costs energy to put an atom on the surface as compared to the bulk
→ origin of **Surface Energy** (γ).
- ❑ The surface wants to minimize its area (wants to shrink)
→ origin of **Surface Tension** (σ).

Let us look at the units of these two quantities

$$\sigma = \frac{\text{Force}}{\text{Length}} = \frac{F}{L} \rightarrow \frac{[N]}{[m]}$$

$$\gamma = \frac{\text{Energy}}{\text{Area}} = \frac{E}{A} \rightarrow \frac{[J]}{[m^2]} = \frac{[Nm]}{[m^2]} = \frac{[N]}{[m]}$$

- Dimensionally γ and σ are identical → Physically they are different type of quantities.
- γ is a scalar while σ is a second order tensor.

LIQUIDS

Surface Energy $\Leftrightarrow$ Surface Tension

In a liquid: surface energy and surface tension are equivalent descriptions of the surface

SOLIDS

Surface Energy $\neq$ Surface Tension

*Except in certain circumstances**

+ Surface energy is Anisotropic

in a solid

* general isotropic solid

- ❑ The surface of a solid is considerably different from the surface of a liquid. If a liquid surface is stretched subsurface atoms can come to the surface and the structure of the surface remains unaltered. In case of a solid this is not the case.
- ❑ The salient differences (& commonalities) between a solid and liquid surface are shown below.

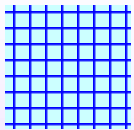


LIQUID SURFACE

- Surface Energy
- Surface Tension

Characterized by one number → the surface density

- *Liquids cannot support shear stresses (hence use of the term surface tension)*



SOLID SURFACE

- Surface Energy
- Surface Stress (*Tensor*)
- Surface Torque

Has a structure and hence more numbers may be needed to characterize a solid surface

- *Crystalline surfaces → all the lattice constants will be required*
- *Amorphous surfaces → Density + a Short Range Order parameter*

- *In the case of solids the term surface tension (which actually should be avoided) refers to surface stresses.*

Surface Energy (γ) $\rightarrow$ is the reversible work required to create an unit area of surface
(at constant V, T & μ_i)

Surface Tension (σ) $\rightarrow$ is the average of surface stresses in two mutually perpendicular directions

$$\sigma = \left(\frac{\sigma_x + \sigma_y}{2} \right)$$

- Surface stress at any point on the surface is the force acting across any line on the surface which passes through this point in the limit the length of the line goes to zero

- Origin of **Surface Torque**

As seen in the previously a solid surface is associated with surface energy, surface stress and surface torque. Surface torque arises from anisotropy in surface energy (i.e. the surface energy depends on the orientation of the surface). If a surface is near an orientation with lower surface energy, then macroscopically it will feel a torque which will tend to rotate it to the orientation with lower energy.

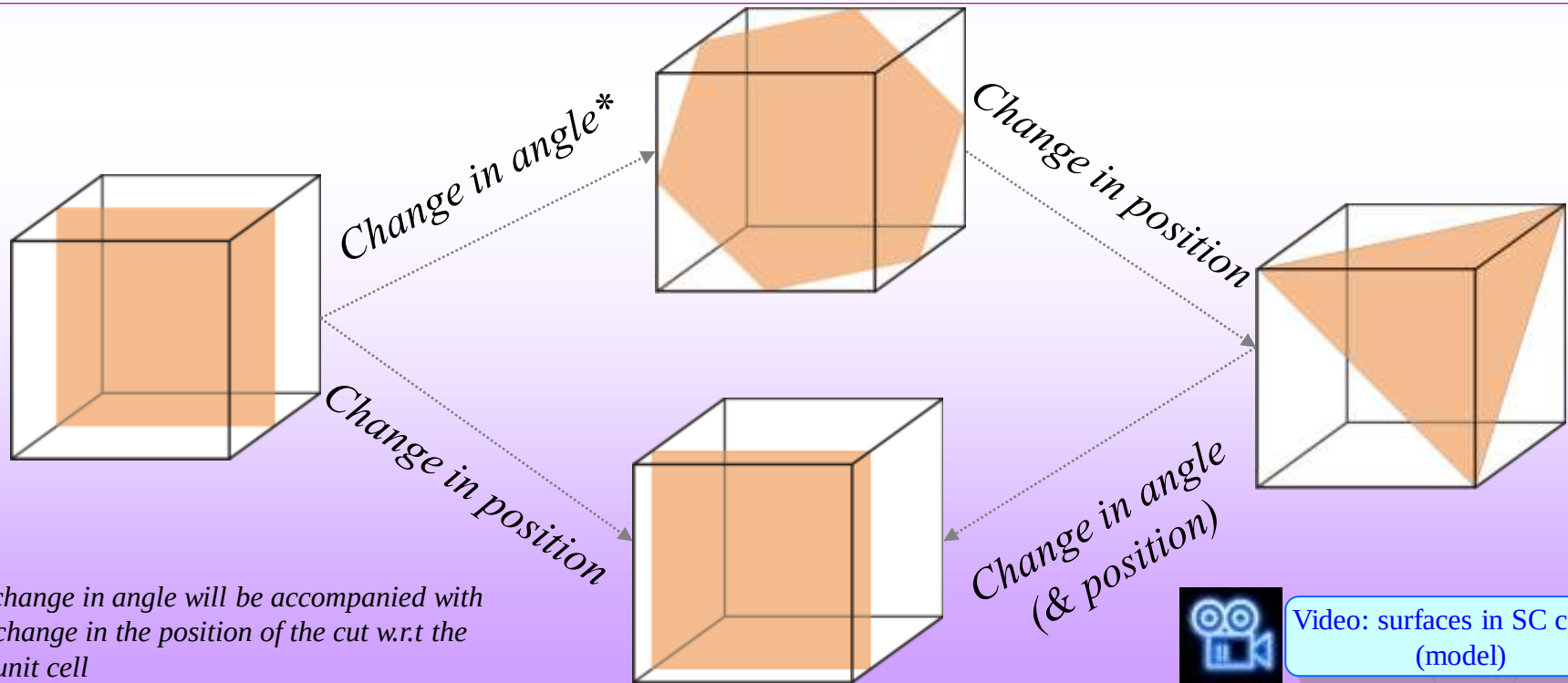
The definition of surface tension in 2D is analogous to the definition of hydrostatic pressure in 3D

[Click here to know more about surface stress](#)

- ❑ Liquid surfaces are characterized by a single parameter: the **density** (atoms / area).
- ❑ At any given instant the liquid surface may have short range order.
- ❑ The short range order in liquids (including their surfaces) is spatio-temporally varying (*varies from place to place and in time*)
→ hence no structure (and no other characteristic) can be assigned to the surface.
- ❑ The surface of a crystalline solid has a definite structure (which could be different from the bulk) and hence additional parameters are required to characterize them.
- ❑ To reiterate, the **order at the surface of a crystal can be different from the bulk.**
- ❑ Amorphous solids have short-range order, but NO long-range order. Under low temperature conditions and short times (i.e. low atomic mobility regimes) the atomic (entity) positions are temporally fixed.
- ❑ As a solid surface can support shear stresses (like the bulk), surface tension is not an adequate measure of the character of the surface → it is surface stress.

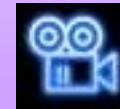
External surface of the crystal

- ❑ A surface (in fact two surfaces) can be created by making a cut in an infinite crystal.
- ❑ Two parameters** characterize this cut:
 - Macroscopic ➤ The **angle of the cut** (w.r.t the crystallographic axis → characterized by Miller indices)
 - Microscopic ➤ The **position of the cut** (within the unit cell, characterized by a shift vector).
- ❑ The angle of cut leads to two degrees of freedom (DOF) and the shift vector is characterized by one DOF (as the direction is normal to the plane).
Total DOF: 2 Macroscopic + 1 Microscopic = 3 DOF.
- ❑ In the case of ionic crystals the cut *may* result in **polar surface**.



* change in angle will be accompanied with change in the position of the cut w.r.t the unit cell

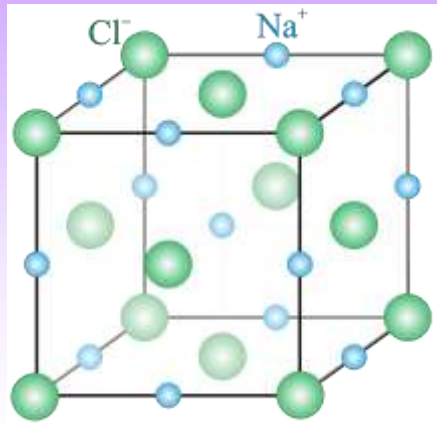
** More about this shortly



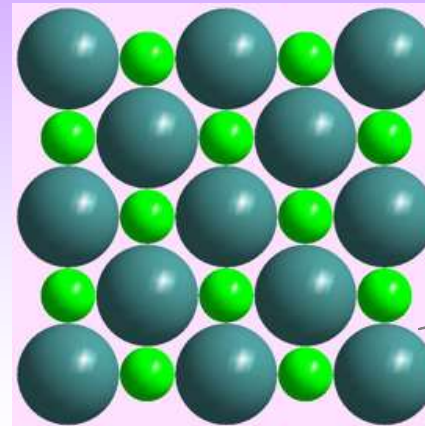
Video: surfaces in SC crystal (model)

E.g. NaCl

Let us use the (111) and (100) cuts in NaCl to understand the origin of polar surfaces.
The (100) surface is non-polar, with equal number of Na^+ and Cl^- ions.
The (111) surface is polar (one side of the cut is Na^+ terminated and the other Cl^- terminated).

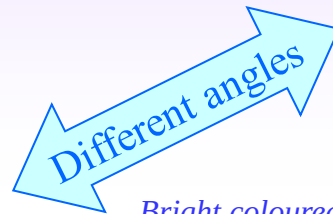


(100) surface



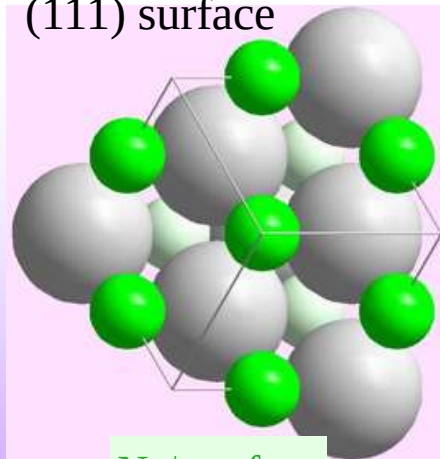
Different position of the (100) cut will give the same configuration of the surface (with shift in origin)

Non-polar surface (100)



Bright coloured atoms are surface atoms and rest are sub-surface

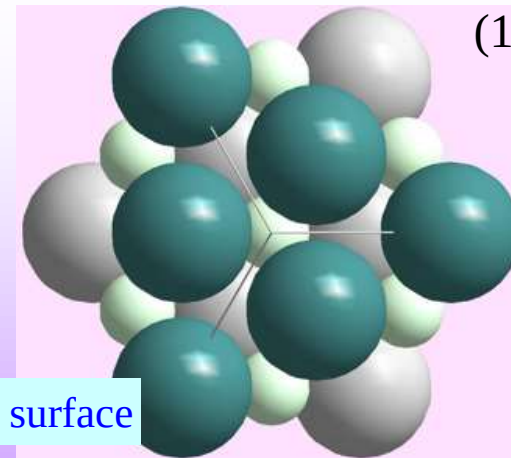
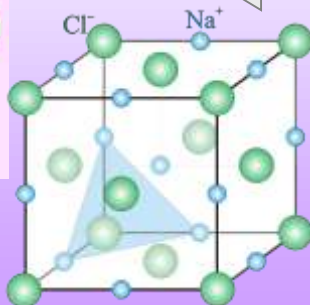
(111) surface



Na^+ surface

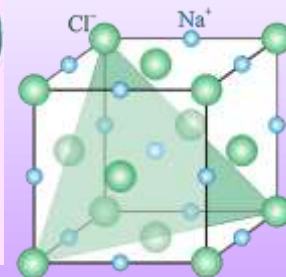


Polar surfaces (111)



Cl^- surface

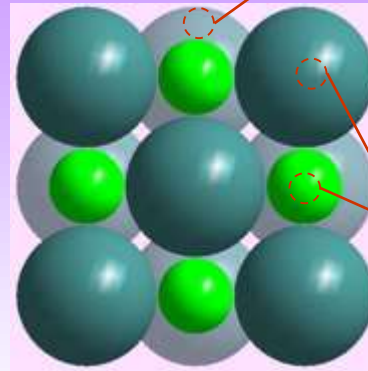
(111) surface



In fact a single cut with a (111) plane will give both these surfaces: one on each side of the cut

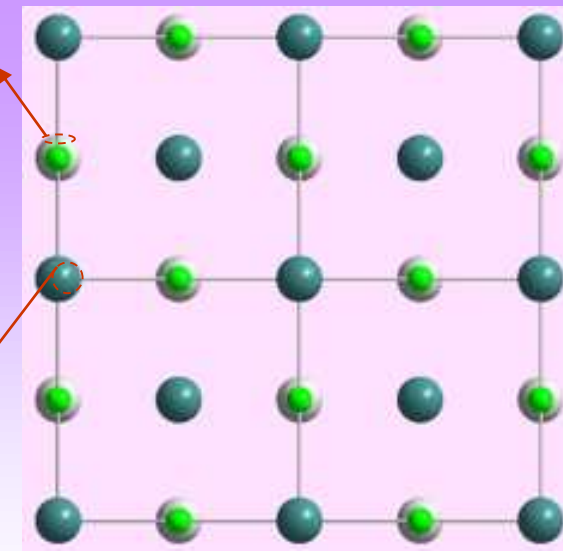
More views & aspects: (100) surface

- The (100) surface is electrically neutral and is 'non polar' $\rightarrow$ equal number of Na^+ and Cl^- ions



Sub-surface ions
(grey)

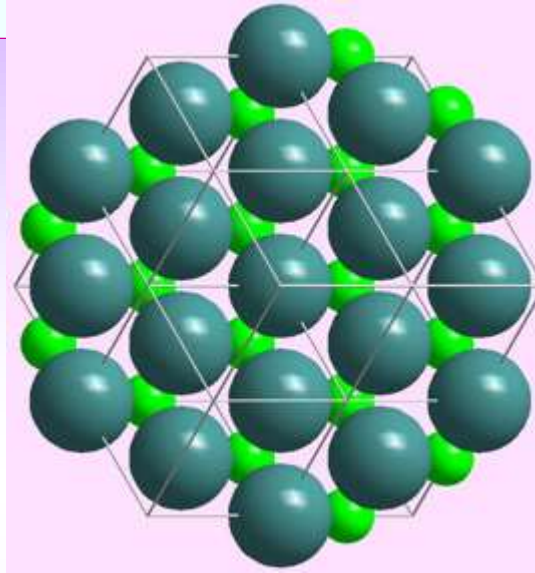
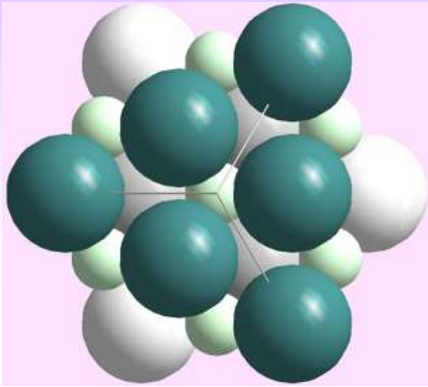
Surface ions



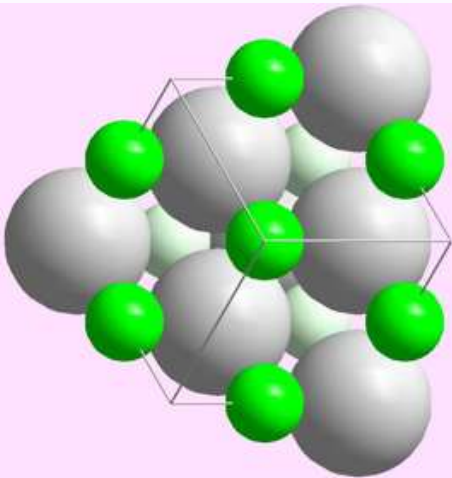
The non-polar (110) surface

More views & aspects: (111) surface

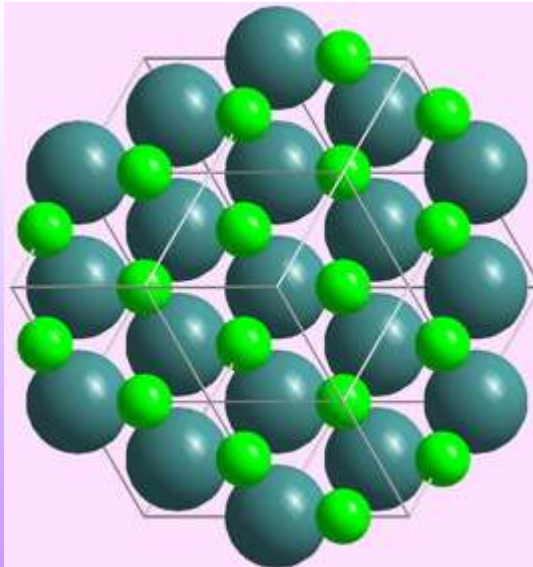
- ❑ The (111) surface is electrically charged and is 'Polar'
→ The Na^+ terminated surface is +vely charged and the Cl^- terminated surface is -vely charged.



Two layers of ions:
(layers below that are hidden for clarity)
dark green (Cl^-) ions are on the surface



Bright green are surface ions and
rest are sub-surface



Two layers of ions:
(layers below that are hidden for clarity)
bright green (Na^+) ions are on the surface

More views using a model: (111) surface



Two (111) cuts



(111) surface consisting of
white balls only
(the white balls on the surface have been
coloured blue for better identification)



(111) surface consisting of
brown balls only
(arising from a deeper cut into
the crystal)

*This structure is similar to the NaCl structure: one major
difference is the size of the spheres here are equal*

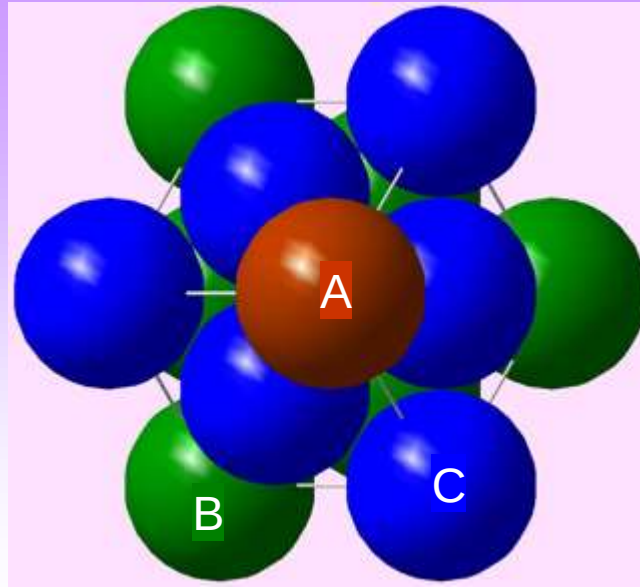
What aspects do we have to consider about the surface?

- ❑ If the surface is neutral or charged (Polar).
 - All cuts in crystals like Cu and Al will lead to non-polar surfaces
 - In NaCl the (100) surface is neutral, while the (111) surface is polar.
- ❑ Though then term **surface** might literally imply only the surface layer, practically we have to *consider a few layers of atoms below the surface* → many aspects of the surface including the surface energy will be determined by the subsurface atomic configuration (*this includes the relaxation after the cut*).
- ❑ We usually (as in this course) consider planar surfaces. In practice a surface could be curved (or may consist of facets or facets with curves). These surfaces are more complex to **treat**. (Few points about faceted surfaces will be considered later).
Another variant of non-planar surfaces are rough surfaces.

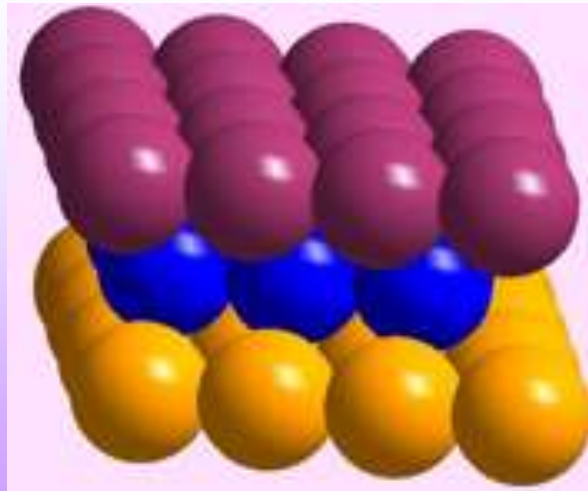
Effect of sub-surface atomic configuration

- ❑ Let us consider examples of two surfaces which have an identical surface configuration of atoms; but, their subsurface configuration (and hence the bonding is different).
- ❑ Let a crystal be constructed of close packed hexagonal layer of atoms →
ABCABCABABCABCABABCABCAB... (refer to topic on [close packing](#) for notation).
- ❑ A cut can be made to create two surfaces →
...ABCABCA | BABCABCAB...
 - Configuration-1 (C1): **ACBACBA...**
 - Configuration-2 (C2): **BABCABCAB...**
- ❑ Both the surfaces are identical if we consider a monolayer, as both of them have hexagonal array of atoms.
- ❑ If we consider two monolayers again there is no difference between C1 & C2
(as $A \rightarrow C$ and $B \rightarrow A$ are identical shifts).
- ❑ If we consider 3 layers the difference between C1 and C2 becomes clear:
 - C1: **ACB...**
 - C2: **BAB...** (the third layer is above the first layer).
- ❑ Hence, the surface energies of these two configurations are expected to be different.

➤ C1: ACB...

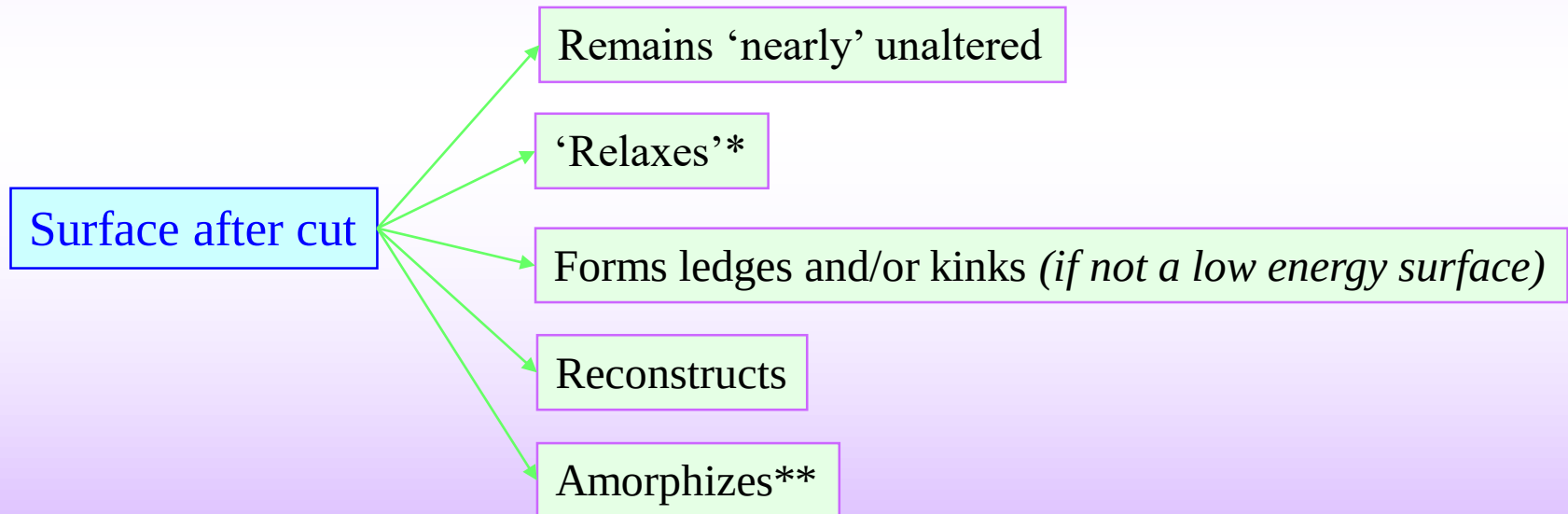


➤ C2: BAB...



What can happen after the cut?

- ❑ After the cut is made, the surface usually does not remain merely as a termination of the bulk.
- ❑ The surface can alter its structure in a few ways so as to reduce the energy. Some of the possibilities are shown below.
- ❑ First we treat the surface as unaltered before considering briefly surface relaxation and reconstruction.



* This implies no structural change but only a change in interatomic spacing (in plane or out of plane)

** Though this is an extreme possibility – it is listed here for completeness

Calculation of the surface energy (case of unaltered surface)

- External surfaces have energy related to the number of bonds broken at the surface → The *broken bond model** of a surface (the surface energy can be calculated using the formula as below). * *The surface remains unaltered after the 'bonds are broken'.*
- As it costs energy to put a surface (Surface energy arising from unsatisfied bonds) the system will want to minimize the surface area → resulting in surface tension (or surface stresses).
- This high energy of the surface (along with the higher freedom for atoms therein) implies that the melting point of the surface will be lower than that of the bulk.
- The diffusivity of species on the surface will also be much higher than in the bulk.

Surface Energy/ unit area (J/m²) → $\gamma = \frac{n_a \cdot n_b \cdot E_b}{2}$

→ No. of atoms/unit area
 → No. of bonds broken/atom
 → Bond energy/bond

As two surfaces are created / bond broken

Understanding the units

$$LHS \rightarrow \frac{[J]}{[m^2]} = \frac{[Nm]}{[m^2]} = \frac{[N]}{[m]}$$

$$RHS \rightarrow \left(\frac{\#atoms}{[m^2]} \right) \left(\frac{\#bonds}{atom} \right) \left(\frac{[J]}{bond} \right) = \frac{[J]}{[m^2]} = \frac{[N]}{[m]}$$

No of bonds broken

Surface free energies (γ) of some crystals (J/m²)

NaCl	LiF	CaF ₂	MgO	Si	Ag	Fe	Au	Cu
0.30	0.34	0.45	1.2	1.24	1.14	1.4	1.4	1.65

Note the values are ~ 1 J/m²

Surface Structure

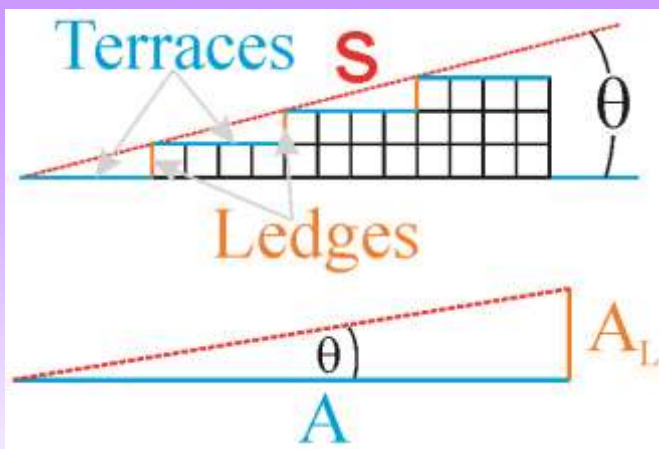
- ❑ Low energy surfaces (usually crystallographically low index planes) can be atomically flat below the roughening temperature*.
- ❑ Surfaces which are at an inclination w.r.t to these low index planes can have a structure consisting of terraces and ledges**. These ledges are *structural ledges* which accommodate the inclination. These ledges can be thought of as a defect in the surface (*i.e. a defect in a defect!*).
- ❑ The surfaces of the terraces and ledges are low energy surfaces → which reduce the energy of the system.
- ❑ Further, a surface having any arbitrary orientation can be made of *Ledges* and *Kinks*. Kinks can be thought of as a defect in the ledge (*i.e. a defect in a defect in a defect!*).
- ❑ These ledges and kinks are sites, where atoms of the growing crystal or foreign atoms can preferentially attach → an atoms attaching to these sites would be better bonded as compared to the flat terrace. *This can increase the crystal growth rate.*

[Click here to know more on the topic of defect in a defect](#)

* A temperature above which the model being described is not valid.

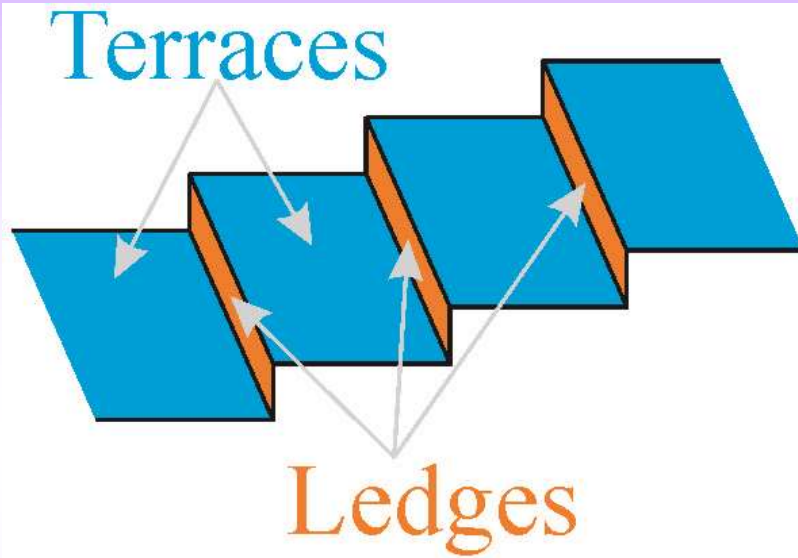
** This is the ledge and kink model of a surface → actual surfaces are made of atoms.

Continued...

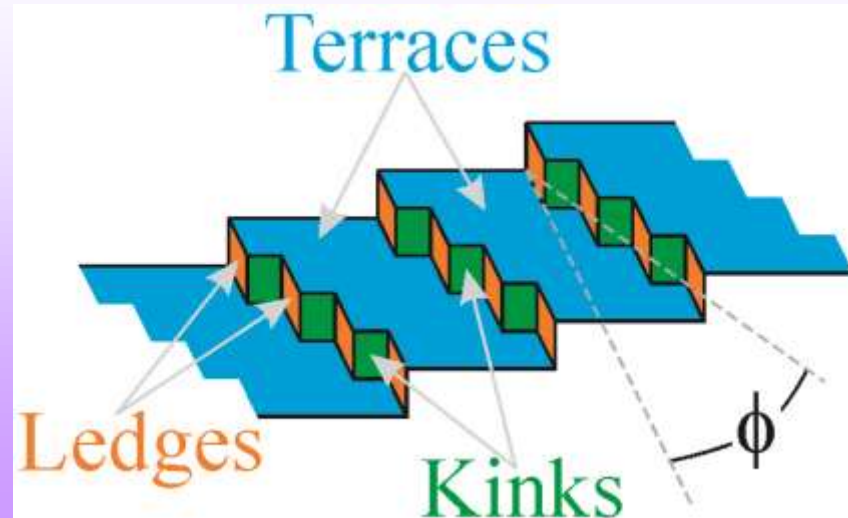


Description of a surface inclined with respect to a low index surface (*expected to be a low energy surface*) in terms of ledges and kinks

Inclination with one tilt



Inclination with double tilt

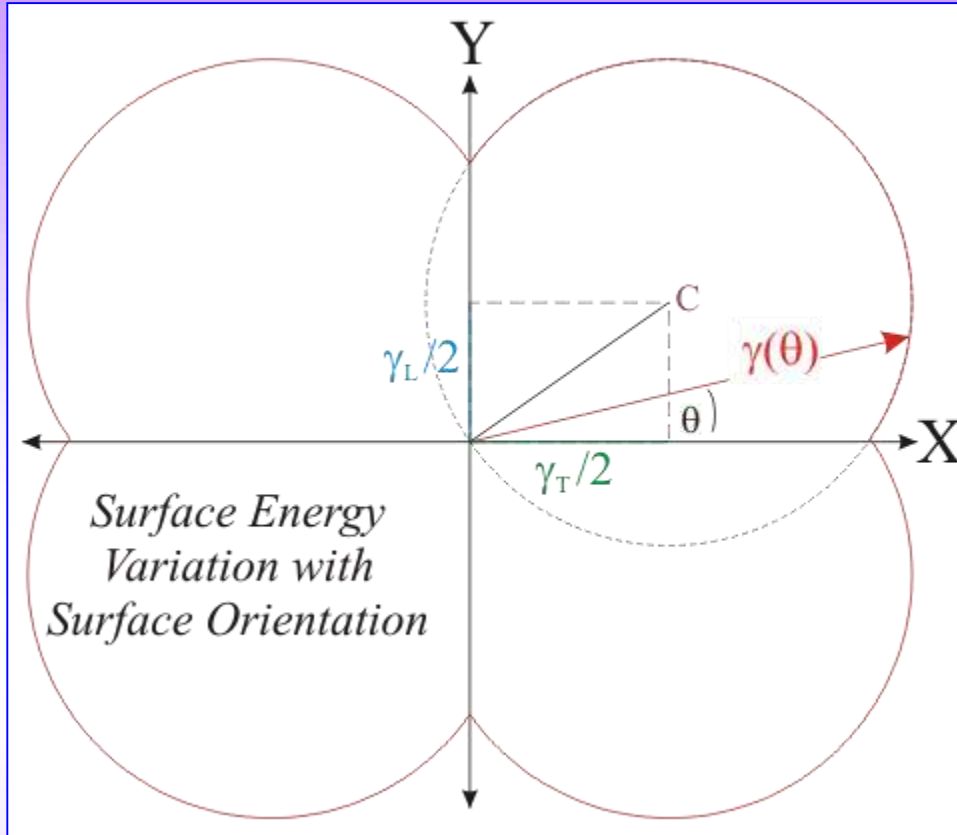


Note: this is a model to visualize surfaces and should not be taken literally

Equilibrium shape of a crystal

- ❑ If the surface energy is isotropic then the equilibrium shape of a crystal will be spherical.
- ❑ Usually the surface energy will be direction dependent (i.e. some crystallographic planes- usually the low index planes- will have lower energy as compared to certain other planes).
- ❑ The changing energy of the planes with changing direction of the plane normals is captured in a polar plot of the surface energy known as the γ -plot. I.e., given the γ -plot one can determine the energy of any plane (if it were the surface).
- ❑ The equilibrium shape of a crystal be bounded by low energy surfaces.
- ❑ The shape of the crystal can be determined by the **Wulff construction** → which is constructed from the polar plot of surface energy (the γ -plot).
- ❑ **This shape will have the point group symmetry of the crystal.**
- ❑ The γ -plot typically has sharp cusps $\Rightarrow$ any orientations (of surface planes) around a low energy plane will have a 'steeply varying' higher energy.
- ❑ In the initial treatment entropic effects are ignored → when included the cusps could be less prominent and could even disappear for high index planes.

γ -plot or Wulff plot



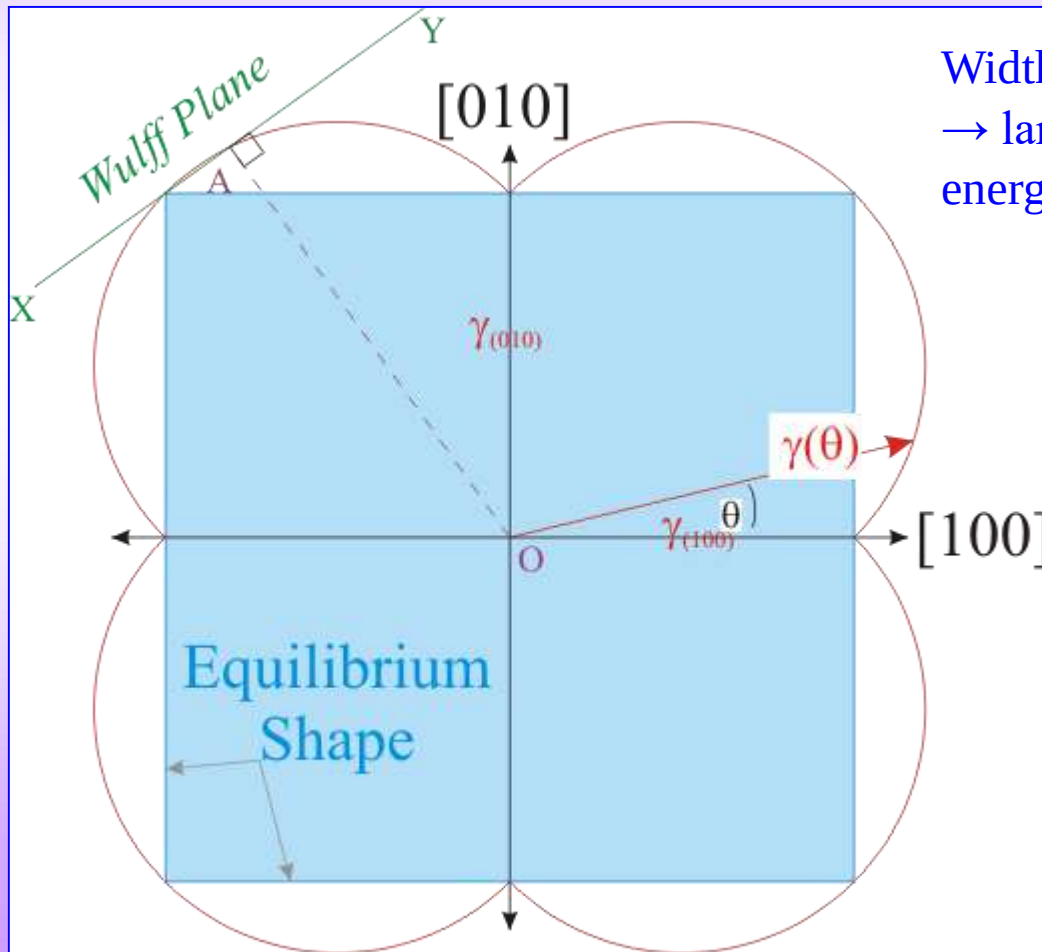
Due to the orientation dependence of surface energy ‘there will be a tendency to rotate’ a high energy surface to a low energy surface $\rightarrow$ the origin of the surface torque

The orientation dependence of γ will tend to rotate the surface to a low energy orientation $\rightarrow$ produce a *torque* on the surface

$$\text{Torque term} \rightarrow \frac{\partial(\gamma)}{\partial(\theta)}$$

From γ -plot to EQUILIBRIUM SHAPE OF CRYSTAL $\rightarrow$ the Wulff construction

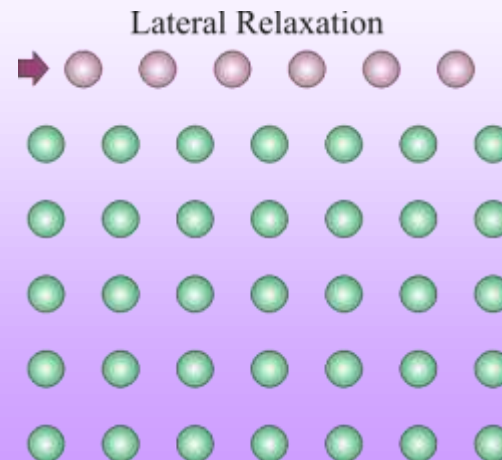
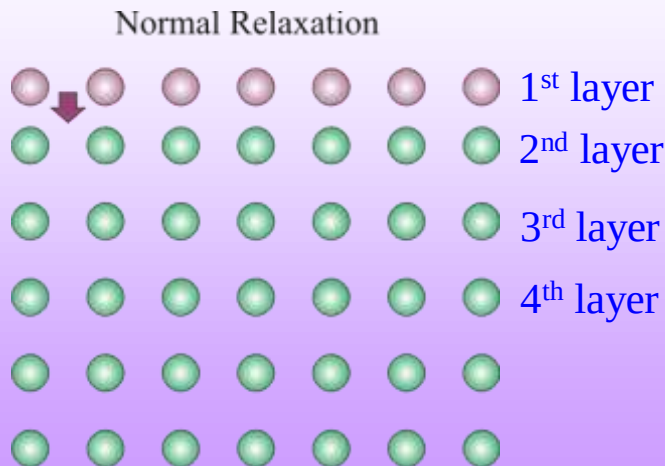
- ❑ Draw radius vectors from the origin to intersect the Wulff plot (OA in Figure)
- ❑ Draw lines $\perp$ to OA at A (line XY)
- ❑ The figure formed by the inner envelope of all the perpendiculars is the equilibrium shape
- ❑ Wulff plot with sharp cusps $\Rightarrow$ equilibrium shape = polyhedron



Width of the crystal facets $\propto 1/(\text{surface energy})$
 $\rightarrow$ largest facets are the ones with lowest energy

Surface Relaxation

- ❑ In relaxation, entire layer(s) of surface atoms move relative to the bulk.
- ❑ This can be normal to the surface (→ resulting in small lattice spacing on the surface as compared to the bulk) or parallel to the surface (or both).
- ❑ Normal to the surface relaxation is most commonly found in metals.
- ❑ The first few layers are affected by the relaxation process (with a progressively decreasing effect inward from the surface).



- ❑ In relaxation the spacing between the 1st and 2nd layer of atoms is reduced. The lower the atomic packing (& density) of the surface, larger will be the magnitude of the contraction. This is a result of the atoms ‘wanting’ to maximize their bonding.
- ❑ While it is clear that there is a contraction between the top two layers, the situation is not clear between the 2nd and 3rd layers. In some cases

Calculated surface relaxation for Cu surface layers

Surface	$\Delta d_1 / d$	$\Delta d_2 / d$	$\Delta d_3 / d$	$\Delta d_4 / d$
(100)	0.129	0.033	0.008	0.001
(110)	0.196	0.047	0.019	0.003
(111)	0.005	0.009	0.001	

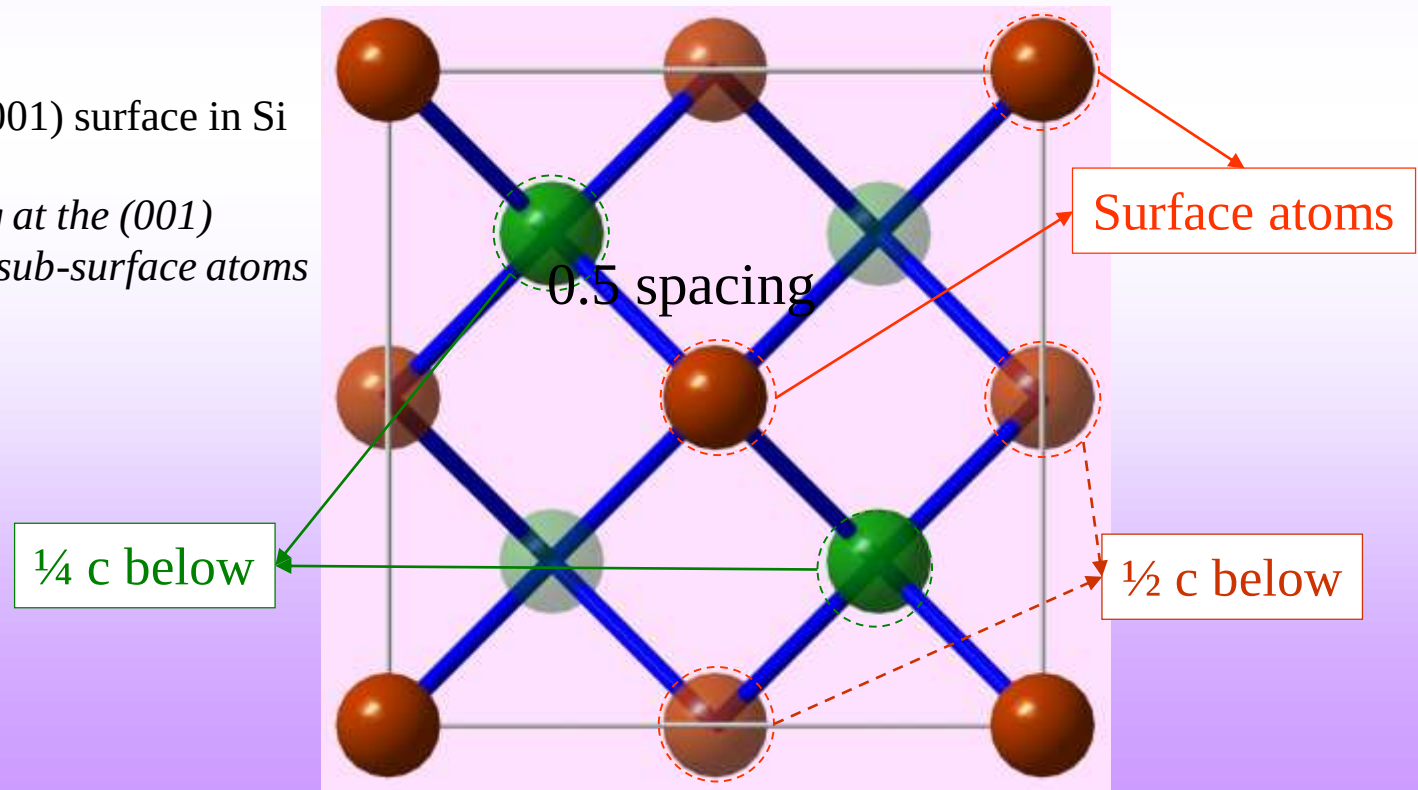
- ❑ This may happen in addition to the relaxation previously mentioned. However, most clean surfaces are not reconstructed. A term often used synonymously with reconstruction is restructuring.
- ❑ Reconstruction may affect one or more layer(s) on the surface. The surface may remain planar after reconstruction or may develop undulations (i.e become 3D).
- ❑ Leads to the change in the structure of the surface layer(s).
- ❑ May also change the symmetry of the surface layer
→ e.g. the Pt (100) surface may reconstruct to a surface with hexagonal symmetry (Pt has cubic symmetry).
- ❑ The reconstructed surface may have equal number of atoms (per unit area) as the bulk (conservative reconstruction) or have more or less atoms than the bulk (non-conservative reconstruction).
- ❑ The situation (of reconstruction) becomes more complicated if surface is not clean (i.e. has adsorbate atoms). Adsorbate may induce reconstruction → “**adsorbate induced reconstruction**”.
- ❑ Important examples of reconstruction include: (100) faces of Pt, Au, W, Mo; (110) faces of Pt, Ir.

Examples of Surface Reconstruction

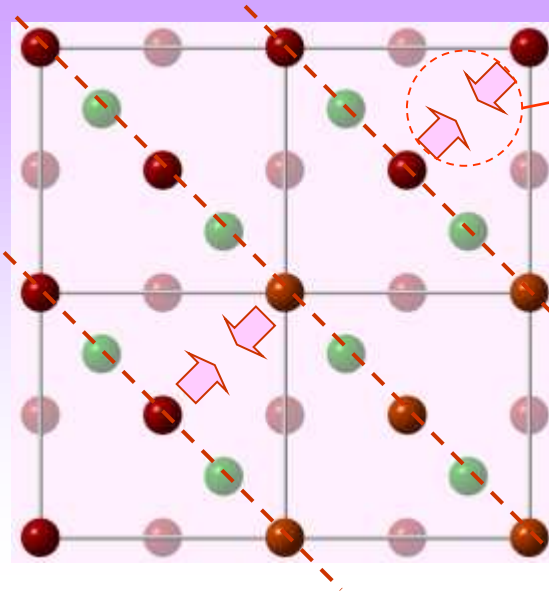
- ❑ Si is a well studied system with regard to reconstructions.
- ❑ Many (~ 5) surface layers may be affected by the reconstruction process.
- ❑ Each Si atom is bonded to 4 other atoms via the sp^3 hybridized bond.
- ❑ If a (100) cut is made to create a surface, 2 out of these 4 bonds are broken $\rightarrow$ costing high energy to the system.
- ❑ In the simplest reconstruction possible (called the 2×1 reconstruction) rows of surface atoms come nearer in pairs to 'increase the bonding'.

Reconstruction of the (001) surface in Si

Let us begin by looking at the (001) surface along with the sub-surface atoms

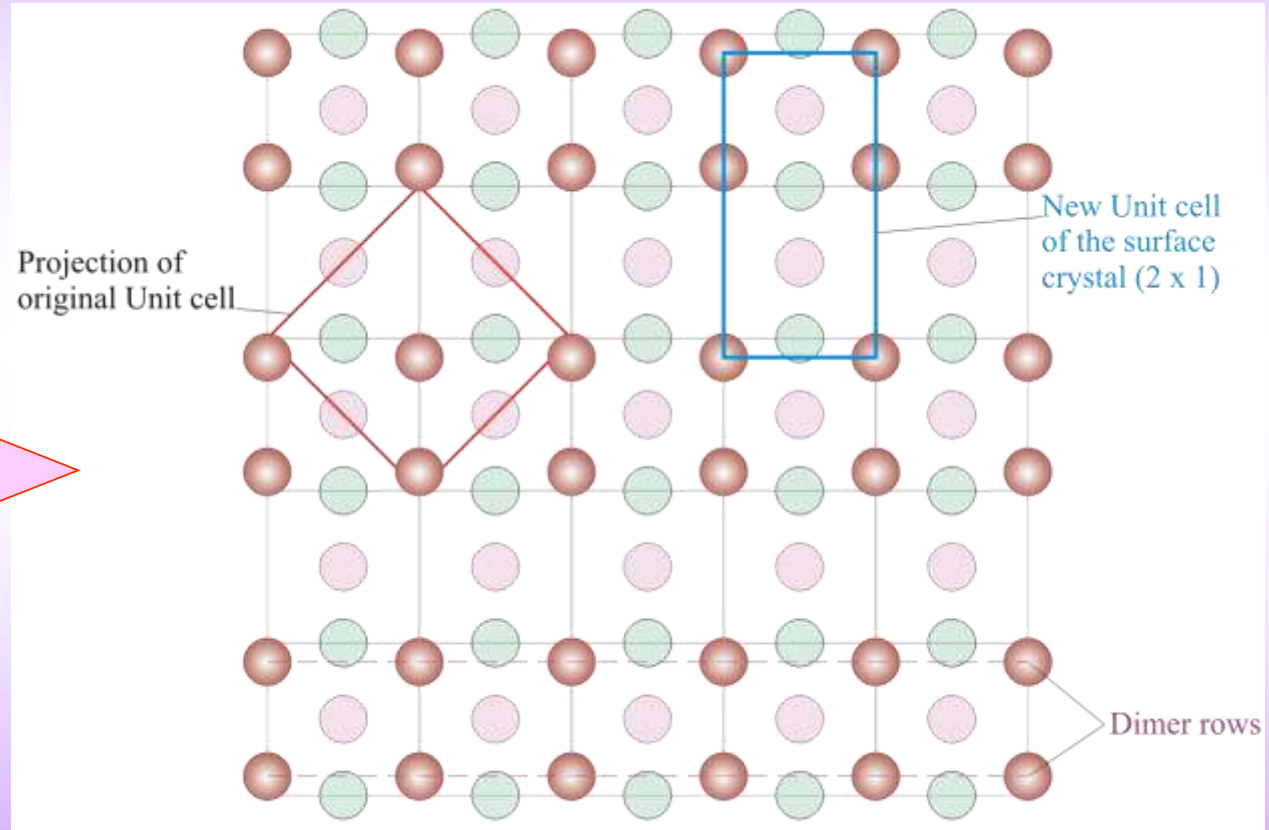


The figure below shows only up to 3 layers of atoms ($z = 0, \frac{1}{4}, \frac{1}{2}$)



Direction of motion of atomic rows

After reconstruction of surface



Only qualitative effect on the surface atoms shown-
in reality a few layers below the surface will also be affected

Warning: only a schematic

INTERFACES

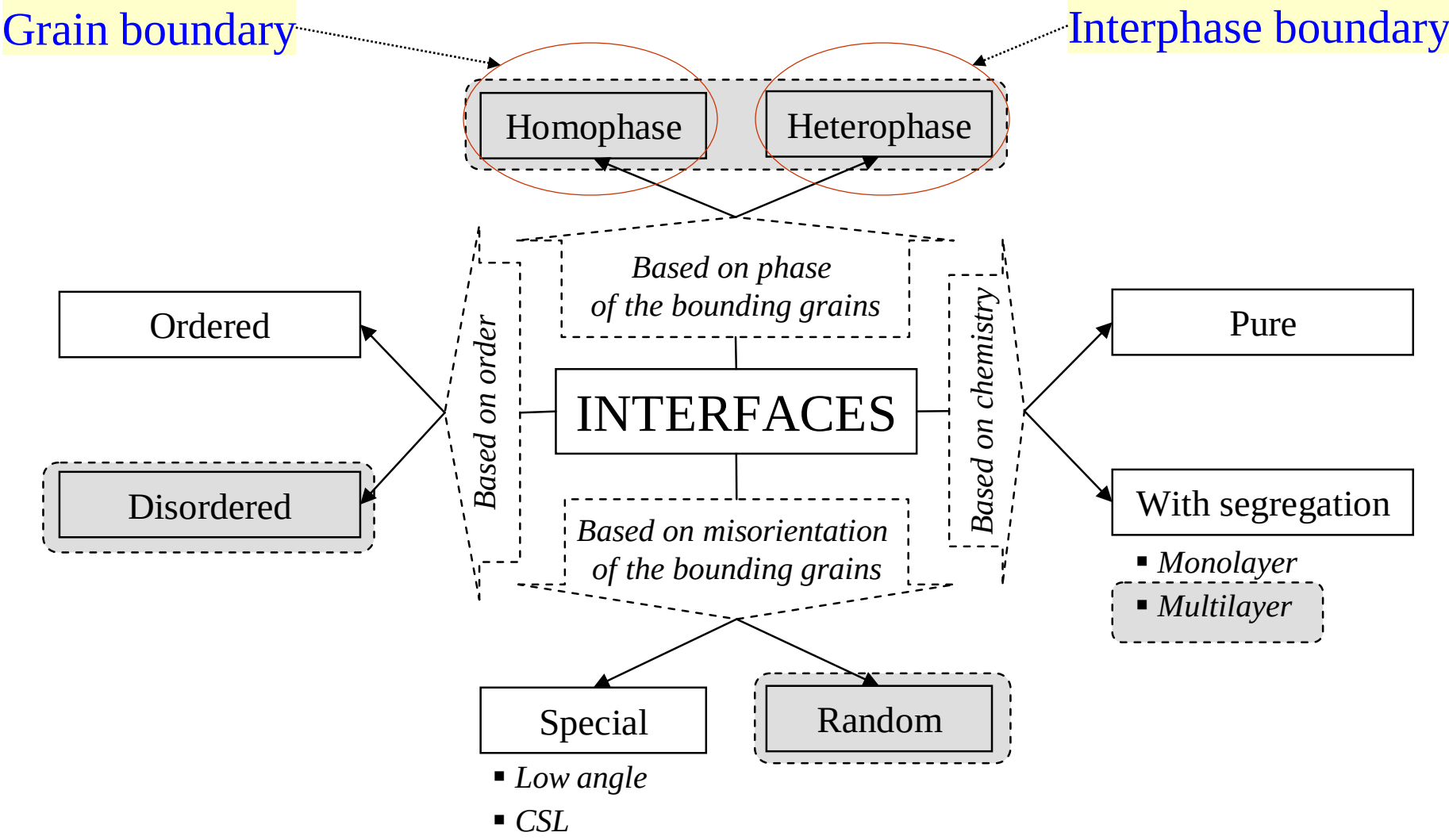
Interfaces

- ❑ An **interface** in general separates two materials *(or material and vacuum)*.
- ❑ A **surface** is a special kind of interface separating a material from vacuum (or air/gas).
- ❑ An interface can be understood from many points of view, as below.
 - If the material on either side of the interface is similar or different
→ **homophase** or **heterophase** interface.
 - If the interface is **ordered** or **disordered**.
 - If the interface is pure or has some ‘**segregated**’ material.
(The material not soluble in either of the two bounding phases will tend to segregate to the interface- the region of high energy).
 - If the misorientation between the bounding crystals is **random/arbitrary** or **special**.
 - If the interface is **Coherent**, **Semi-coherent** or **Incoherent**.

Diagram related to
this on next slide

- ❑ Internal interfaces play a very important role in determining the behaviour of materials. Some examples of the importance of internal interfaces are as below.
 - **Hot shortness** (S segregation to grain boundary region with the formation of FeS → brittle in hot rolling conditions).
 - **Enhanced diffusivity** of Ga along the grain boundaries in Al
(When polycrystalline Al foil sample in tension is exposed to Ga (liquid) → it separates along grain boundaries).
 - **Creep** caused by grain boundary sliding.

A classification of interfaces based on multiple viewpoints



Grain Boundary

- ❑ A polycrystalline material consists of single crystalline regions bonded to each other. The interface between the two **misoriented** crystallites is called the **Grain Boundary**. The formation of the polycrystalline material along with the grain boundary is best visualized during the crystallization of a melt\$.
- ❑ The grain boundary region may be distorted with atoms belonging to neither crystal → costs energy to the system (*the **grain boundary energy***).
- ❑ The ‘thickness’# may be of the order of few atomic diameters.
- ❑ The crystal orientation changes abruptly at the grain boundary*.
- ❑ In a low angle boundary the orientation difference is $< 10^\circ$
(usually this is in the range of $10\text{-}15^\circ$ - the reason for this limitation in angles will be considered later).
- ❑ In the low angle boundary the orientation change is not so drastic as the high-angle boundary → can be described as an **array of dislocations**.
- ❑ Grain boundary energy is responsible for grain growth on heating $\sim (>0.5T_m)$ → Large grains grow at the expense of smaller ones**.
- ❑ Grain boundaries usually have an order of their own (*different from the order in bounding grains*); however, in special cases they may even be amorphous.

the concept of a grain boundary being a 2D defect is an idealization

* at a high angle grain boundary

** more about this in the chapter on Phase Transformations

\$



Video: Dendritic growth of crystal from melt

How many parameters characterize grain boundary in a material?

- ❑ A grain boundary can be created by *(in a thought experiment)*:
taking two identical crystals → mis-orienting one crystal w.r.t to another → cutting both the crystals with a plane → joining them.
- ❑ Number of Degrees of Freedom (DOF) of a grain boundary:
 - 5 Macroscopic (at the geometric level)
 - 4 Microscopic (at the unit cell/atomic level).
- ❑ **Macroscopic DOF:**
 - Misorientation between the two crystals can be created by taking a **rotation axis** and rotating one crystal w.r.t to another by a certain **angle** ($\theta_{\text{misorientation}}$). The axis is defined by **two** independent degrees of freedom* + angle of rotation (θ) being **one** DOF = **three** DOF for rotation.
 - The orientation of the interface plane (grain boundary plane) is designated by its normal. The plane normal is a line with **two** DOF*.
 - *Total (Macroscopic) DOF = 3 for misorientation + 2 for plane normal = 5.*
- ❑ **Microscopic DOF:**
 - Having fixed the orientation of the interface plane, the origin of one crystal (atomic level) can be displaced with respect to another by a **vector** (having **2** DOF)
 - Further, the interface plane can be positioned anywhere within the crystal by moving it along the normal vector (without changing its orientation) → this gives us (another) **2** DOF (*one for each of the two crystals*).
 - ➔ *4 Microscopic DOF.*
- ❑ *Grant total DOF for a grain boundary = 5 macroscopic + 4 microscopic = 9 DOF.*

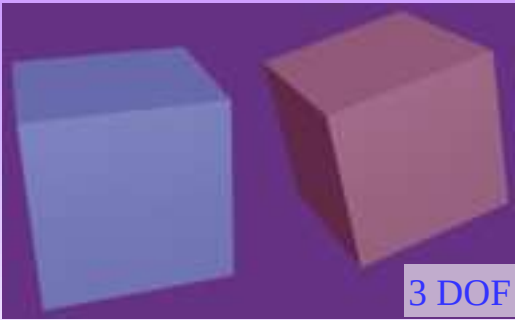
DOF implies the number of parameters we need to specify.

* A direction can be defined by the three direction cosines (α, β, γ), there is one equation connecting the direction cosines ($\text{Cos}^2\alpha + \text{Cos}^2\beta + \text{Cos}^2\gamma = 1$) ⇒ only two of them are independent.

Macroscopic DOF



Choose two identical crystals



Rotate one w.r.t. to another

3 DOF

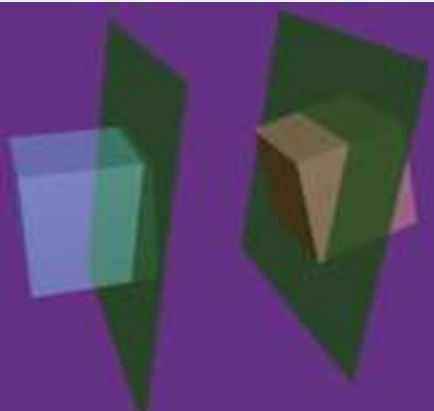


Rotation Axis



2 DOF

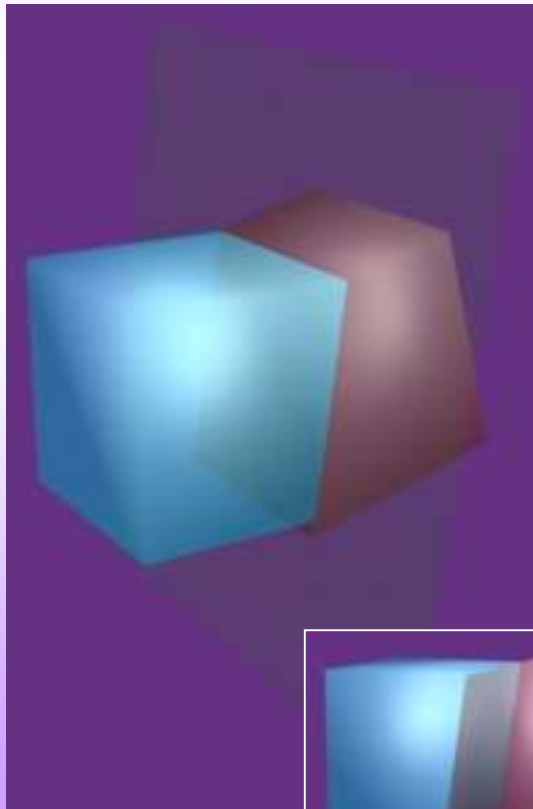
Chose a plane of cut



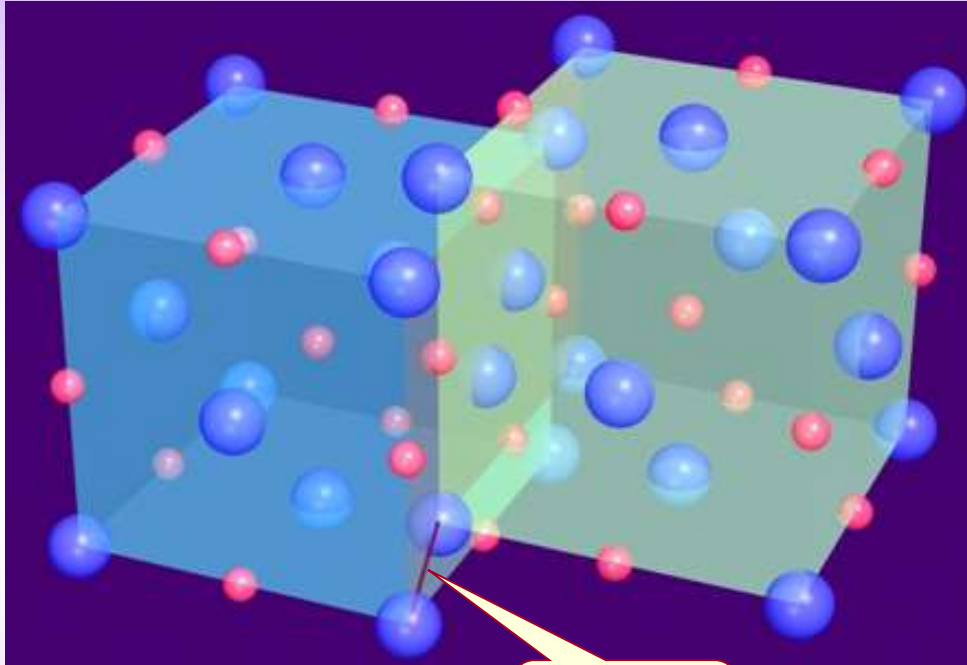
Cut both the crystals using this plane



Join the crystals to create a grain boundary

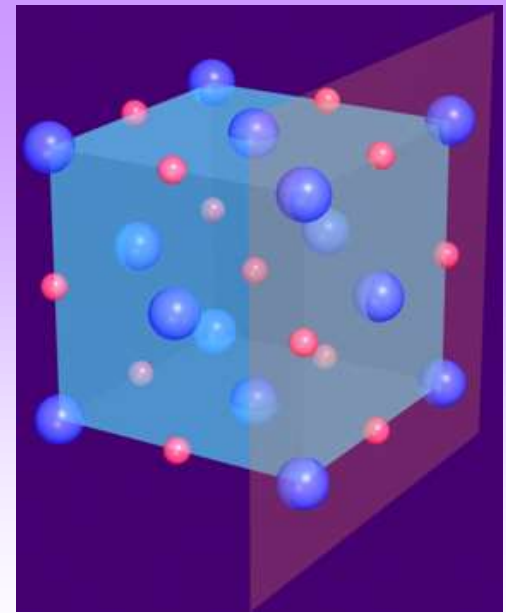


Microscopic DOF:

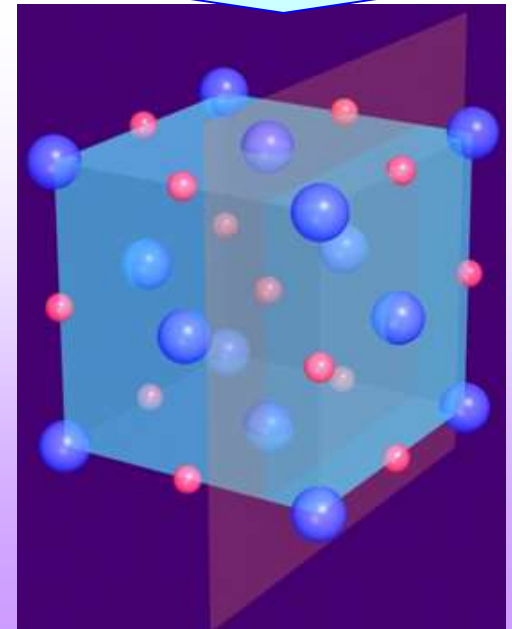


Shift along
interface
giving 2DOF

Warning: *this is purely a schematic to illustrate the shift vector*

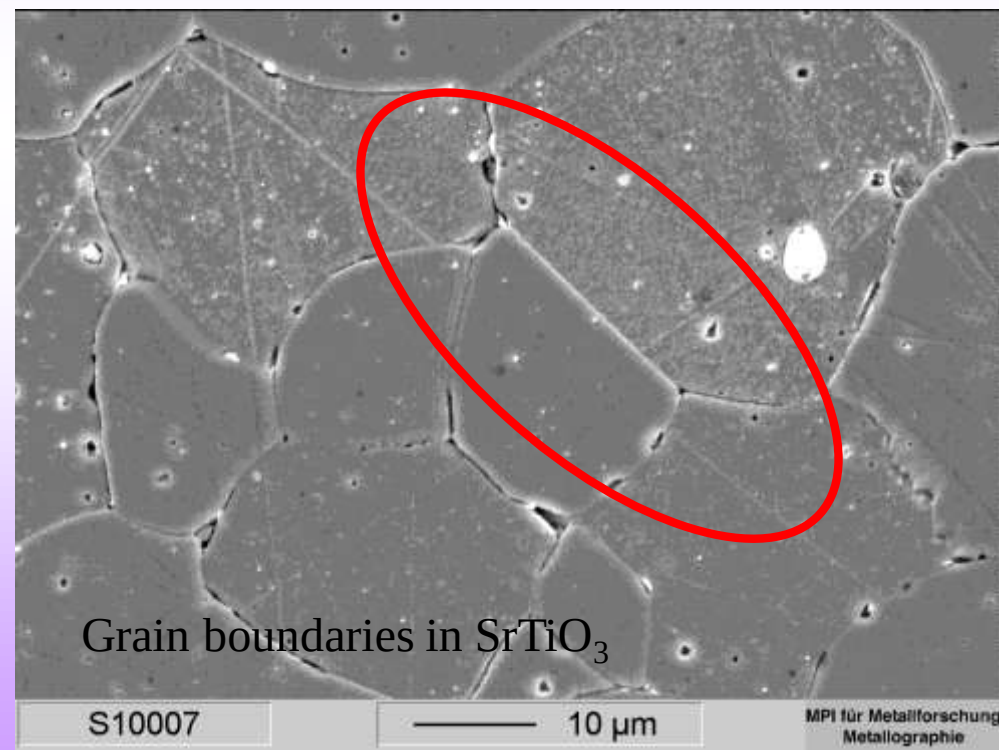


Shift of cut plane
perpendicular to the
interface: 1DOF per crystal



- ❑ As an example the grain boundary structure in SrTiO_3 (a ceramic) is shown in the micrograph below).
- ❑ Some typical values of GB energies of metals are listed in the tables below. They are of the order of $0.1\text{-}1 \text{ J/m}^2$.

Type of boundary	Energy (J/m^2)
Grain boundary between BCC crystals	0.89
Grain boundary between FCC crystals	0.85
Interface between BCC and FCC crystals	0.63



Crystal	γ_β [mJ m^{-2}]	T [$^\circ\text{C}$]	γ_b/γ_{sv}
Sn	164	223	0.24
Al	324	450	0.30
Ag	375	950	0.33
Au	378	1000	0.27
Cu	625	925	0.36
$\gamma\text{-Fe}$	756	1350	0.40
$\delta\text{-Fe}$	468	1450	0.23
Pt	660	1300	0.69
W	1080	2000	0.41

- ❑ The average no. of nearest neighbours for an atom in the grain boundary of a close packed crystal is less than 12 (*i.e. atoms are 'bonded less efficiently' as compared to the bulk*).
- ❑ Grain boundaries hence have a higher 'free volume' as compared to the crystal interior $\Rightarrow$ species not soluble in the crystal tend to segregate to the grain boundaries. (Further the segregated atoms could form a separate phase along the grain boundaries).
- ❑ This segregation (with or without the formation of a separate phase) can often result in undesirable properties (especially during deformation at high temperatures). **Hot shortness*** is an example of this.
- ❑ During phase transformation grain boundaries (just like many other defects) can be preferred sites for heterogeneous nucleation.
- ❑ A striking example of this process leading to drastic alteration in the properties of the material is the precipitation of cementite in hypereutectoid steel (say 0.9%C steel). During transformation of hypereutectoid steel the proeutectoid cementite precipitates all along the grain boundaries and can form a continuous film. As cementite is a hard but brittle phase, this provides a continuous pathway for crack propagation (say during impact loading of the material). Making the steel very brittle $\rightarrow$ in spite of the small volume fraction of the brittle phase.

* *Hot shortness: If Sulphur is present in steel (even in low percentages) it combines with iron to form a low melting iron sulphide $\rightarrow$ leading to the material literally breaking apart while hot working (say rolling). This problem is usually alleviated by the addition of manganese which has a higher affinity for S than iron and forms manganese sulphide (which has a higher melting point than the hot rolling temperature of steel)*

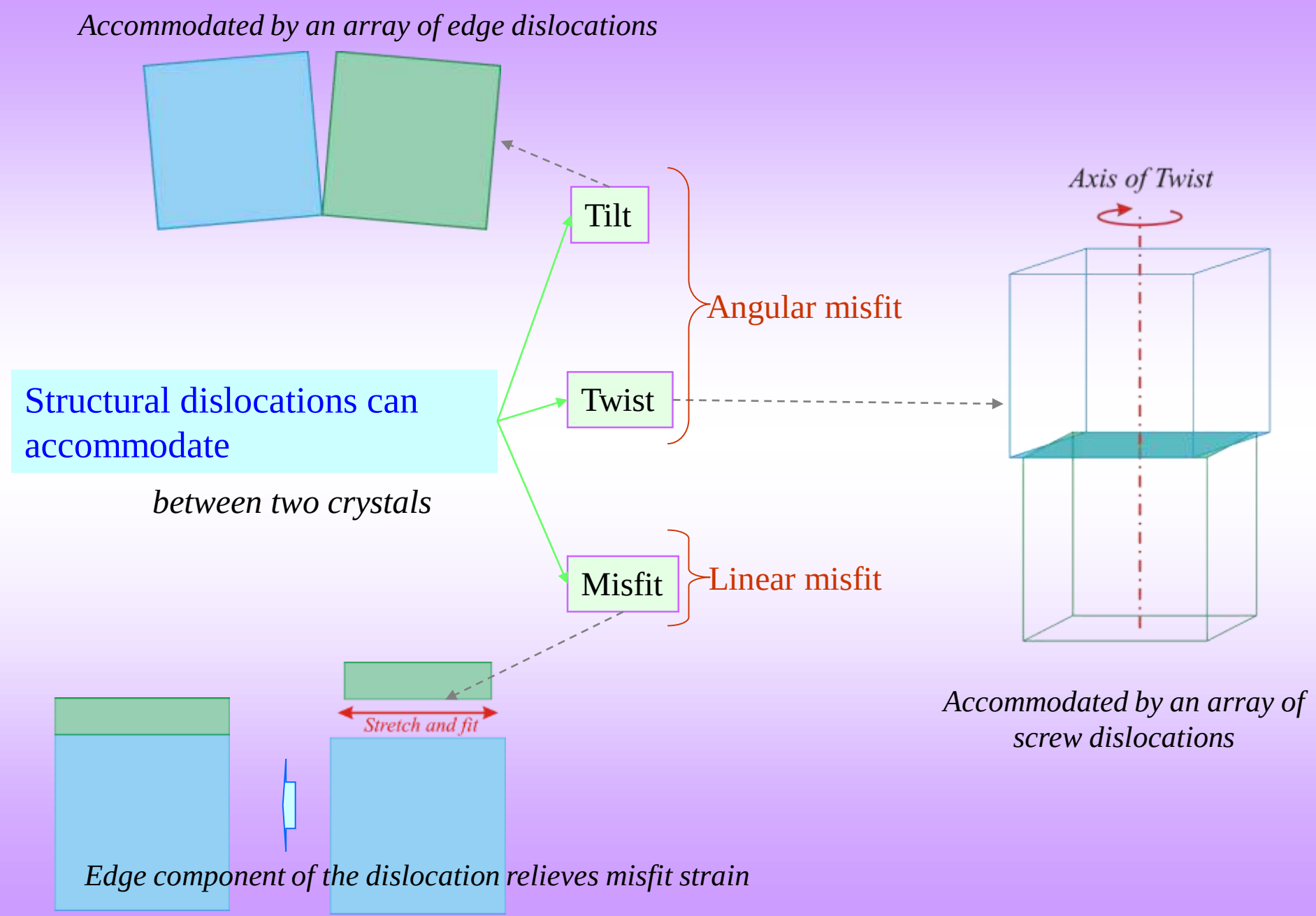
Accommodation of angular or linear misfit by Structural Dislocations

- ❑ One crystal may not match with an adjacent crystal (across an interface) in two basic ways:
 - the two crystals may be rotated w.r.t each other (angular misfit → misorientation)
→ e.g. low angle grain boundary
 - the crystals may not have the same lattice parameter (linear misfit)
→ e.g. epitaxial films.
- ❑ 'Ofcourse' we can have a combination of the above two effects.
- ❑ Small misfits (angular or linear) between crystals can be accommodated by **Structural Dislocations***. (Misorientation angle less than about 10°)
- ❑ In the presence of dislocations the interface breaks up 'roughly' into two regions:
 - **regions of good fit**- where lattice planes are continuous and
 - **regions of bad fit**- which are the dislocation lines/cores.

➔ The overall energy of the system is reduced by this method.
- ❑ If the lattice mismatch between a precipitate and the matrix is small, the interface can be coherent, when the size of the precipitate is small. As the precipitate grows, the misfit strain energy becomes large and the interface gets decorated with dislocation loops (i.e. the misfit is accommodated by interfacial dislocations).
- ❑ In addition, structural dislocations arise for example in the case where, the mirror plane of a twin is not the habit plane (*incoherent twins*).

** Structural defects (including dislocations) play a role in the structure or constitution of the material. They are typically localized to certain regions. The other kind of dislocations are 'random' or 'statistically stored' dislocations (which can be randomly present throughout the grain).*

Structural dislocations can accommodate angular (tilt, twist or a combination) or linear misfits



- ❑ In a low angle grain boundary the misfit is accommodated by an array of edge or screw dislocations.
- ❑ A general boundary could have tilt and twist components.

Low angle grain boundaries (*misorientation angle* $< 10^\circ$)

Two extremes

TILT

An array of edge dislocations

- Rotation axis lies on the boundary plane

TWIST

An array of screw dislocations

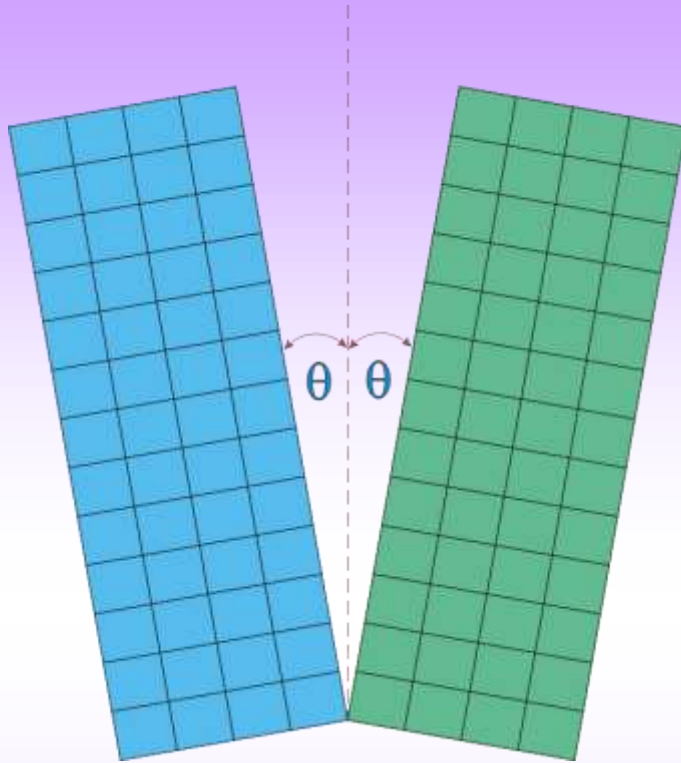
- Rotation axis lies $\perp$ to the boundary plane

Note: these are **structural dislocations*** and their role in plasticity by slip is very different as compared to **random** (“statistically stored”) dislocations

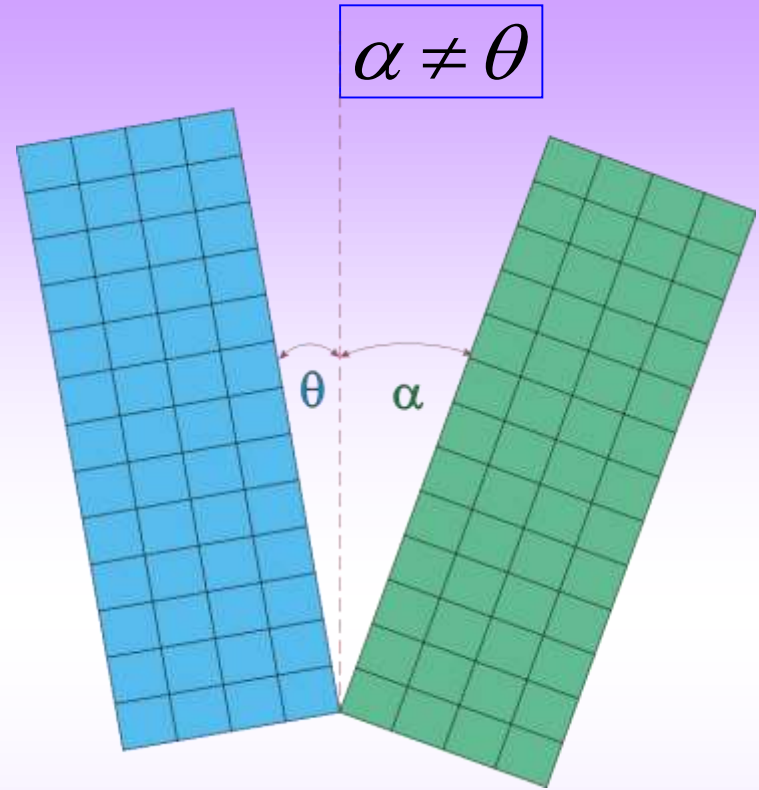
* With respect to interfacial misfit dislocations, Vellinga et al. (Acta Mater. 45 (1997) 1525) have said (which is an interesting viewpoint as well):

“These dislocations, unlike the dislocations in the bulk, are not defects but an integral part of the interfacial structure”.

Symmetric and Asymmetric tilt boundaries



Symmetric tilt boundary



Asymmetric tilt boundary

- In this chapter we will only consider symmetric tilt boundaries (i.e. the semi-angle of tilt *with respect to the interface plane*, is equal for both the crystals).
- Similarly we can have symmetric and asymmetric twist boundaries.
- The reason for the distinction between symmetric and asymmetric tilt (or twist) will become clear soon (based on the interface dislocation structure).

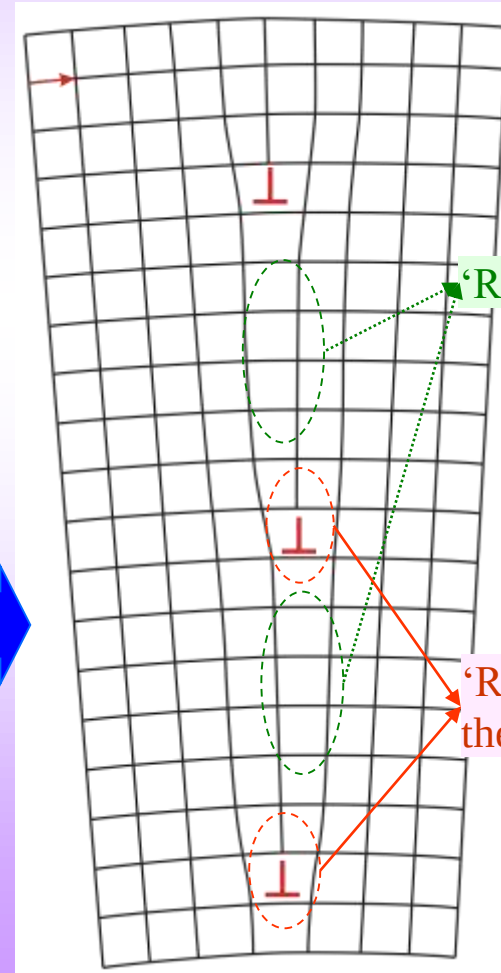
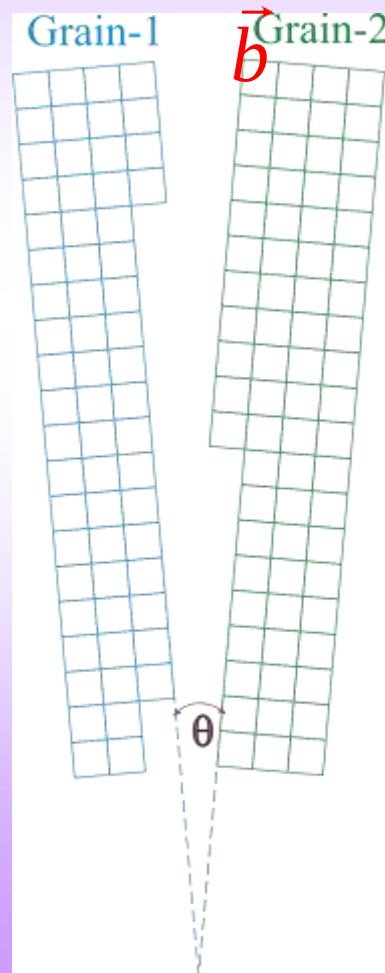
Low angle tilt grain boundary (LAGB) (symmetric)

- ❑ A grain boundary can be constructed by truncating two crystals and putting them together.
- ❑ The structure of low angle grain boundaries can be visualized as an array of dislocations:
 - Tilt boundaries consist of an array of **edge dislocations***
 - Twist boundaries → array of **screw dislocations**.

* There are alternate models based on an array of disclinations.

The original dislocation model is due to W.G. Burgers (brother of J.M. Burgers)

These dislocations are also considered as stress-annihilator crystal lattice dislocations or anti-coherency dislocations.



‘Regions of good fit’

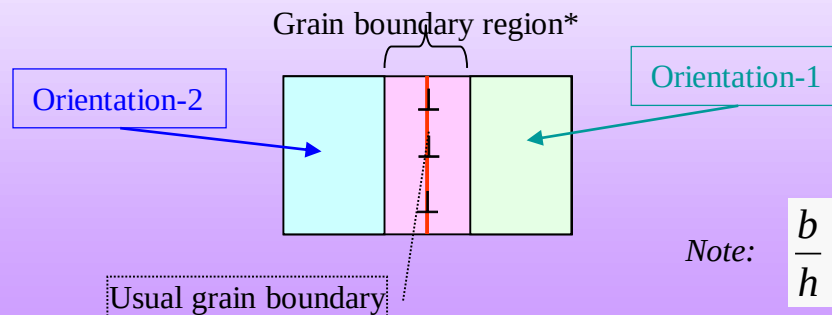
‘Regions of bad fit’- the dislocation cores

Note here we have drawn the **b** close to the dislocation line (has to far away!!)

To construct the triangle to calculate the dislocation spacing given a misorientation, we have to use two successive dislocations provided by the same crystal (crystal on the left in the figure). Hence, this is twice the spacing between the dislocations.

$$\frac{b}{2h} = \sin\left(\frac{\theta}{2}\right) \quad \frac{b}{h} \sim \theta$$

- As $\theta \uparrow \rightarrow h \downarrow$ (as the misorientation angle increases the dislocations get closer).
- As the dislocations come very near the cores of the dislocations will overlap and their individual identity will be blurred.
- This usually happens when $\theta \sim 10-15^\circ$. Hence, grain boundaries with misorientation angle ($> 10-15^\circ$) are called *High-angle Grain boundaries* and cannot be visualized as an array of dislocations.

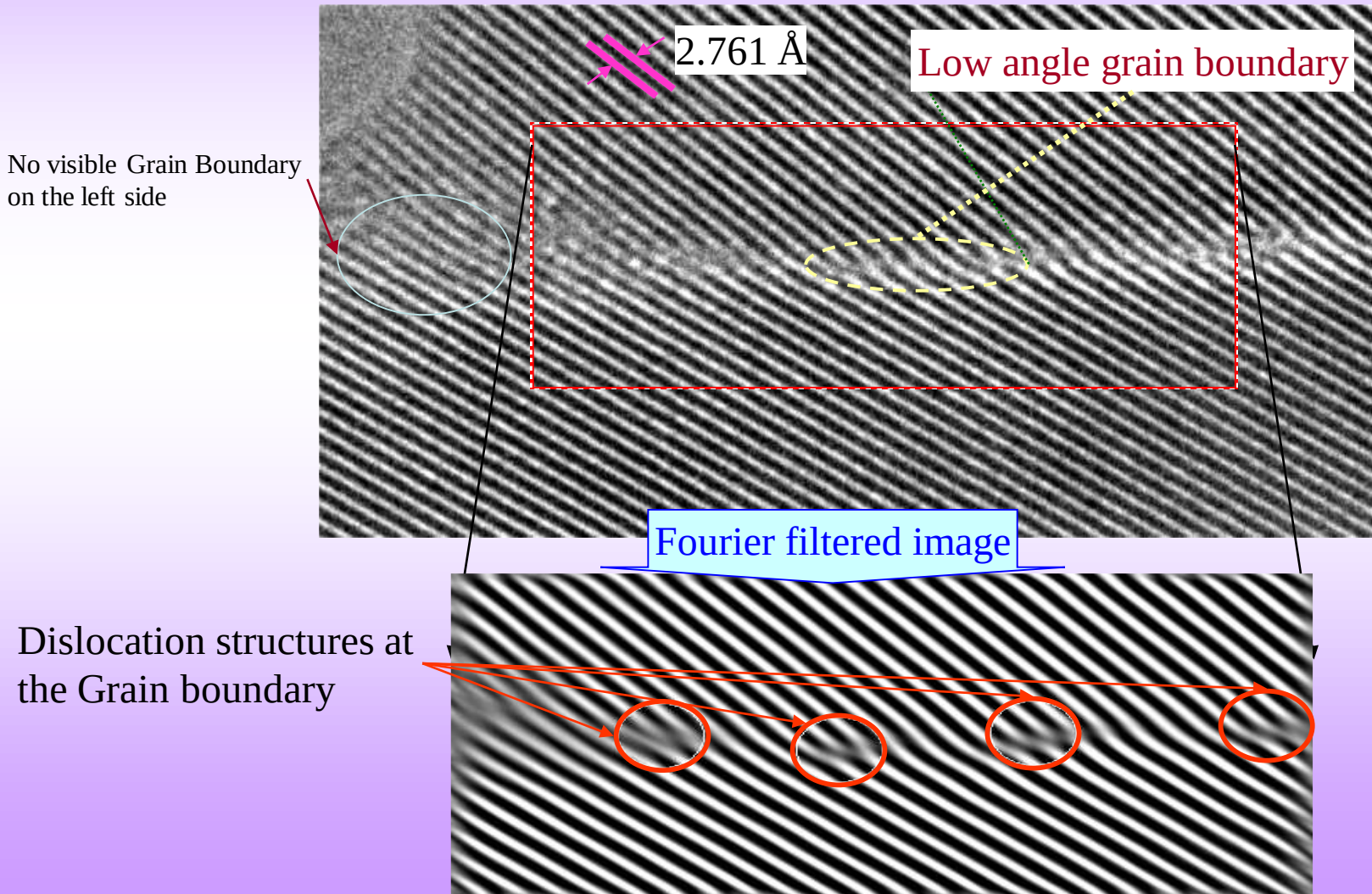


Note: $\frac{b}{h} = \tan \theta$

* Where most of the misorientation and energy is 'stored' (Courtesy: Dr. Anurag Gupta)

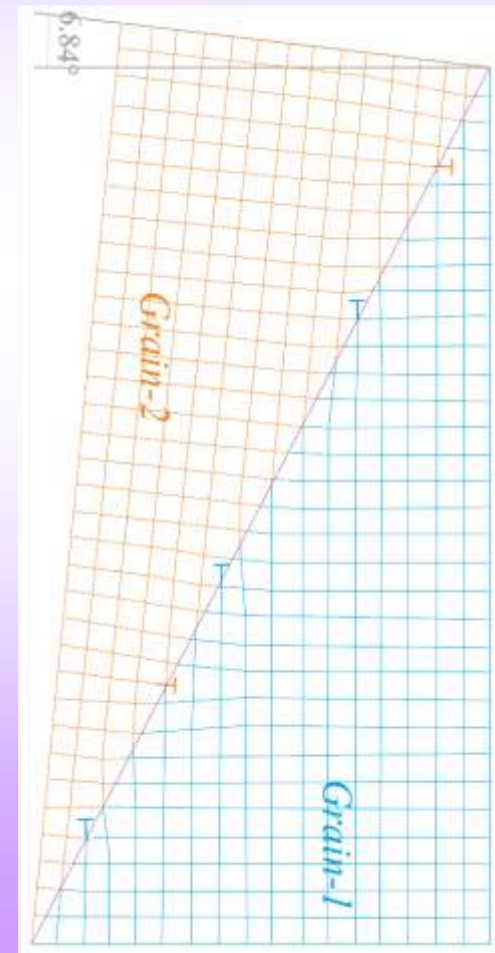
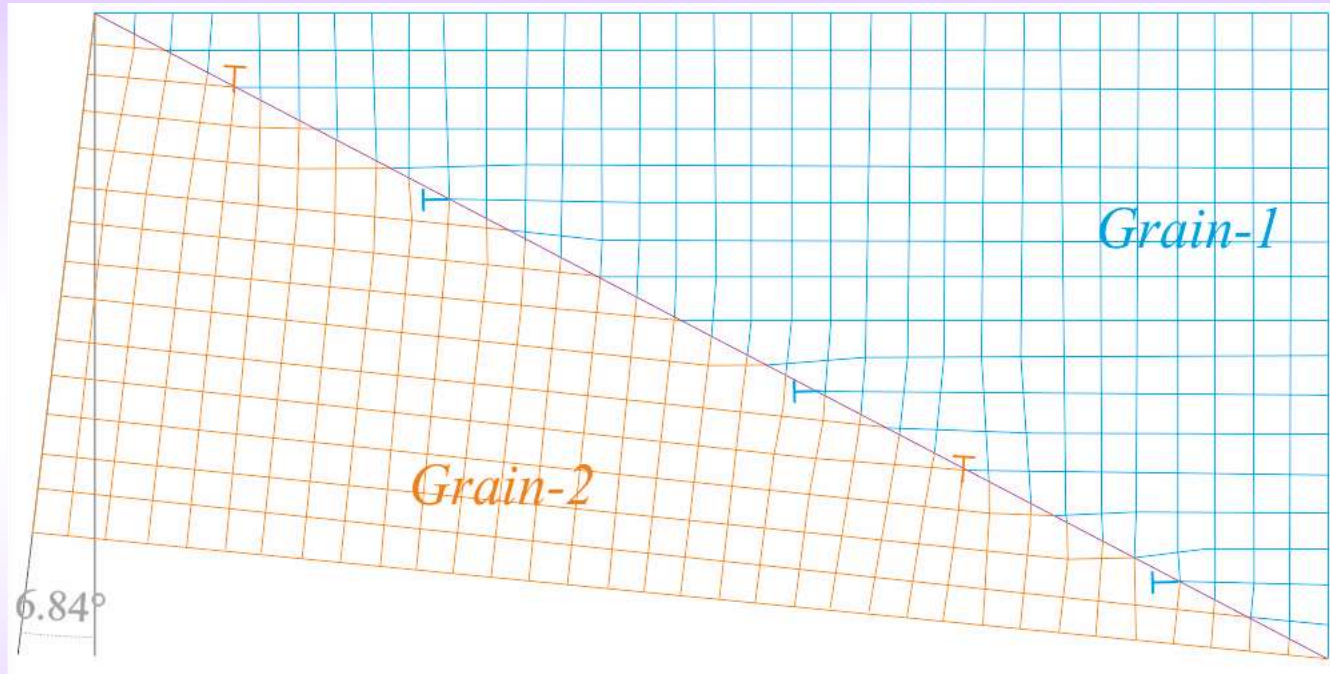
~8° TILT BOUNDARY IN SrTiO₃ POLYCRYSTAL

- The figure below shows a high-resolution lattice fringe image (HRLFI) taken in a transmission electron microscope (TEM). The bright lines correspond to atomic planes.
- The 'Fourier filtered image' shows the existence of dislocations clearly.



Low angle tilt grain boundary (asymmetric)

- ❑ In this case there are two sets of dislocations accommodating the angular misfit between the two grains.

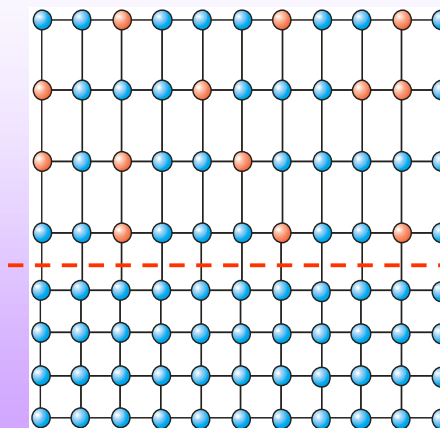
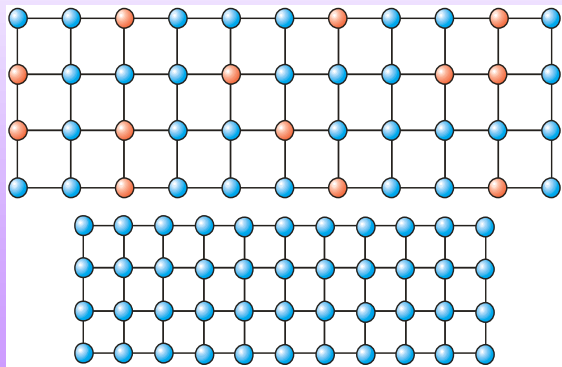


Structural dislocations accommodating misfit

- Let us consider the example of Ge-Si system. There is a 4.2% difference in the lattice constants of Si and Ge. Hence, $\text{Si}_{1-x}\text{Ge}_x$ has a bulk relaxed lattice constant which is larger than Si.
- Therefore when a layer of $\text{Si}_{1-x}\text{Ge}_x$ is *epitaxially* grown on top of Si, it is tetragonally strained. This causes strain energy to the system.
- The term epitaxy implies that there is an atom to atom matching across the interface (i.e. there is one to one correspondence of atoms across the interface → for each atom in the film, there is an atom in the substrate).
- Such an interface is termed as coherent. Epitaxial growth occurs (in the simple sense), when the lattice mismatch between the film and substrate is small (~ few %).
- Similar to the case here, during precipitation of a second phase from a matrix, the interface may be coherent (if the misfit is small).

Let us consider an 'infinitely' thick substrate, with a thin epitaxial film.

$\text{Si}_{1-x}\text{Ge}_x$ on Si



Strained $\text{Si}_{1-x}\text{Ge}_x$ on Si

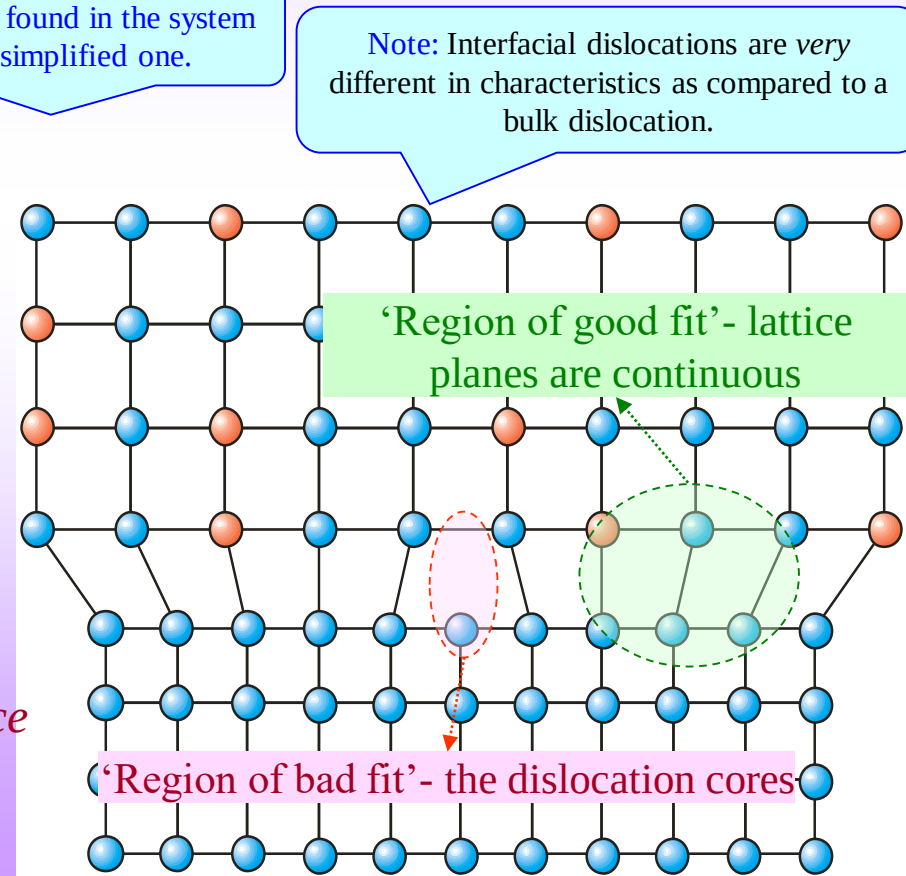
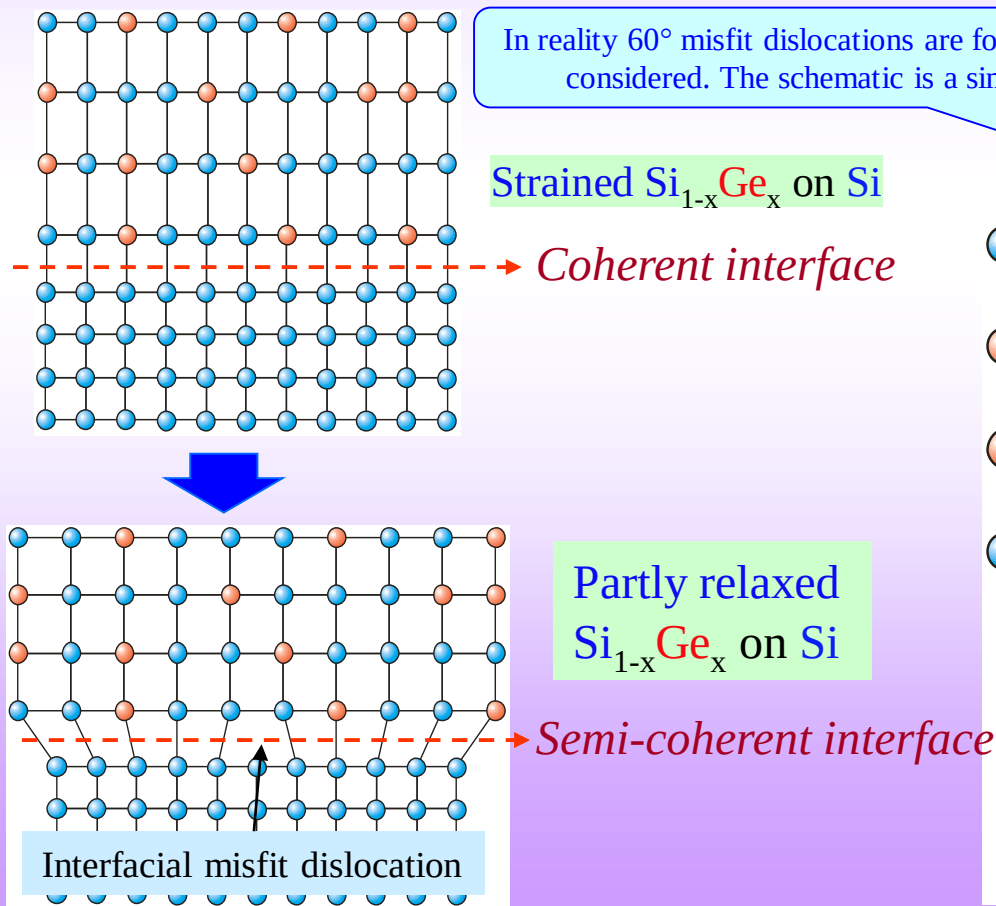
Schematics

Coherent interface

If layers are grown below the critical thickness then they become strained with the lattice symmetry changing from cubic to tetragonal.

The semi-coherent epitaxial interface

- When the thickness of the film becomes larger than a *critical thickness*, the ‘coherency strain energy cost becomes ‘too much’ and the interface becomes *semi-coherent* → **misfit dislocations** decorate the interface (which now becomes semi-coherent), which act to partly relieve the strain in the epitaxial film. Only one dislocation is shown in the schematic, in practice the interface is characterized by a dislocation for every ‘n’ atomic positions (or length).
- The semi-coherent interface can be visualized as regions of ‘good fit’ and ‘bad fit’ (dislocation cores).



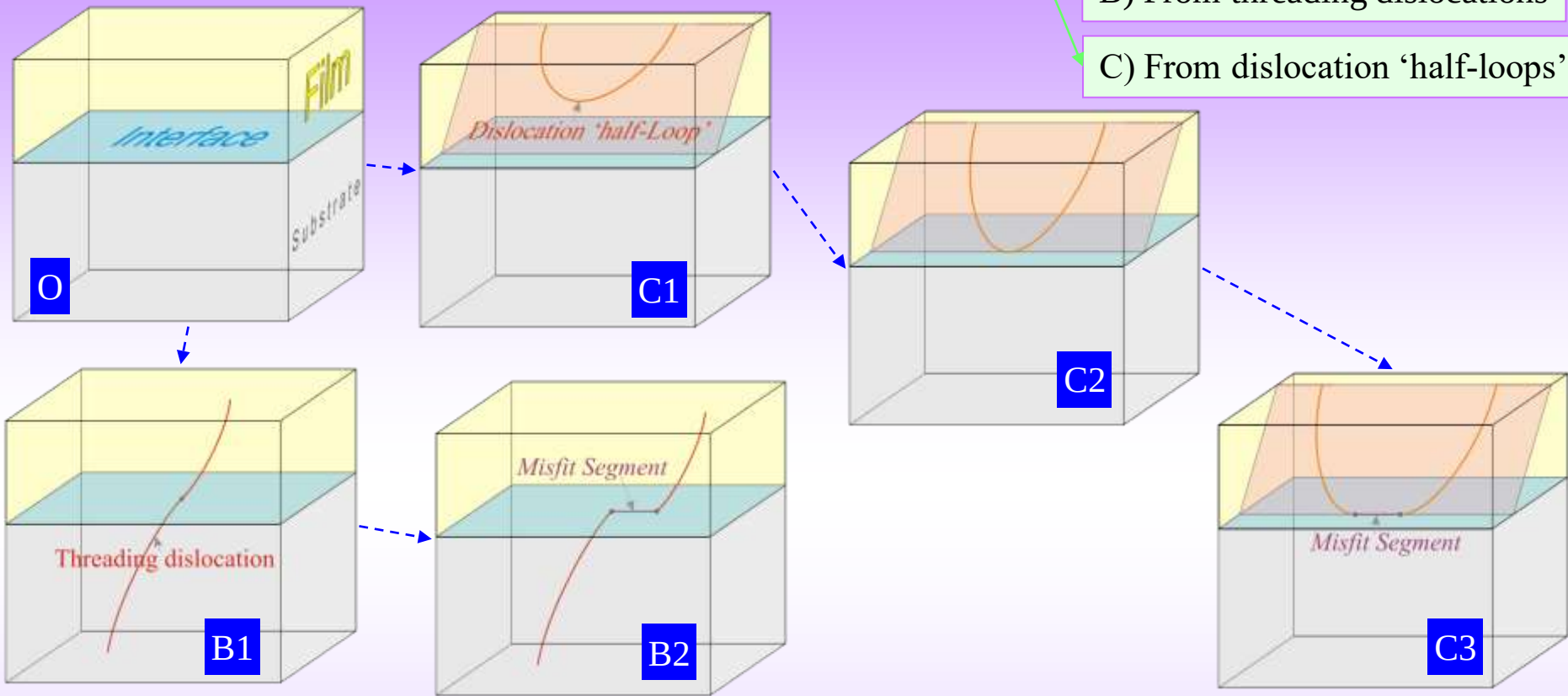
How can misfit dislocations form?

Misfit dislocations (/segments)

A) Nucleation

B) From threading dislocations

C) From dislocation 'half-loops'



- Let us consider an epitaxial film over a substrate with an interface (O).
- The mechanism of nucleation of a misfit dislocation (A) is usually not feasible.
- In mechanism-B, a threading dislocation extends across the interface (B1). On growth beyond a critical thickness, the epitaxial stresses lead to the formation of misfit segment residing at the interface (B2).
- In mechanism-C, a dislocation half loop is nucleated from the free top surface (C1)– after a critical thickness of the film. Under the effect of epitaxial stresses, this loop extends to the interface (C2) and further forms a misfit segment (C3).

What are the modes of growth of epitaxial overlayers?

- ❑ Three modes of growth exist for epitaxial overlayers.
 - Layer by layer (Frank-van der Merwe growth mode)
 - Islands on substrates (Volmer-Weber growth mode)
 - Formation of a film, followed by growth of islands on these films (Stranski-Krastanov mode).

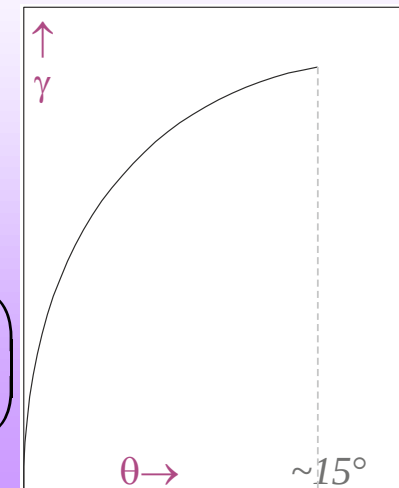
Grain boundary energy and its variation with misorientation

- ❑ The grain boundary energy is a strong function of the misorientation angle between the grains.
- ❑ In the low angle regime the Read & Shockley model can be used for the computation of the GB energy, which is a function of the dislocation spacing. The formula is for isotropic materials.
- ❑ The formula has good fit with experiments, sometimes even to high-angles, wherein the theory is expected to break down.
- ❑ Entropic effects are ignored in the Read-Shockley analysis. An issue of the existence of secondary cusps (overlaid on the main curve) in LAGB is a topic of advanced study.

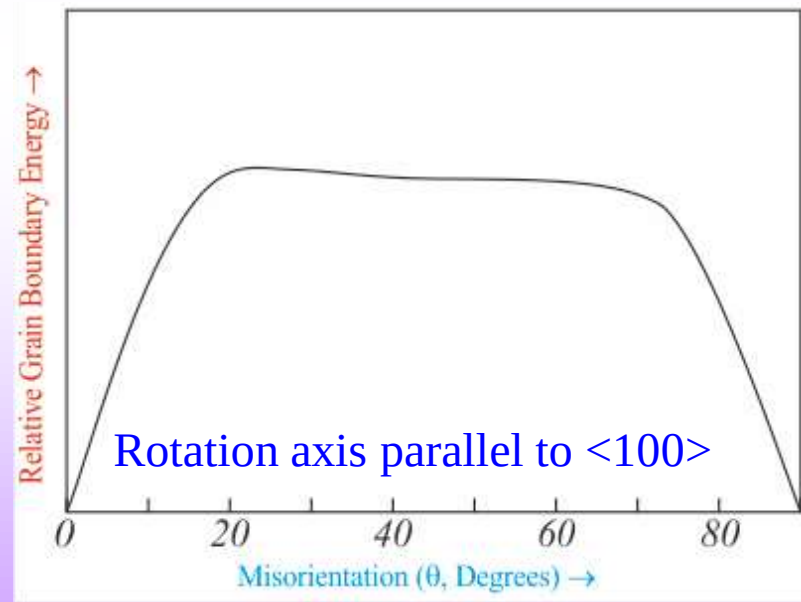
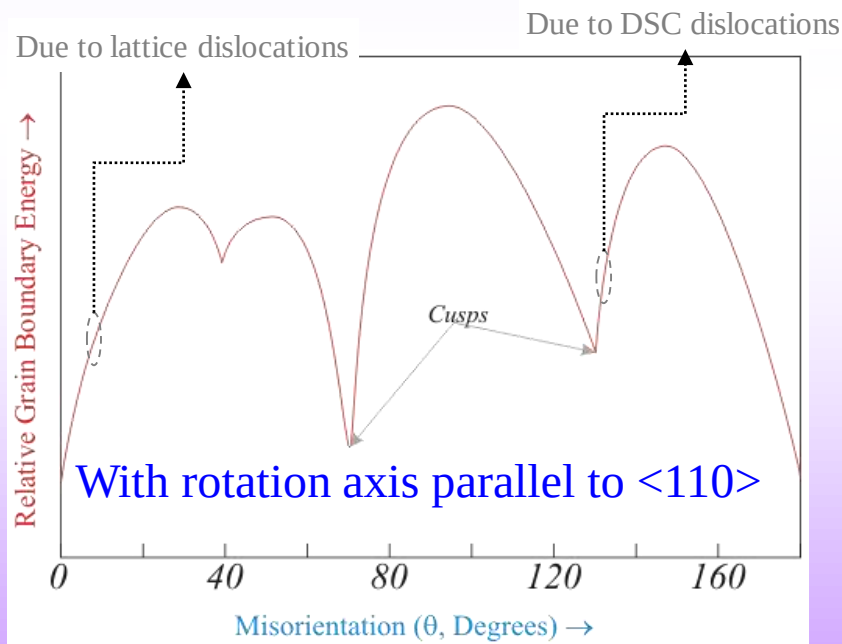
The energy of the LAGB is given by $\gamma = \gamma_0 \theta (A - \ln \theta)$

γ_0 is a function only of the elastic constants. $\gamma_0 = \frac{Gb}{4\pi(1-\nu)}$

‘A’ additionally involves parameters related to the core of the dislocation. E_c is the core energy & r_c is the core radius. $A = \left(\frac{4\pi(1-\nu)E_c}{Gb^2} \right) - \ln\left(\frac{r_c}{b} \right)$



- ❑ The plot of energy of the grain boundary vs misorientation angle may show sharp cusps at special angles (Fig. below) → e.g. symmetric tilt boundaries in Al, when the rotation axis is parallel to $\langle 110 \rangle$. These angles correspond to **low Σ numbers** (in the **Coincident Site Lattice (CSL)** model of grain boundaries). These sharp cusps may be absent for other tilt angles (e.g. symmetric tilt boundary in Al, when tilt axis is parallel to $\langle 100 \rangle$).
- ❑ Clearly these cusps must correspond to some ‘good bonded’ states, which arises from coincidence of lattice sites.
- ❑ The first cusp ($\theta \rightarrow 0^\circ$) is due to GB dislocations. The energy variation around the other cusps at large angles can be understood as dislocations in the DSC lattice (to be considered later).

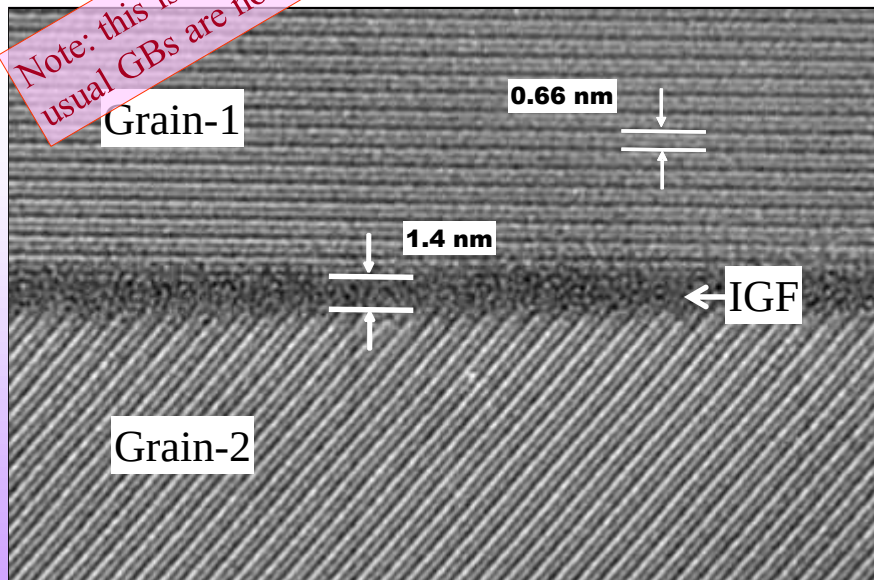


Variation of Grain boundary energy with misorientation for symmetric tilt boundaries in Al

Amorphous Grain boundaries

- ❑ During the ‘early years’ it was assumed that the grain boundaries have an ‘amorphous’ character → there is a region around the grain boundary which is disordered.
- ❑ It was realized later that the arrangement of atoms in bounding grains can be nearly perfect ‘almost’ up to the grain boundary.
- ❑ However, there are special cases of grain boundaries in materials like Si_3N_4 , SiC , SrTiO_3 , Al_2O_3 , etc., wherein the grain boundary region can have an amorphous film of about 1–2 nm thickness (the Intergranular Glassy Film (IGF)).

Note: this is a special case-
usual GBs are not like this



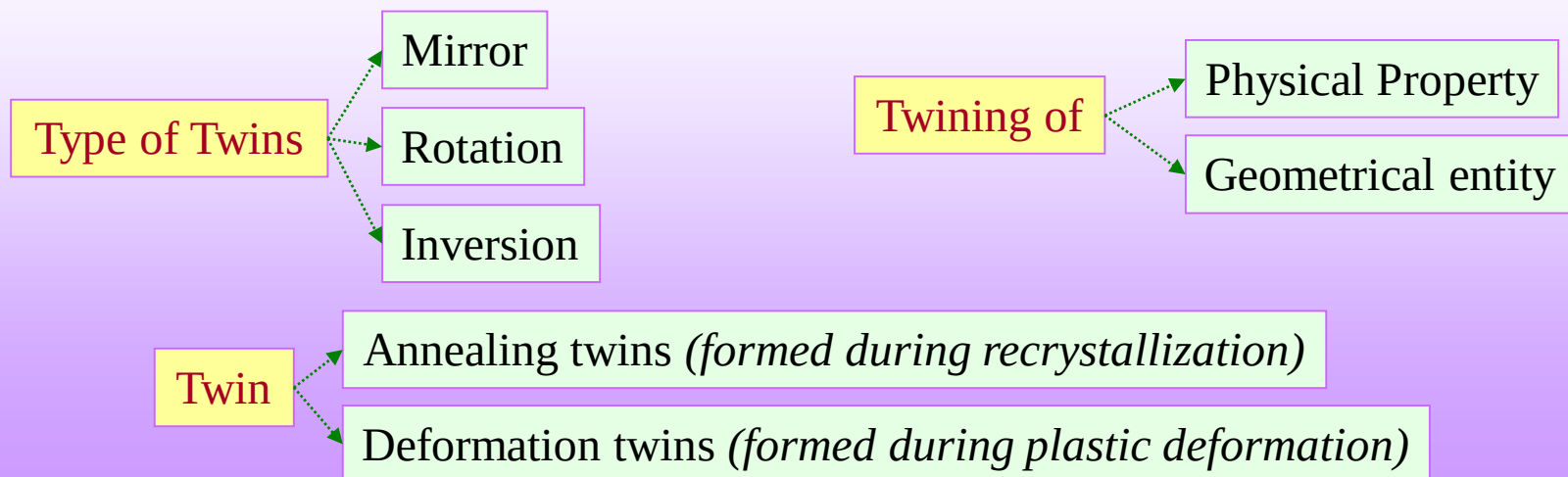
] Amorphous region

High-resolution micrograph from a Lu-Mg doped Si_3N_4 sample showing the presence of an Intergranular Glassy Film (IGF).

Twin Boundary

- ❑ The atomic arrangement on one side of the twin boundary is related to the other side by a **symmetry operation** (usually a mirror).
- ❑ The symmetry operator defining a twin cannot be a symmetry operator of the crystal (e.g. the mirror plane defining a mirror twin cannot be part of the point group symmetry of the crystal → *If this were so then the whole entity would be the same crystal*).
- ❑ Twin boundaries usually occur in pairs such that the orientation difference introduced by one is restored by the other.
- ❑ The region between the regions is called the twinned region (or a **Twin**).
- ❑ Twinning could be w.r.t to a **geometrical entity** or a **physical property** (or both) → if the crystal is defined w.r.t a geometrical entity (e.g. atoms), then the twinning could be of the atomic positions. On the other hand if the twin is defined w.r.t to a physical property (e.g. magnetization vector) then the twinning could be w.r.t to that (orientation of the magnetization vector).
- ❑ Two or more crystallites related by the symmetry operator are called the **variants** of the twin. The plane of the twin boundary is sometimes referred to as the **habit plane**. Unlike other interfaces, the twin boundary is usually *atomically sharp*.
- ❑ The mirror plane (twinning plane) need not be the same as the twin boundary. If the two are the same, then the twin is referred to as a **coherent twin**, else as in **incoherent twin**.
- ❑ In materials wherein slip is limited (e.g. BCC Fe at low T), **twinning can accommodate some plastic deformation**. (TWIP steels: steels where twinning induces plasticity).

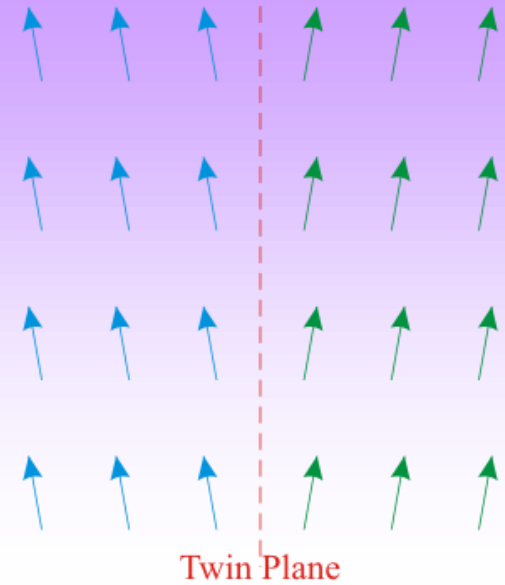
- ❑ Twins can be classified based on various perspectives.
- ❑ The symmetry operation connecting the two twin variants can be: (i) mirror, (ii) rotation or (iii) inversion.
Correspondingly we will have mirror, rotation or inversion twins.
- ❑ Twinning can be of: (i) geometrical entity or (ii) physical property.
- ❑ Twins can form during: (i) recrystallization (during annealing of a deformed material) or (ii) during deformation.
- ❑ Deformation twinning is an important mechanism of [plastic deformation](#). The process can give some ductility in materials, which inherently poor in this regard (e.g. BCC metals like Fe at low temperatures).
- ❑ Materials with low [stacking fault energy](#) (SFE) tend to readily twin (twins can be found in Cu as compared to Al).
- ❑ Twinning involves a shift of atomic positions with respect to a single crystal, by a fraction of the unit cell dimensions (in the CCP (111) twin the shift vector is $\{1/6\langle 112 \rangle\}$, which is the vector associated with the stacking fault).



Mirror twins

- ❑ The most common type of twin is a mirror twin.
- ❑ The twinning can be of a physical property like magnetic moments or of geometrical entities like atoms.
- ❑ As we shall see soon, the mirror plane need not coincide with the interface plane (leading to the concept of coherent and incoherent twins).

Twinning of magnetization vectors

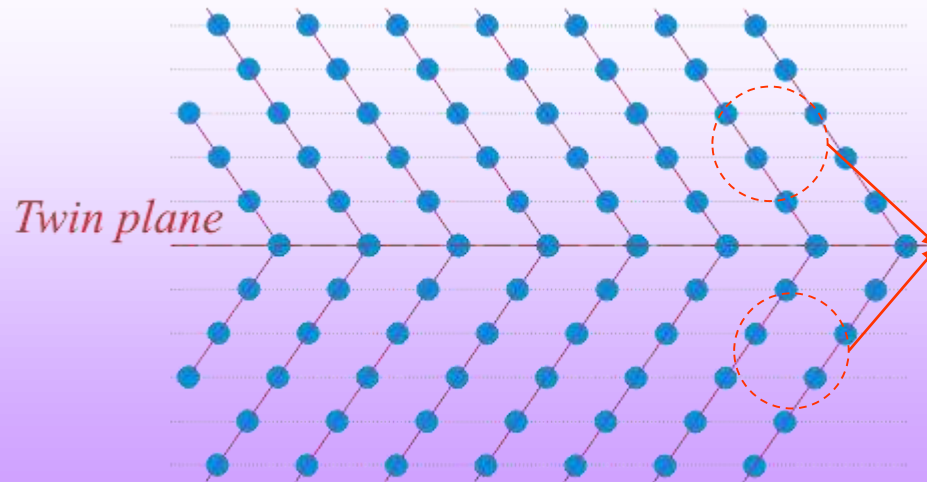


Twining of

Physical Property

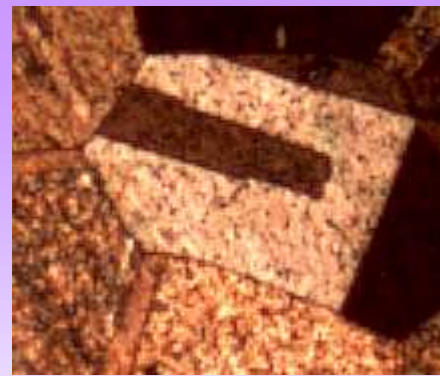
Geometrical entity

Twinning of atomic positions



Two 'variants' of the twin related by a mirror plane

Note: these are schematics



Twin terminating within the crystal



Twin extending from one grain boundary to another

Cold worked and recrystallized Cu Sample

Note how orientation change introduced by twinning also influences the contrast on etching

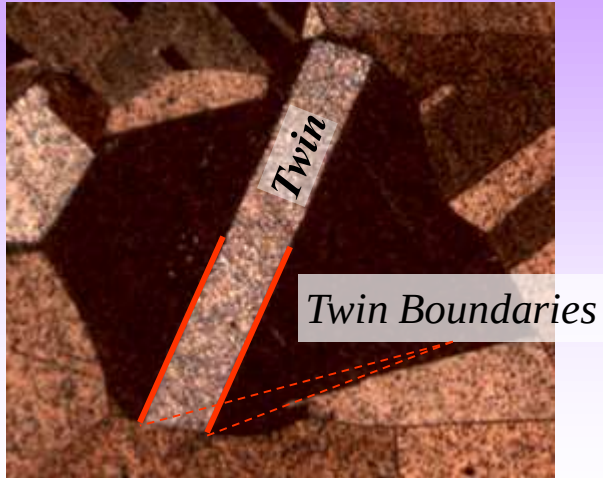
- Note how straight the twin boundaries are.
- Twin boundaries can be very sharp (one of the rare examples of an nearly atomically sharp interface!).
- Also note how twin boundaries are occurring in pairs in the sample.

Examples of twinning

	Examples	Twin plane	Twin Direction
CCP	Ag, Au, Cu (Low stacking fault energy)	{111}	$\langle 112 \rangle$
BCC	α -Fe, Ta	{112}	$\langle 111 \rangle$
HCP	Zn, Cd, Mg, Ti	{10-12}	$\langle -1011 \rangle$

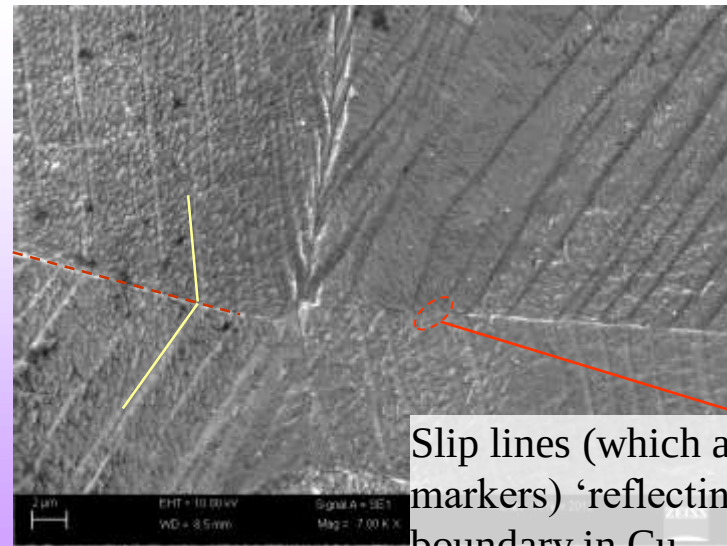
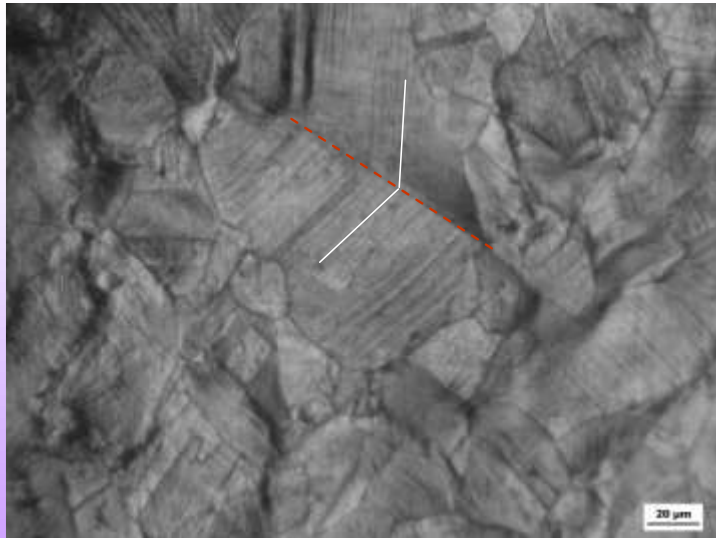
Twin Boundaries

❑ How do we know if the straight lines are twin boundaries (in the micrograph below)?



- We could obtain a HRLFI (High-Resolution Lattice Fringe Image) in a TEM and actually ‘see’ that the atomic planes reflect across the twin boundaries.
- In the case of Copper (and some other materials with **low stacking fault energy**- wherein, twinning is more common and are ductile as well) one could use slip lines* as *crystallographic markers* to identify twins (even in an optical microscope).

Below are two micrographs (left is a light optical micrograph and right is a scanning electron micrograph) showing slip lines ‘mirrored’ by the twin planes.

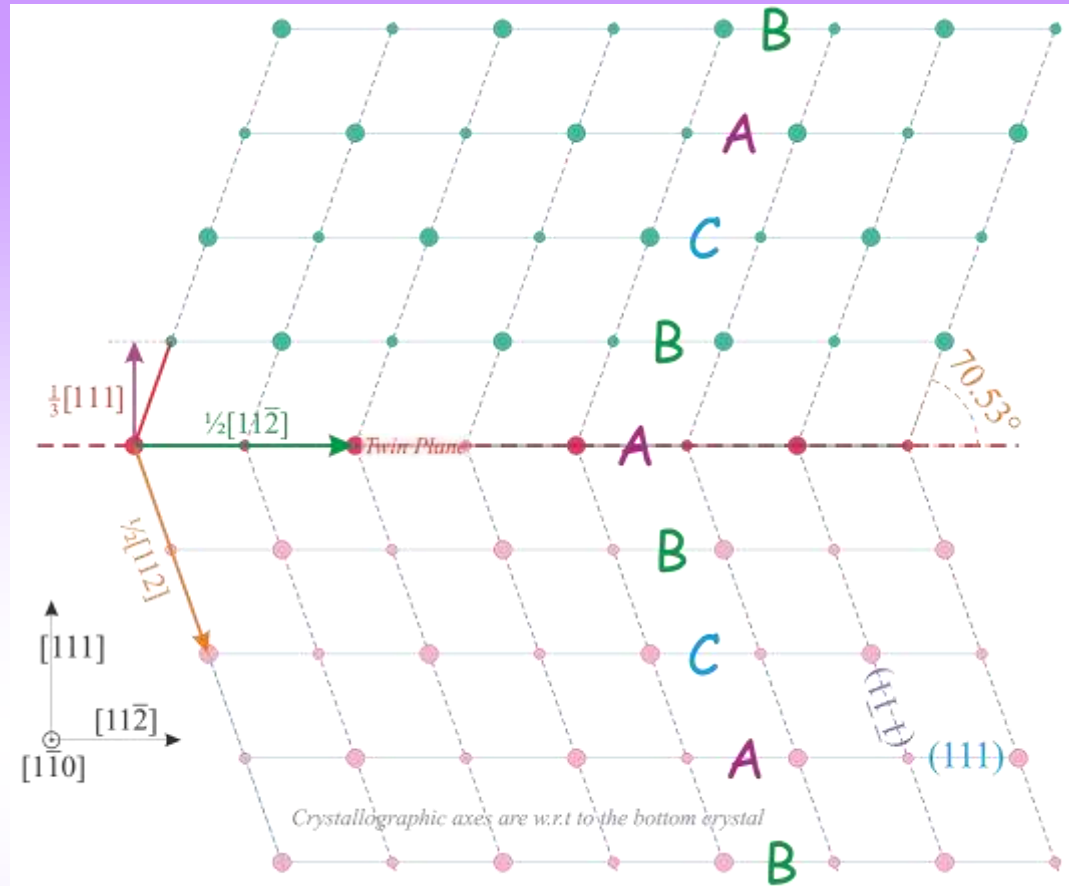


Slip lines (which are crystallographic markers) ‘reflecting across’ a twin boundary in Cu

* Slip line is a step created by many dislocations operating on a single slip plane → leaving the crystal surface to create a step of height **nb**.

Mirror twin in CCP

- Twinning takes place on (111) planes with a shear along the $[11\bar{2}]$ direction.
- The shift vector associated with the twin is the $\{1/6\langle 112\rangle\}$ kind of vector (the one which connects 'B' to 'C' positions & this is the one associated with a stacking fault).
- The angle between the two types of $\langle 111\rangle$ planes drawn in the figure is $\text{Cos}^{-1}(1/3)$.



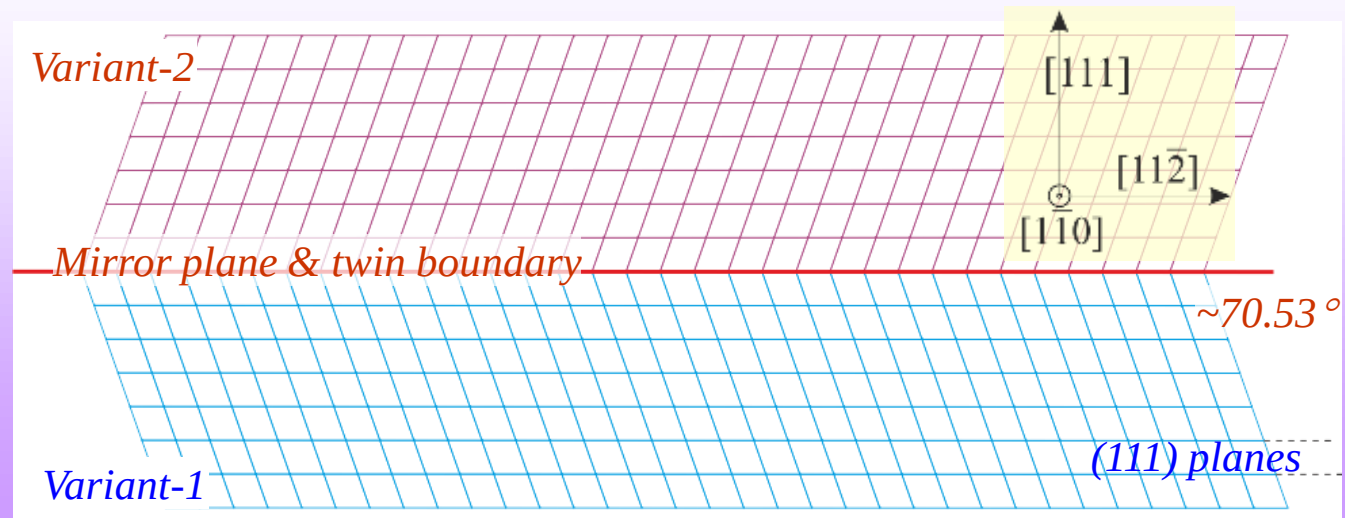
Some measures

$$\angle[(111), (11\bar{1})] = \text{Cos}^{-1}\left(\frac{1}{3}\right) = 70.528^\circ \approx 70.53^\circ$$

$$\left|\frac{1}{3}[111]\right| = \left(\frac{1}{\sqrt{3}}\right) \approx 0.577$$

$$\left|\frac{1}{2}[11\bar{2}]\right| = \left(\frac{\sqrt{3}}{2}\right) \approx 1.225$$

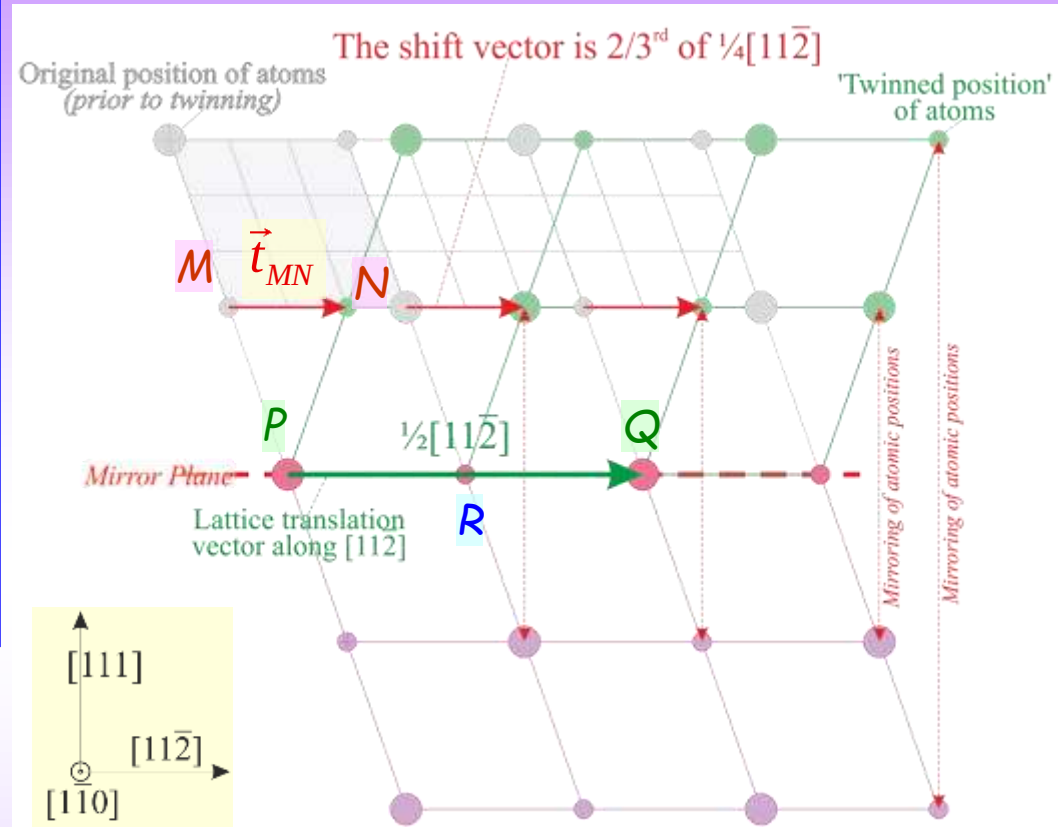
$$\left|\frac{1}{6}[11\bar{2}]\right| = \left(\frac{1}{\sqrt{6}}\right) \approx 0.408$$



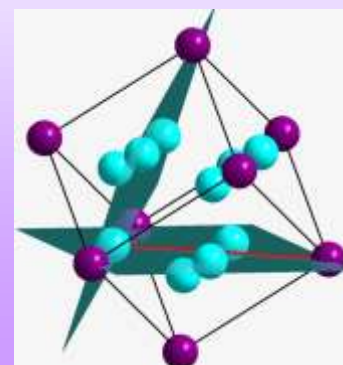
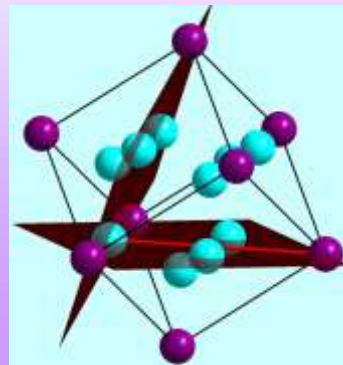
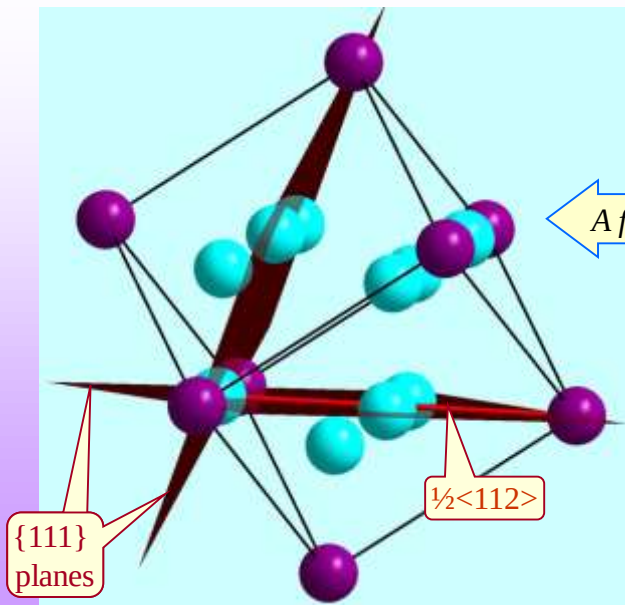
How does atomic shift lead to the twinned structure?

$$\text{Shift vector} = \vec{t}_{MN} = \frac{2}{3} \left\{ \frac{1}{2} \left(\frac{1}{2} [11\bar{2}] \right) \right\} = \frac{1}{6} [11\bar{2}]$$

- We saw that the shift vector in twinning is a fraction of the lattice spacing. In the case of the CCP structure it is the $\{1/6\langle 112 \rangle\}$ kind of vector.
- This vector can be understood as follows. The lattice translation vector along $[11\bar{2}]$ is $\vec{PQ}$ and has a magnitude $|\frac{1}{2} [11\bar{2}]|$. The shift vector $\vec{MN}$ (connecting grey atom to green atom) is $\frac{1}{2}$ of $\vec{PR}$. $\vec{PR}$ is $\frac{1}{2}$ of $\vec{PQ}$.
- $\vec{MN}$ is $\frac{2}{3}$ rd $\vec{PR}$.
- Later on when we discuss Coincident Site Lattices (CSL) we will refer to such a twin as the $\Sigma = 3$ CSL boundary.

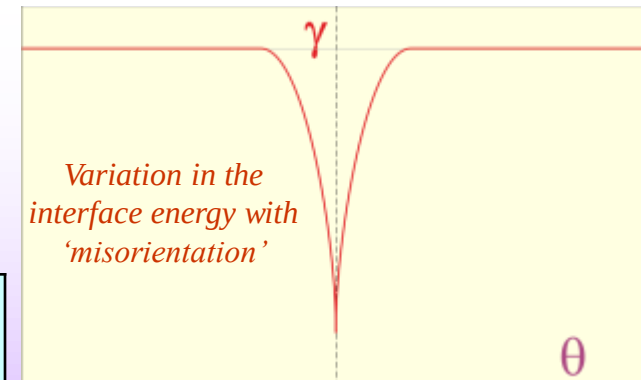
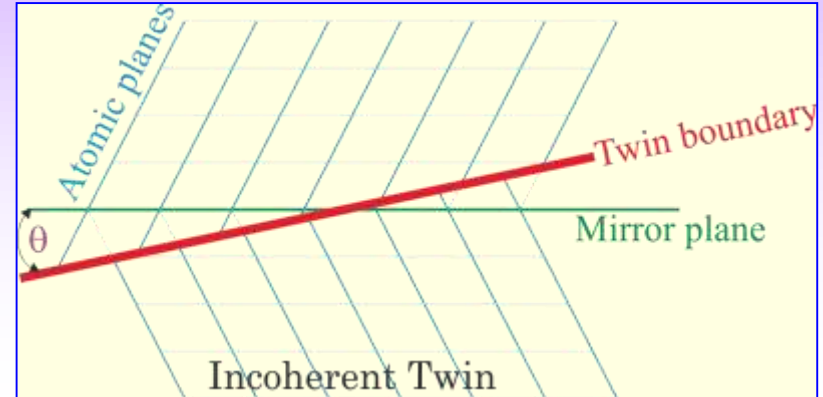
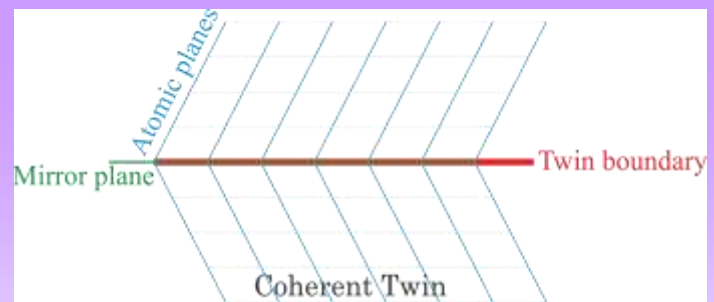


A figure slightly tilted about the $\langle 110 \rangle$ direction to understand the twin structure.



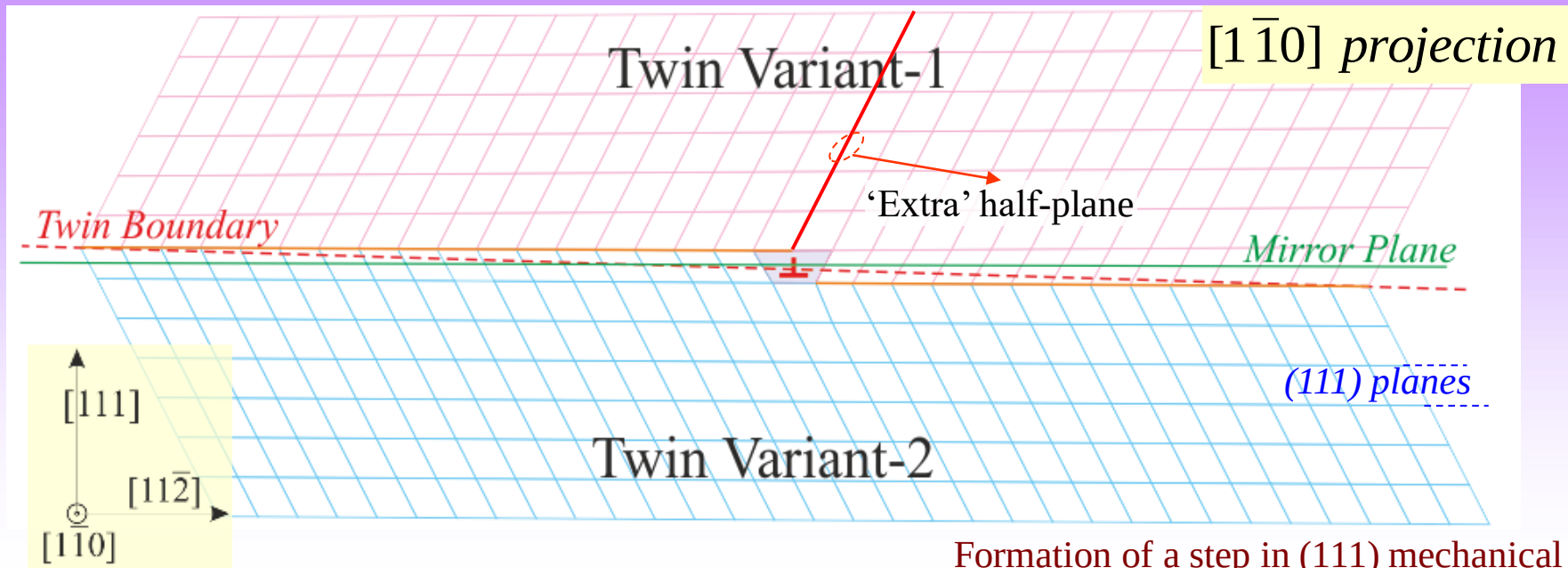
Coherent and incoherent twins

- In the case of the coherent twins, the twin boundary coincides with the mirror plane (i.e. the plane of reflection of the atomic planes).
- In the case of the incoherent twin, these planes are distinct. This ‘mismatch’ is accommodated by dislocations (more about this sooner).
- The above implies that the coherent twin boundary is lower in energy (γ) than the incoherent one. A schematic of a plot of γ versus θ is shown here, which is reminiscent of the energy variation close to a Σ -boundary (CSL).
- The ‘misorientation’ is between the mirror plane and the twin boundary.



Crystal	$\gamma_{\text{Coherent Twin boundary}}$ [mJ/m ²]	$\gamma_{\text{Incoherent Twin boundary}}$ [mJ/m ²]	$\gamma_{\text{Grain boundary}}$ [mJ/m ²]
Cu	21	498	623
Ag	8	126	377
Fe-Cr-Ni (SS304)	19	209	835

Dislocations at incoherent twin boundaries



- In the case that the mirror plane does not coincide with the habit plane (interface plane) of the twin variants, a series of dislocations accommodate this 'mismatch'. The 'extra' half-plane (as in the figure) is truly speaking not extra!
- The interface can be visualized as a series of steps (orange line), which are separated by dislocations. The average orientation generated by these dislocations is given as the red dashed line (the twin boundary).
- The Burgers vector (**b**) corresponds to that of a Shockley partial dislocation.

Formation of a step in (111) mechanical twin in a FCC crystal. The Burgers vector of the 'coherency dislocation' is as below.

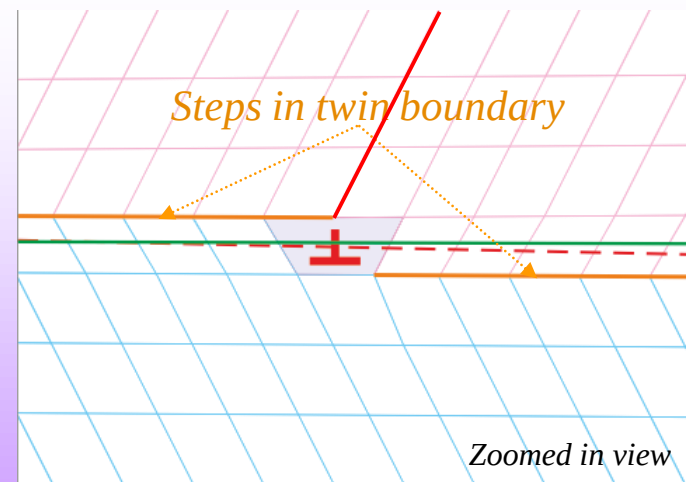
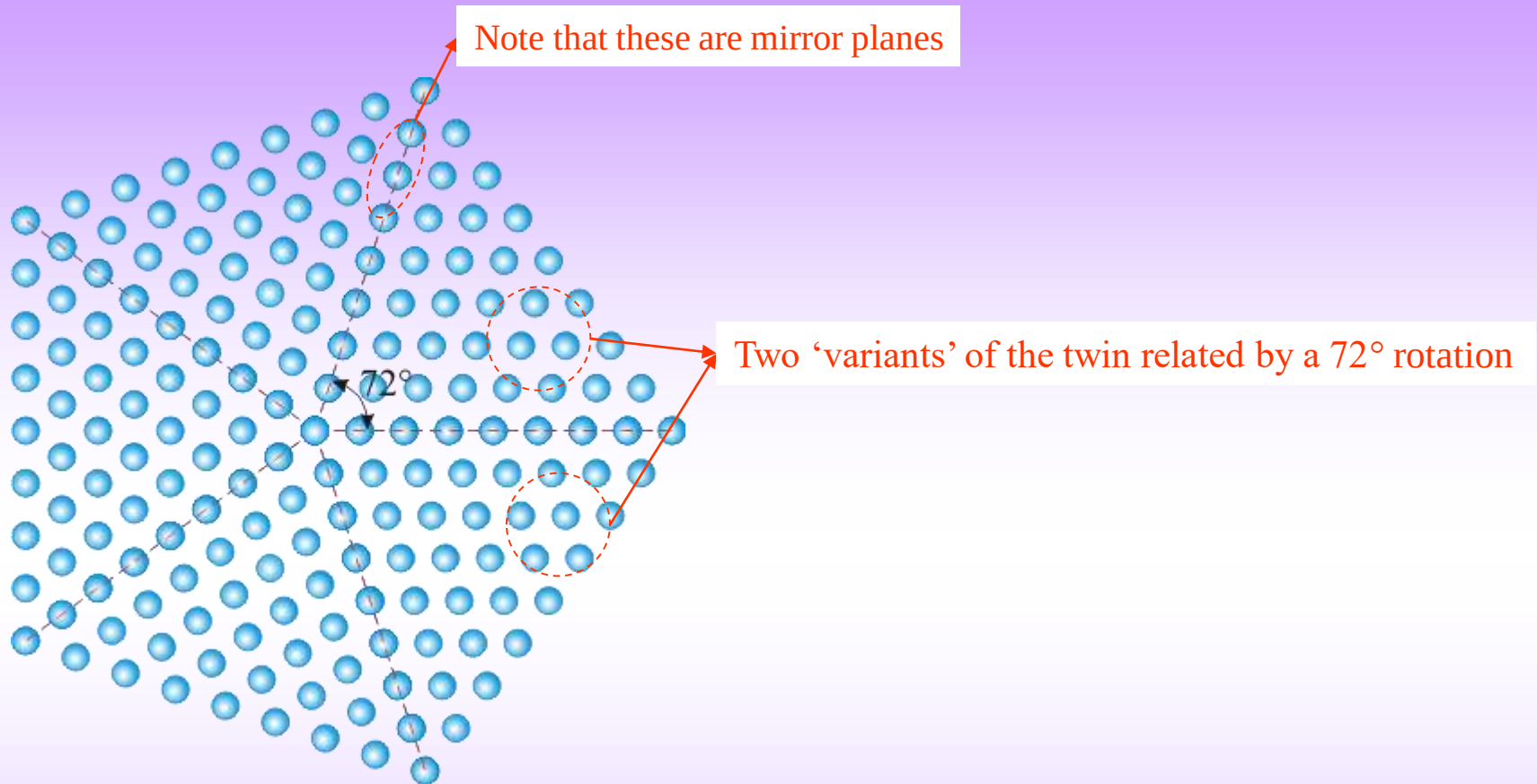


Fig.IT

Burgers vector of the twin $\bar{b} = \frac{1}{6} [11\bar{2}]$

Rotation twins



Schematic of a rotation twin with a combined 5-fold symmetry

Note: In practical examples of such 5-fold twinning the angles may all not be equal to each other (& hence $\neq 72^\circ$)

e.g. (i) in twinning of nanocrystalline Cu, the angles found are 71.2°, 70.9°, 73.4°, 71.5° & 73°*

*(ii) in twinning in twinning in a diamond film**, the angles are : 71.1°, 71.4°, 72.9°, 73.1° & 71.5°*

* P. Huang, G. Q. Dai, F. Wang, K. W. Xu, Y. H. Li, Applied Physics Letters 95, 203101 (2009)

** Hidetaka Sawada, Hideki Ichinose, Diamond & Related Materials 14 (2005) 109–112

Stacking Fault

- ❑ Stacking faults are typically described for close packed structures.
- ❑ An error in the sequence of stacking atomic planes (close packed planes) leads to a Stacking fault. (It is actually not a good idea to think of this as an inclusion or removal of a plane).
- ❑ A stacking fault is defined by a shift vector. In the example below the shift vector is from the C-plane atom position to the A-plane atom position.
- ❑ In a stacking fault the configuration of the nearest neighbours remains undistorted, while the next nearest neighbour configurations are different than in the perfect crystal.
- ❑ Stacking fault in HCP can lead to thin region of CCP kind of stacking and vice-versa.
- ❑ In CCP crystals when the perfect dislocation splits into partials, the included region in the slip plane is stacking fault (i.e. the stacking fault is bounded by two partials).

CCP stacking → ...ABC ABC ABC ABC...

*Thin region of HCP
type of stacking*



CCP stacking
with a stacking fault → ...ABC AB AB ABC...

HCP stacking → ...AB AB AB AB...

*Thin region of FCC
type of stacking*

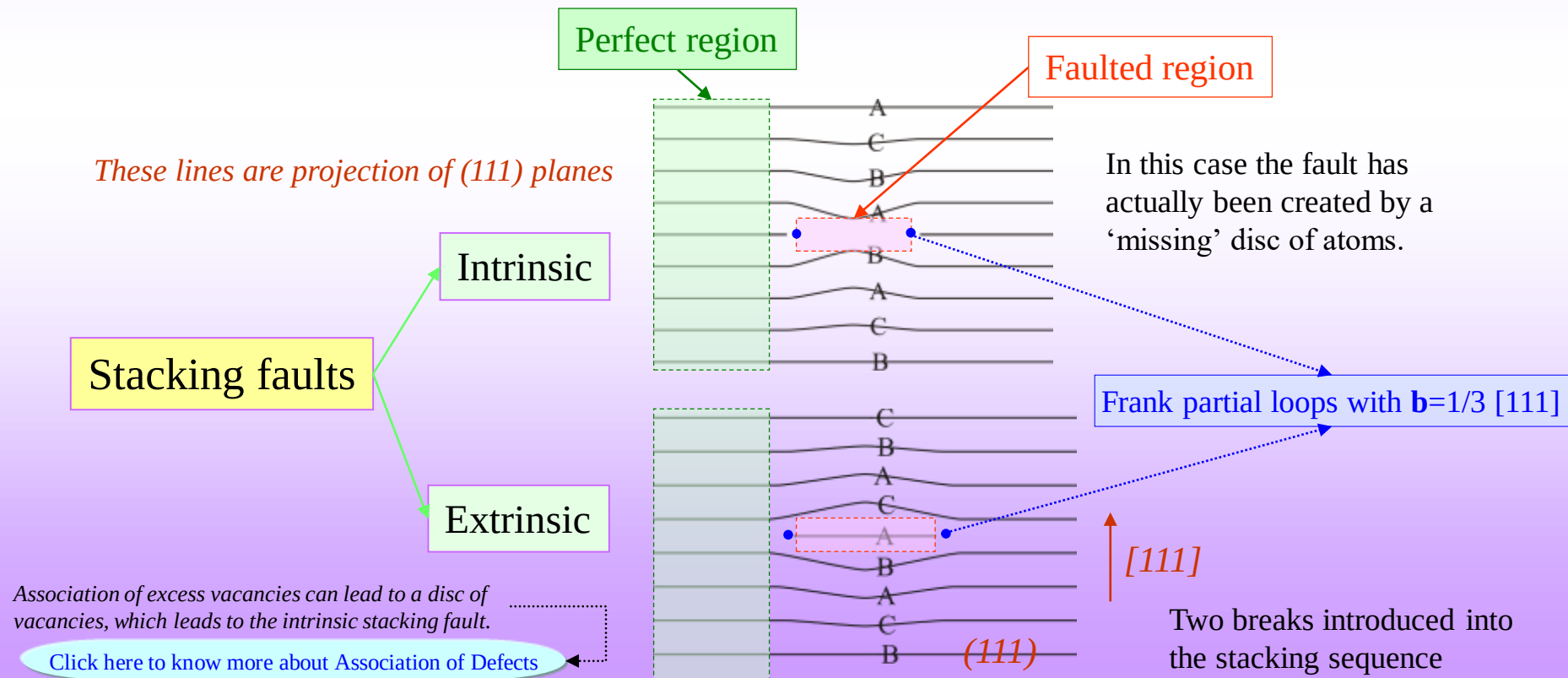


HCP with a
stacking fault → ...AB ABC AB AB...

- ❑ Stacking faults can be right across the crystal or can be limited to a certain region (as in the figures in the next slide). In the case of limited faults, they can be visualized as bounded by edge dislocations. In 3D this is a dislocation loop. In CCP crystals the dislocation is a Frank partial dislocation with $\mathbf{b} = 1/3 [111]$. [Click here to know more about Frank partial dislocations in CCP crystals.](#)
- ❑ The energy per unit area of the stacking fault is the stacking fault energy (SFE). Stacking fault energy $\sim 0.01 - 0.05 \text{ J/m}^2$.
- ❑ SFE in $\text{mJ/m}^2 \rightarrow$ Brass <10 , 303 SS ~ 8 , 304 SS ~ 20 , 310 SS ~ 45 , Ag ~ 25 , Au ~ 50 , Cu ~ 80 , Ni ~ 150 , Al ~ 200 . Note that the composition of SS changes the SFE considerably!

Frank partial dislocation loop bounding a stacking fault in CCP crystal

This shows that pure edge dislocation loop can exist (but a pure screw loop cannot exist)



- ☐ In in the materials community, the term intrinsic means '*nothing added from outside*' and extrinsic means '*something has been added from outside*'.
- ☐ These terms often have to be understood from a '*local perspective*'.
- ☐ In the case of the stacking fault, the intrinsic stacking fault *conceptually* involves removal of matter (a disc of atoms). The extrinsic stacking fault involves the addition of a disc of atoms.
- ☐ Another example is of 'intrinsic' and 'extrinsic' semiconductors. In intrinsic semiconductors the conductivity is due to excitation of electrons from the valence band to the conduction band, thus giving rise to electron-hole pairs; which both contribute to conductivity. In the case of extrinsic semiconductors, a dopant is added in small concentrations (~20 parts per billion) and these contribute to the conductivity at low temperatures. E.g. doping with 'P' leads to a a n-type semiconductor, wherein the majority carriers are electrons.

Comparison of Energy of Various 2D Defects	Type of boundary	Energy (J/m ²)
	Surface	~ 0.89
	Grain boundary	~0.85
	Twin Boundary	~ 0.63, 0.498 (Cu)
	Stacking Fault	0.08 (Cu), 0.2 (Al)

Metal	Surface	Solid/ Liquid	Grain Boundary	Twin Boundary	Stacking Fault
	(J/m ²)				
					mJ/m ²
Gold	1370	132	364	~10	~50
Silver	1140	126	790	-	17, ~25
Platinum	1310	240	1000	196	~95
Nickel	1860	255	690	-	~150
Aluminium	1140	-	625	120	~200
Copper	1750	177	646	44	73, ~80
Iron	1950	204	780	190	-
Tin	680	54.5	-	-	-

Repeat



Video: Models to understand the formation of defects in crystals

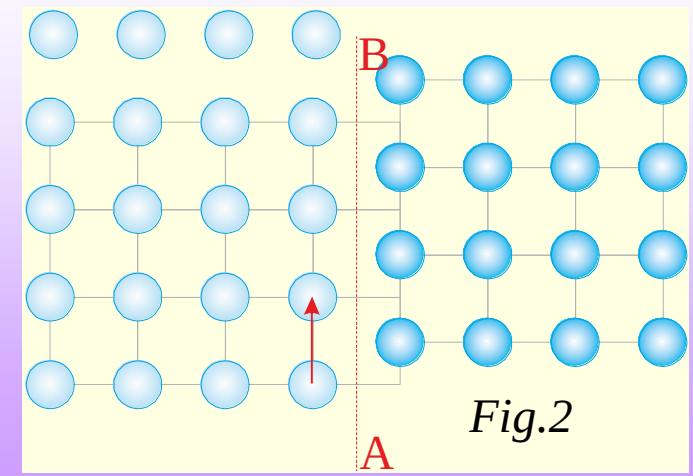
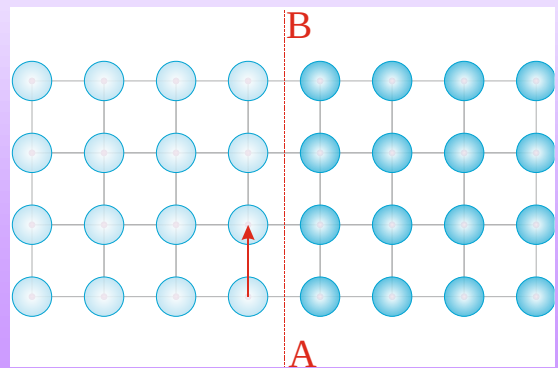
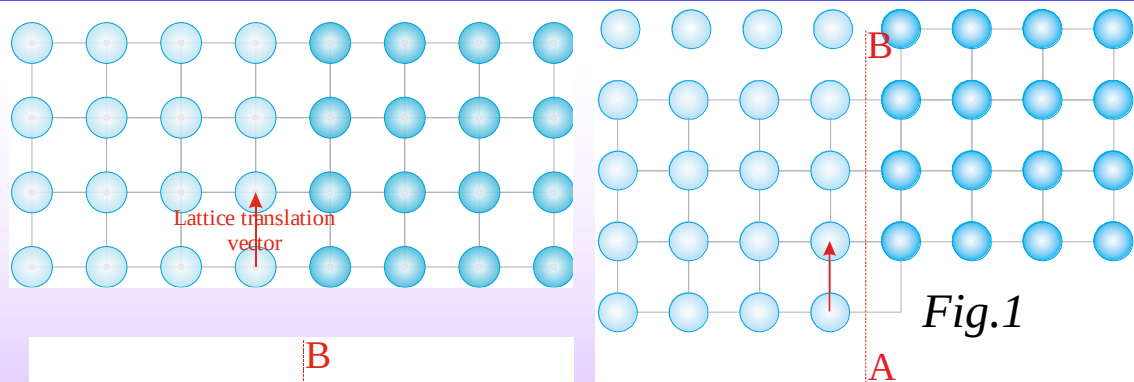
- ☐ *Vacancies*
- ☐ *Grain boundaries*
- ☐ *Surface [(100), (111), (110)]*
- ☐ *Disclination*
- ☐ *Edge dislocation*

Click here
for solved
example

Calculate the surface enthalpy of Cu crystal, when the external surface is of {100}, (110}, {111} type.

Why can't the a transformation defining a defect be a symmetry operator of the crystal?

- ❑ We had noted that the mirror plane of a twin cannot be the mirror plane of the crystal.
- ❑ In general the a transformation defining a defect be a symmetry operator of the crystal.
- ❑ To understand this let us consider a crystal as below. The atoms have been shaded differently for visualization purpose.
- ❑ Suppose we want to create a defect across the plane AB. If translate one part of the crystal w.r.t to another part by a lattice translation vector, then we do **not** create a defect (Fig.1).
- ❑ On the other hand if we shift the RHS of the crystal w.r.t the left hand side by some fraction of the lattice translation vector, then a interface is created between the two crystals.
- ❑ The same is true for other symmetry operations of the crystal.



Advanced Topics

Grain boundary models.

- ☐ Coincident Site Lattice (CSL) model
- ☐ Displacement Shift Complete (DSC) model
- ☐ O-lattice model.

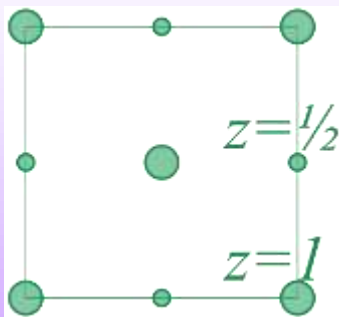
Grain Boundary Models

- ❑ We had seen some elementary concepts related to grain boundaries before.
- ❑ Multiple models and tools have been used to understand the structure of grain boundaries. These include:
 - Models: Coincident Site Lattice (CSL) model, O-lattice model, Displacement Shift Lattice (DSC) model, Structural Unit model, etc.
 - Experimental tools: High-Resolution Lattice Fringe Imaging (HRLFI), Energy Dispersive X-ray (EDX) Analysis, Electron Energy Loss Spectroscopy (EELS), Fresnel contrast imaging, etc.
- ❑ The broad area of crystallography dealing with such structures is termed as: *bicrystallography*.
- ❑ Here we will try to understand ‘just a bit’ of these advanced topics.

Coincident Site Lattice (CSL)

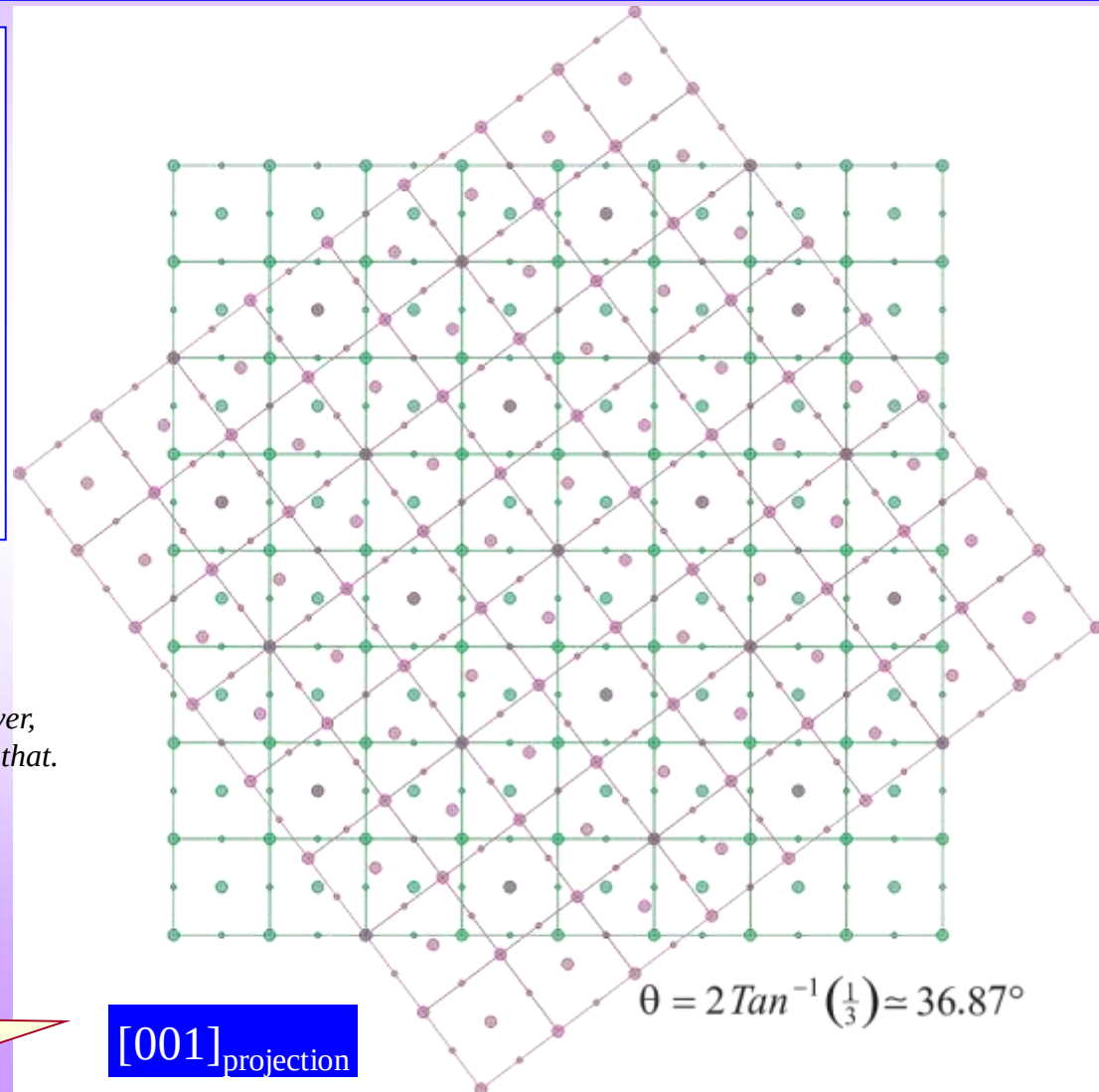
- In the plot of grain boundary energy with misorientation angle (along specific axes), we had noted that there are sharp cusps (i.e. there are low energy grain boundaries at certain angles). This implies a good bonding at the interface. A natural model to explain this observation is the CSL model.

- Let us take the example of the FCC lattice and consider the [001] projection (Fig.1).
- Further, let us consider multiple cells and create two interpenetrating FCC lattices (green and maroon)– one rotated with respect to the other by an angle of $36.87^\circ (=2\tan^{-1}(1/3))$.



*The large circles are in top layer,
while the small ones are below that.*

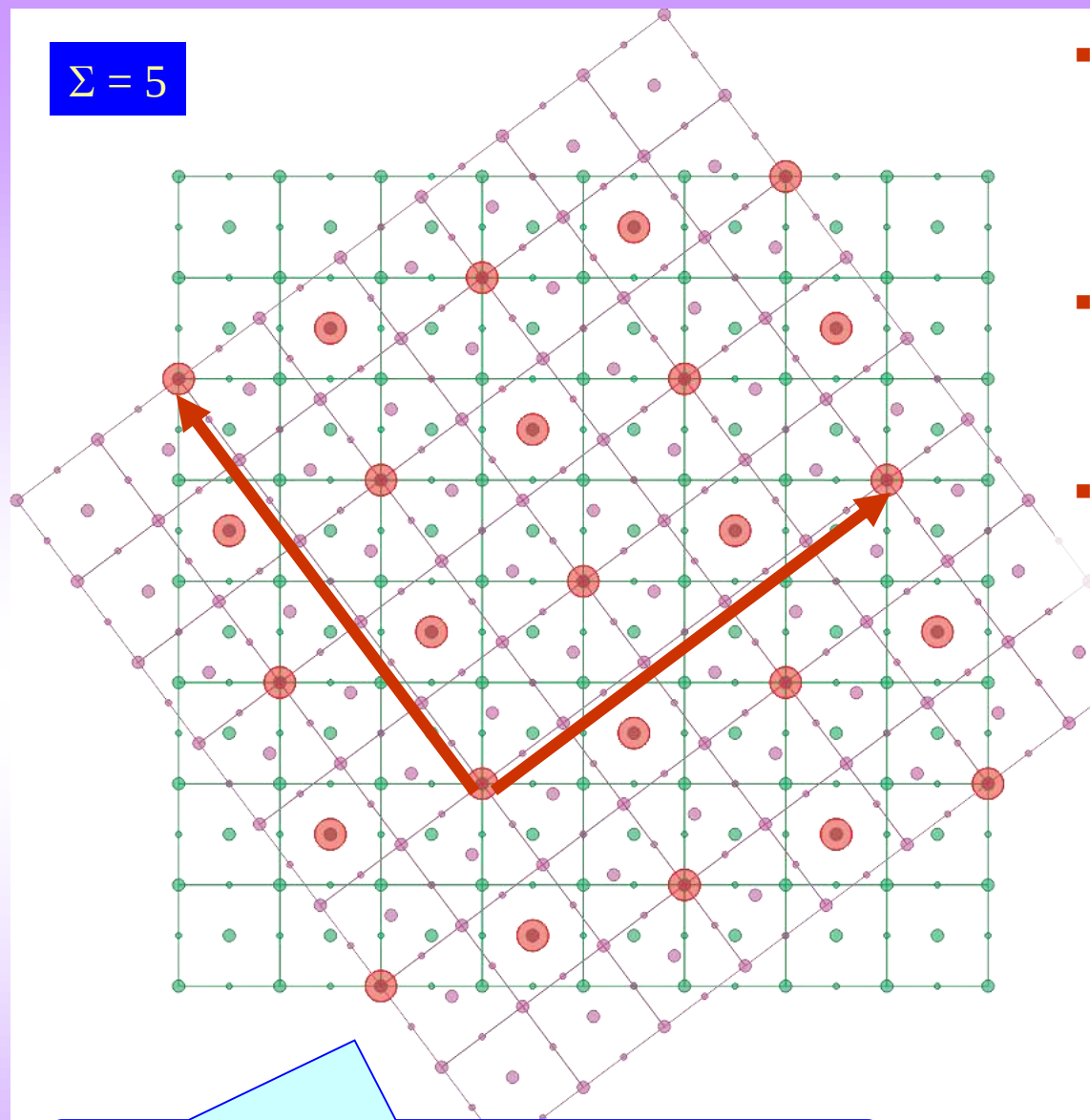
Note that this partial coincidence of atomic sites occur at a very specific angle of misorientation



$$\theta = 2 \tan^{-1} \left(\frac{1}{3} \right) \approx 36.87^\circ$$

[001] projection

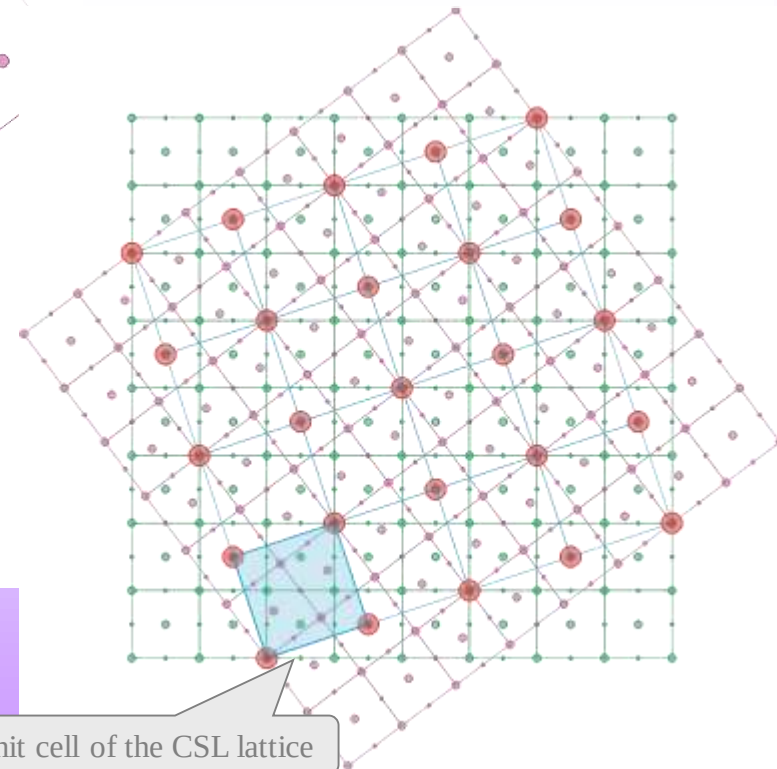
$$\Sigma = 5$$



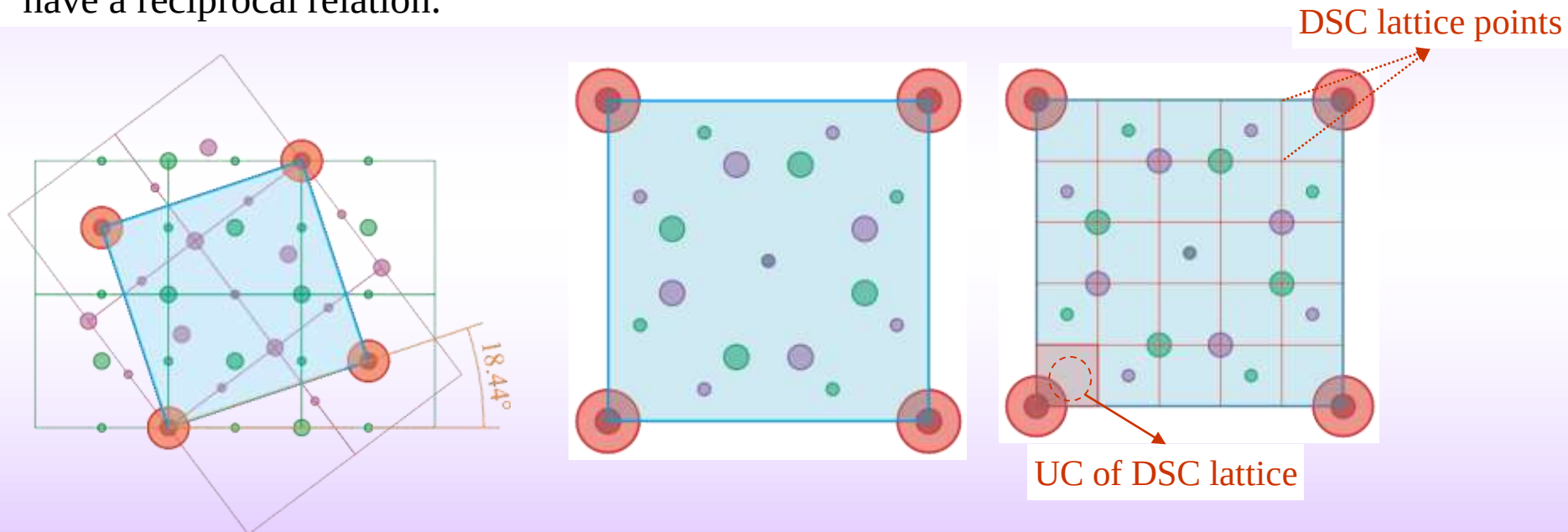
Along x and y, one in 5 lattice points coincide

- The basis vectors of the CSL unit cell bisect the angle made by the basis vectors of the two original lattices.

- This leads to a coincidence of *one in 5 sites* (marked with large red circles)– i.e. the formation of a **Coincident Site Lattice (CSL)**.
- The symbol used is for the coincident fraction is Σ & in this example $\Sigma = 5$.
- We can draw a unit cell for this CSL lattice (blue).



- Let us now focus on this CSL unit cell (blue one). This is rotated with respect to the 1st lattice by an angle of $\sim 18.44^\circ$. This unit cell has atoms from both crystals and 2 effective coincident lattice points in the two layers considered (the one in the centre is also coincident at $z = \frac{1}{2}$).
- A grid can be drawn within the unit cell, which gives rise to the DSC (Displacement Shift Complete) lattice (it is the coarsest lattice which contains the original lattices– rotated with respect to another).
- The DSC lattice has a lattice parameter of $1/\Sigma$ ($1/5$ of the blue CSL cell). → The DSC and CSL have a reciprocal relation.



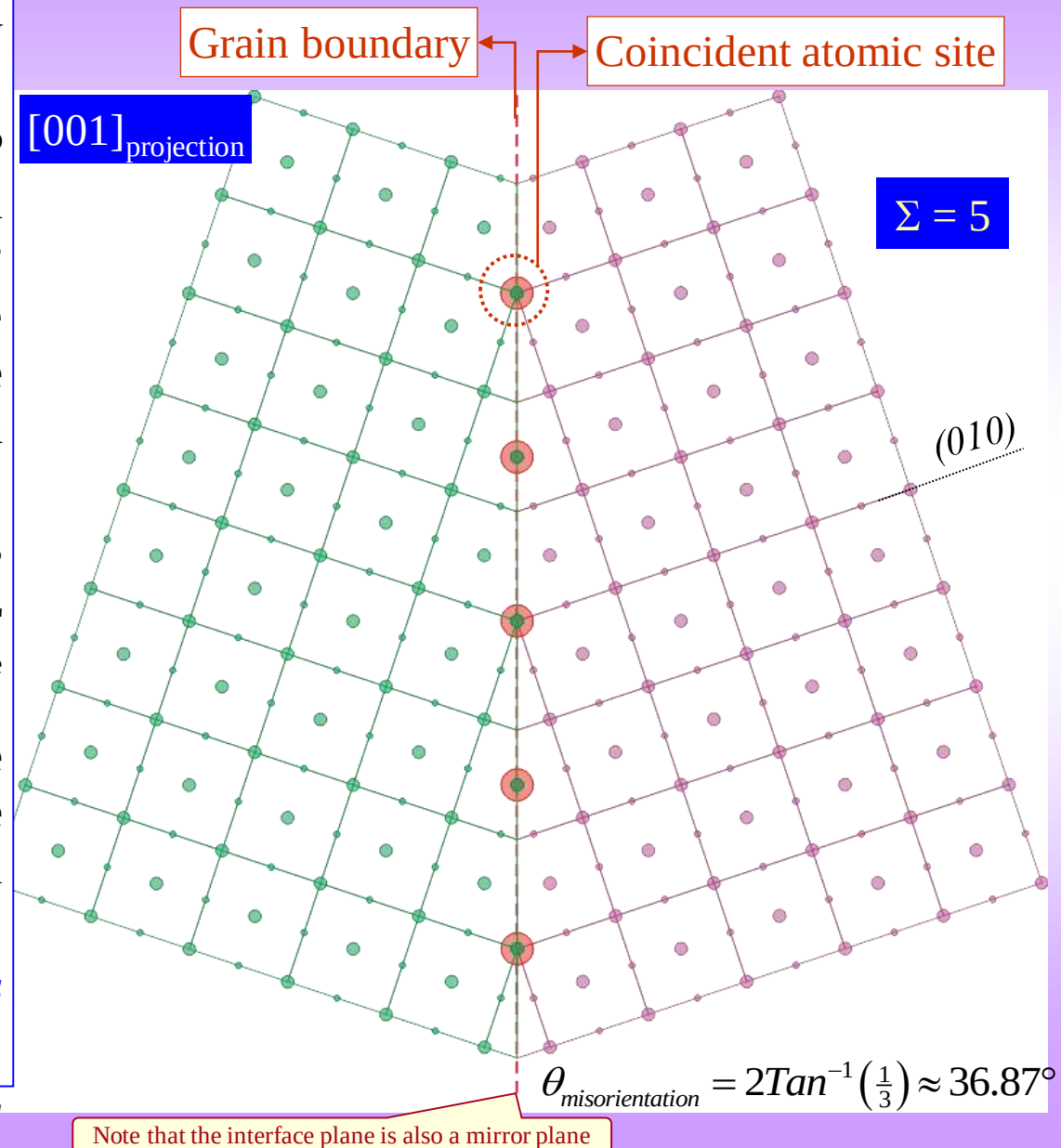
- Note that the DSC lattice does not lead to the formation of a crystal in the usual sense (i.e. there is no unique motif). The DSC lattice (& UC) is tilted equally w.r.t to the two original lattices.
- The DSC lattice being 'finer' has a smaller Burgers vector. Dislocations in this lattice accommodate the misorientation with respect to the CSL orientations (where the low energy cusps occur in the GB energy with respect to misorientation).

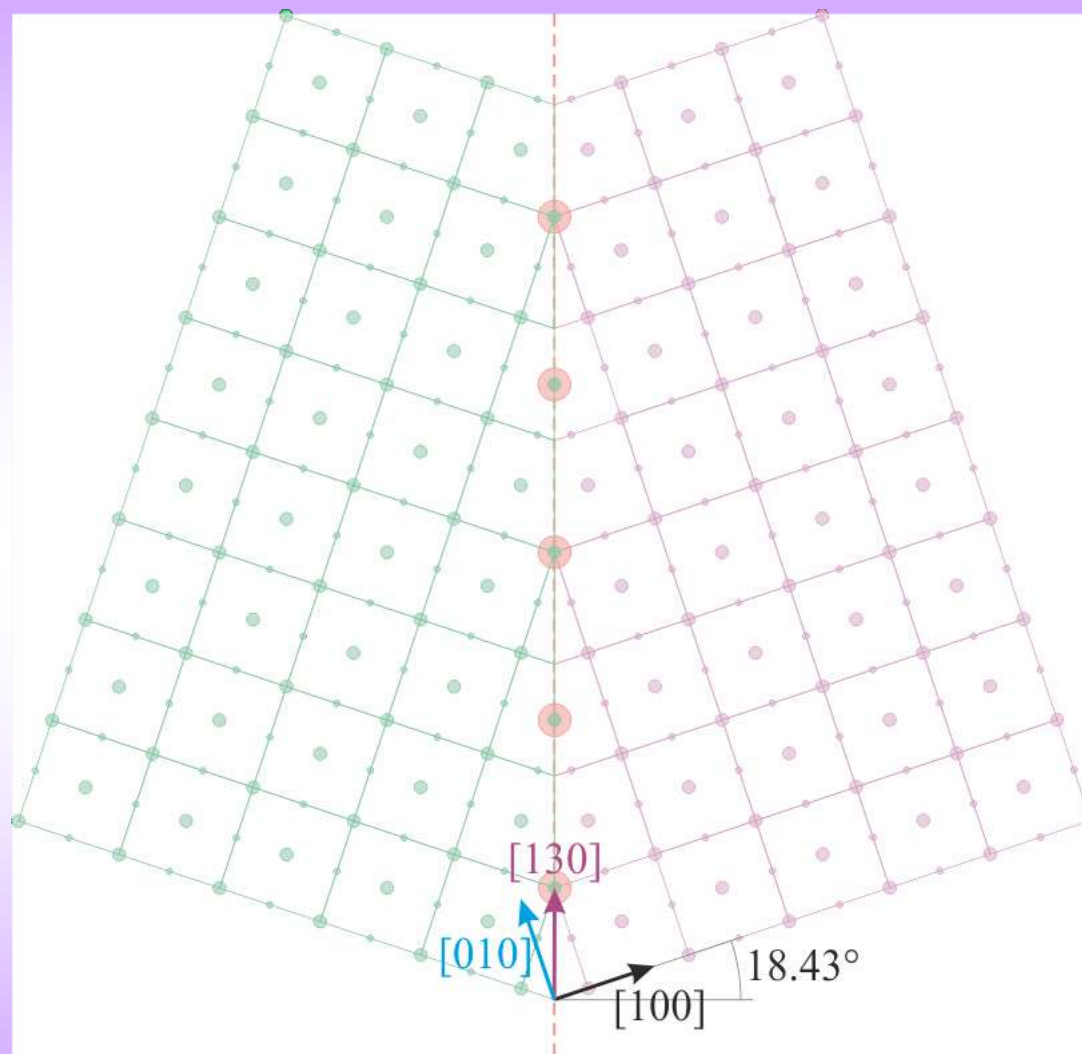
A boundary with CSL points

- Now that we have understood the concept of a CSL, let us see how the boundary looks like.
- As before we consider two CCP crystals in [001] projection which are misoriented by 36.87° ($=2\tan^{-1}(\frac{1}{3})$). However, this time let us not allow overlap of the crystals and let us construct a symmetric tilt boundary.
- The boundary now has atoms which perfectly match (the CSL 'atoms'). For rest of the atoms the bonds have been distorted.
- This matching (i.e. atoms at the boundary are common to both the bounding crystals) gives rise to a lower grain boundary energy.
- Note the reflection of planes across the interface.*

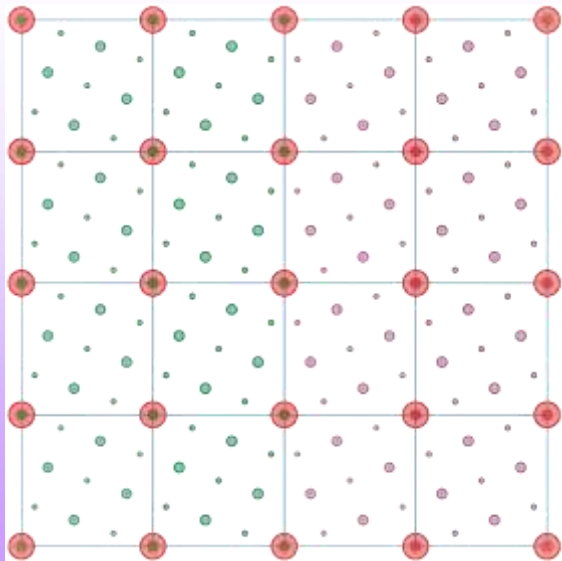
This time we cannot draw the CSL unit cell

Note that the frequency of matching is higher along the boundary as compared to the x & y axis of the participating crystals

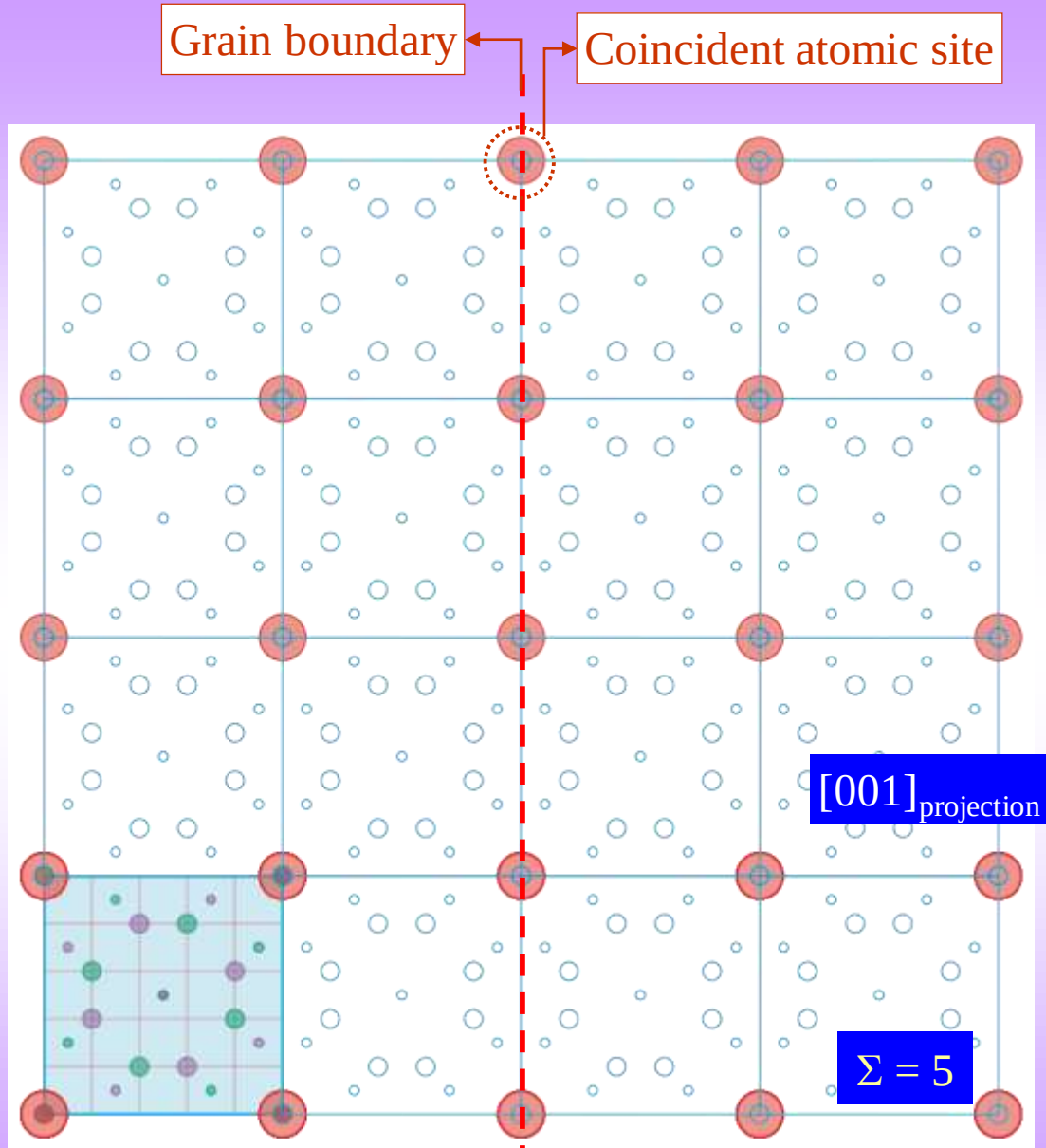




- The CSL, by its very construction, is ‘equally oriented’ (or equally misoriented) with respect to the two crystals. As mentioned before, the UC has atoms from both crystals.
- However, (more technically) we should draw only the crystal-1 (maroon) atoms to the right and only the crystal-2 (green atoms) to the left of the boundary (which is what we had in the first place!).



Now let us see the same $\Sigma = 5$ boundary in terms of the CSL.



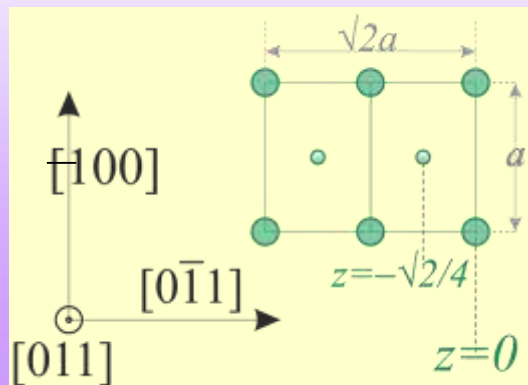
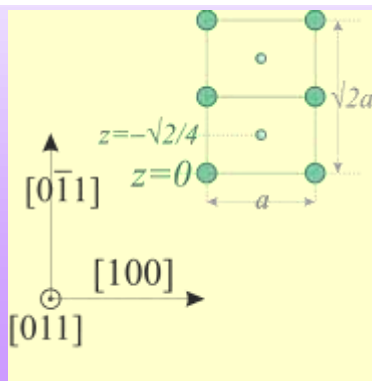
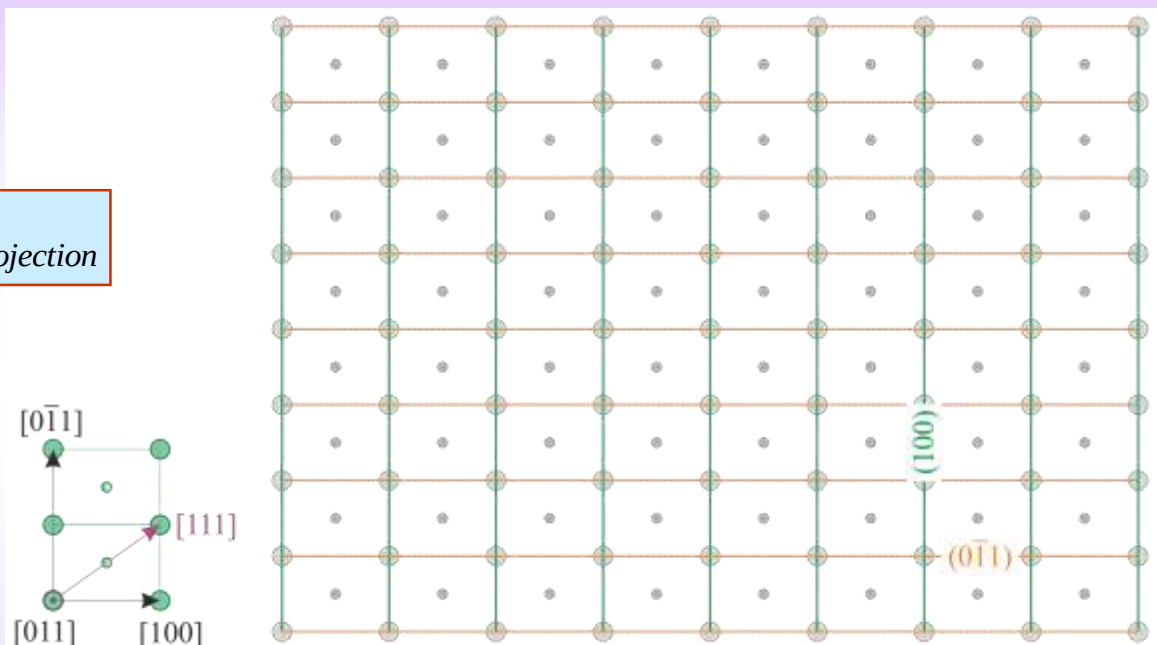
Let us construct another CSL (with $\Sigma = 3$, i.e. matching of one in three lattice points)

- To understand the $\Sigma = 3$ boundary, let us start with the $[011]$ project.
- We consider two layers of atoms (the larger circles are the top layer and the smaller ones are in the layer below).

The large circles are in top layer, while the small ones are below that.

The rectangular grid consists of $\{100\}$ and $\{110\}$ type of planes

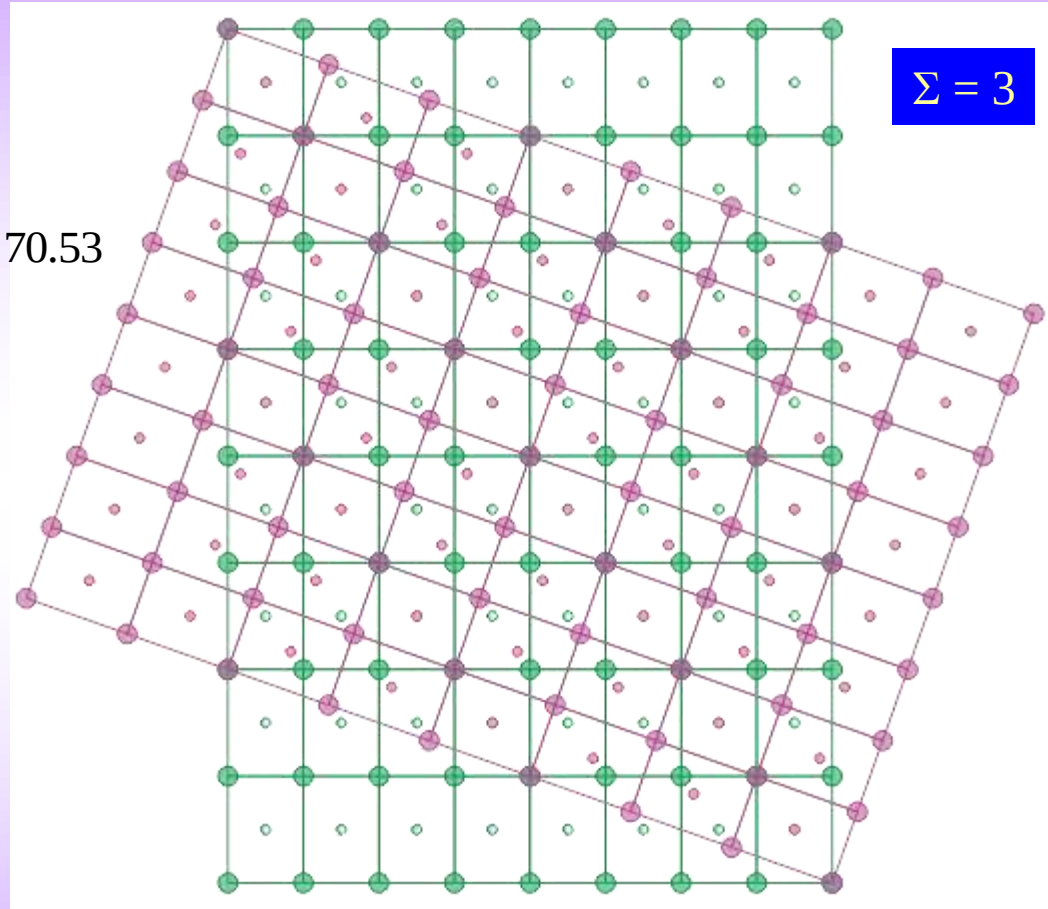
$[1\bar{1}0]$ projection



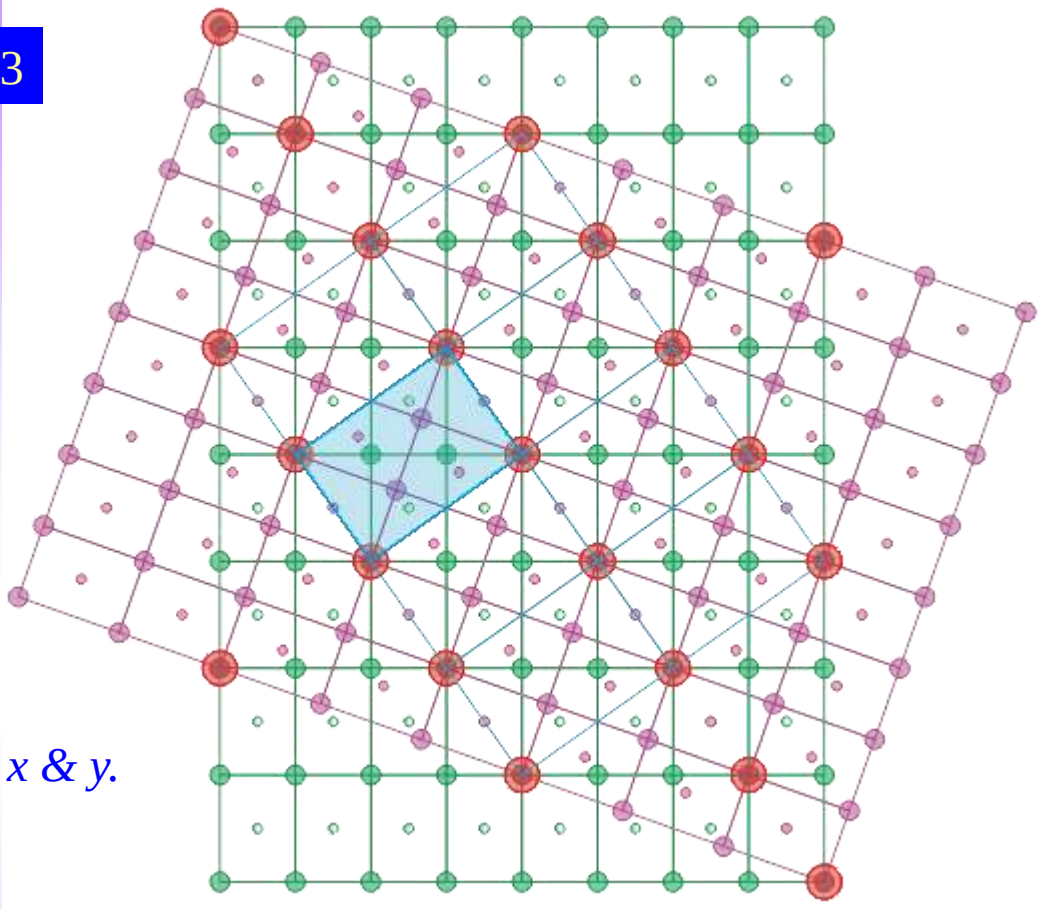
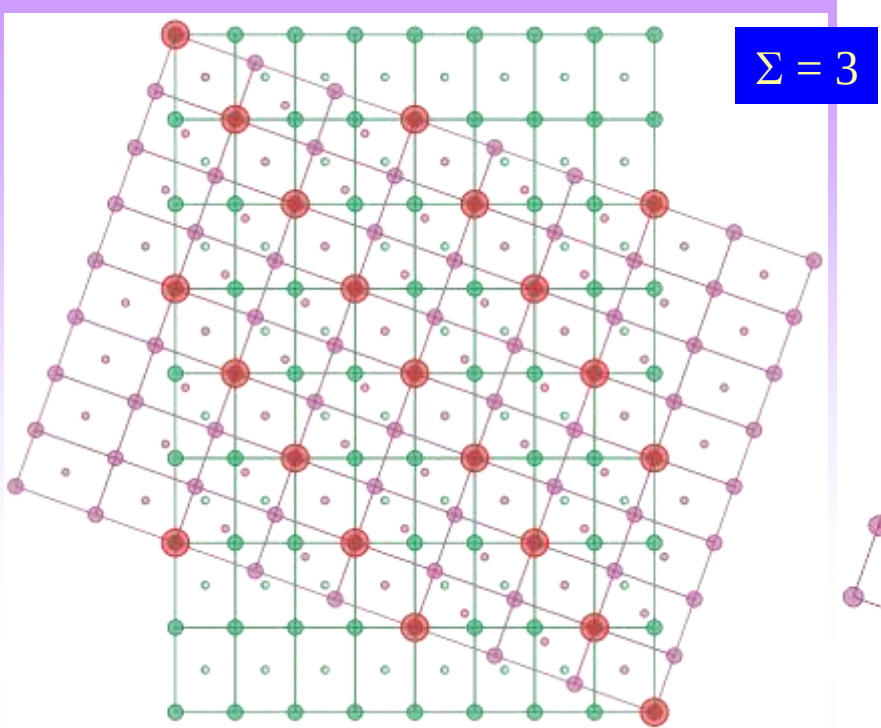
Atoms touch along the $\langle 011 \rangle$ directions

□ Take a CCP crystal with two layers of atoms in [1-10] projection. Misorient this crystal with an identical crystal by an angle of $\sim 70.53^\circ$.

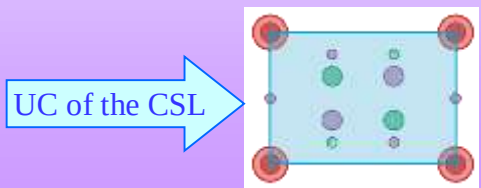
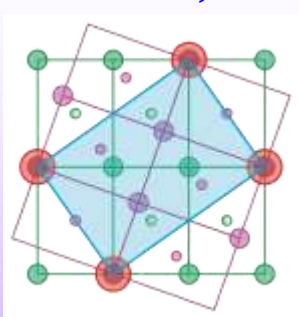
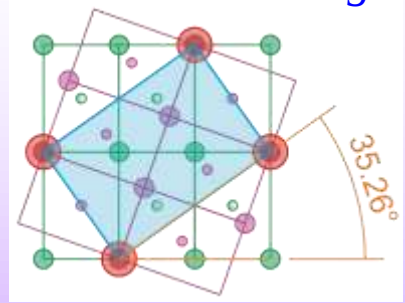
$$\theta = \cos^{-1}\left(\frac{1}{3}\right) \approx 70.53$$



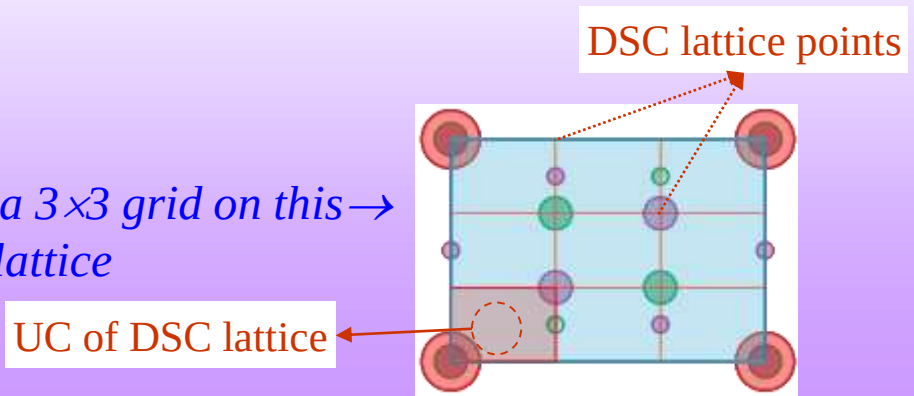
Locate the coincident lattice points and locate the unit cell (in blue).



Note a matching of 1 in 3 atoms, along x & y.



We can overlay a 3x3 grid on this → this is the DSC lattice

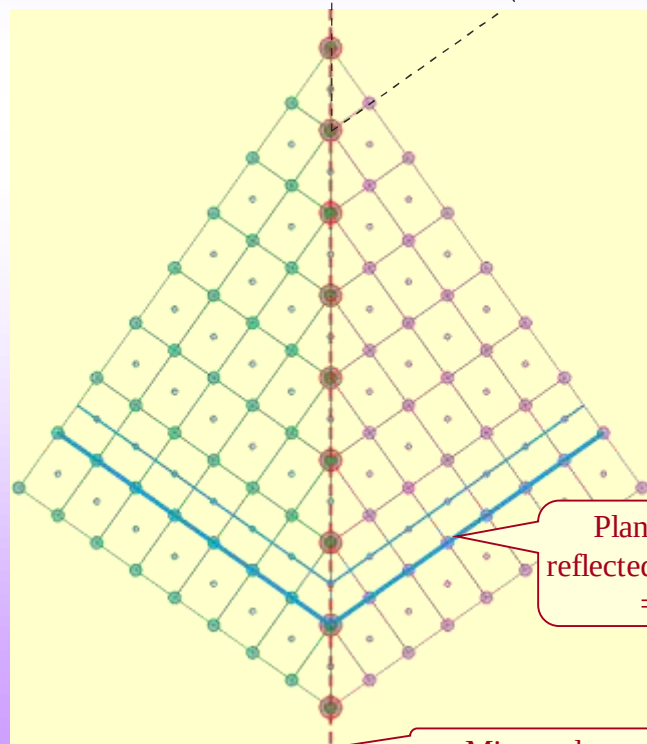


Like for the case of the $\Sigma = 5$ boundary, let us create a $\Sigma = 3$ boundary.

- ❑ This is the standard mirror twin boundary in a CCP material. I.e. this grain boundary becomes a special one— a twin boundary. This changes the significance and the character of the boundary.

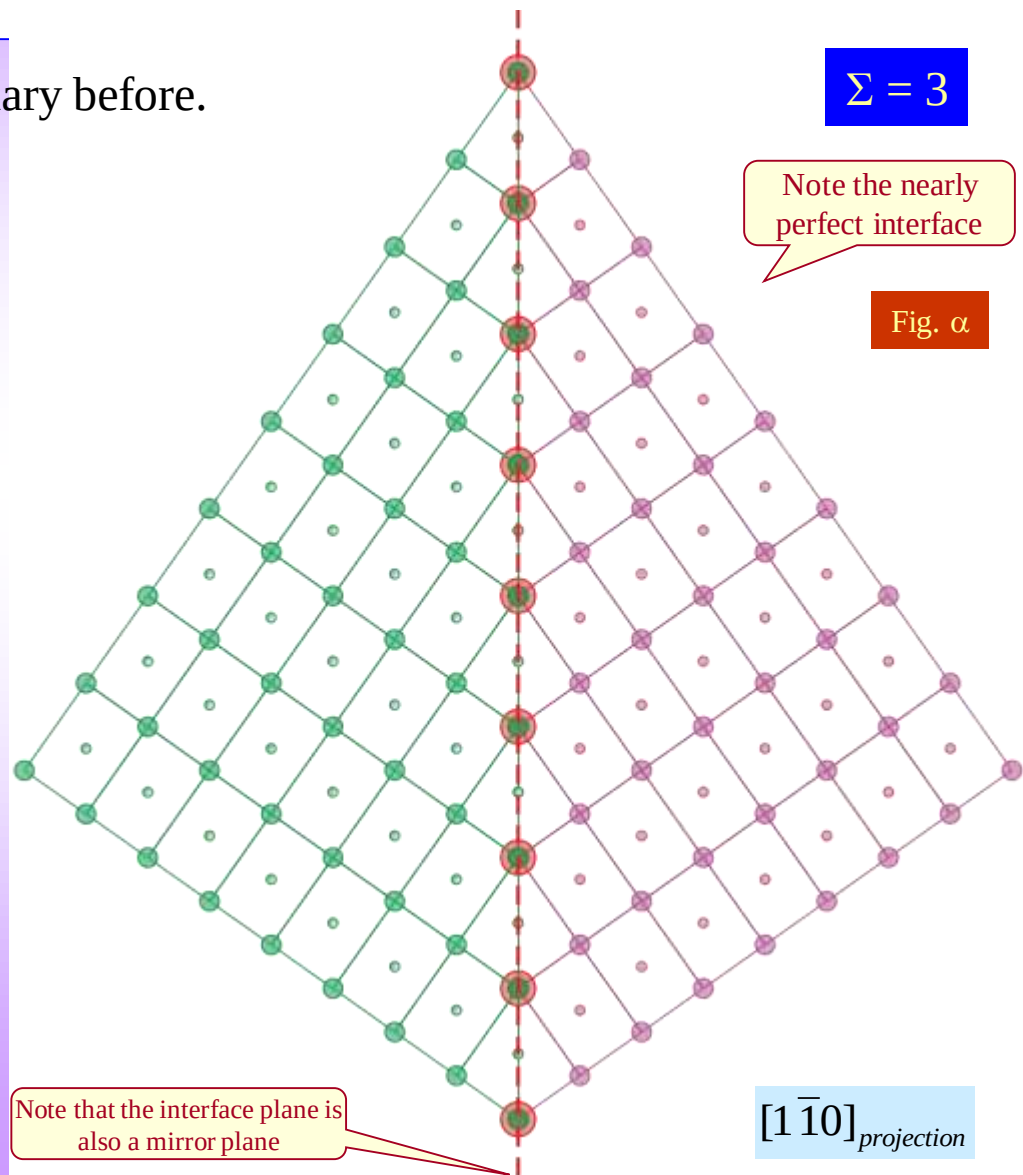
We had seen a different picture of the boundary before.
Next we will try to connect these figures.

$$\cos^{-1}\left(\frac{1}{\sqrt{3}}\right) = 54.73^\circ$$



Planes being reflected across GB = TB

Mirror plane



$\Sigma = 3$

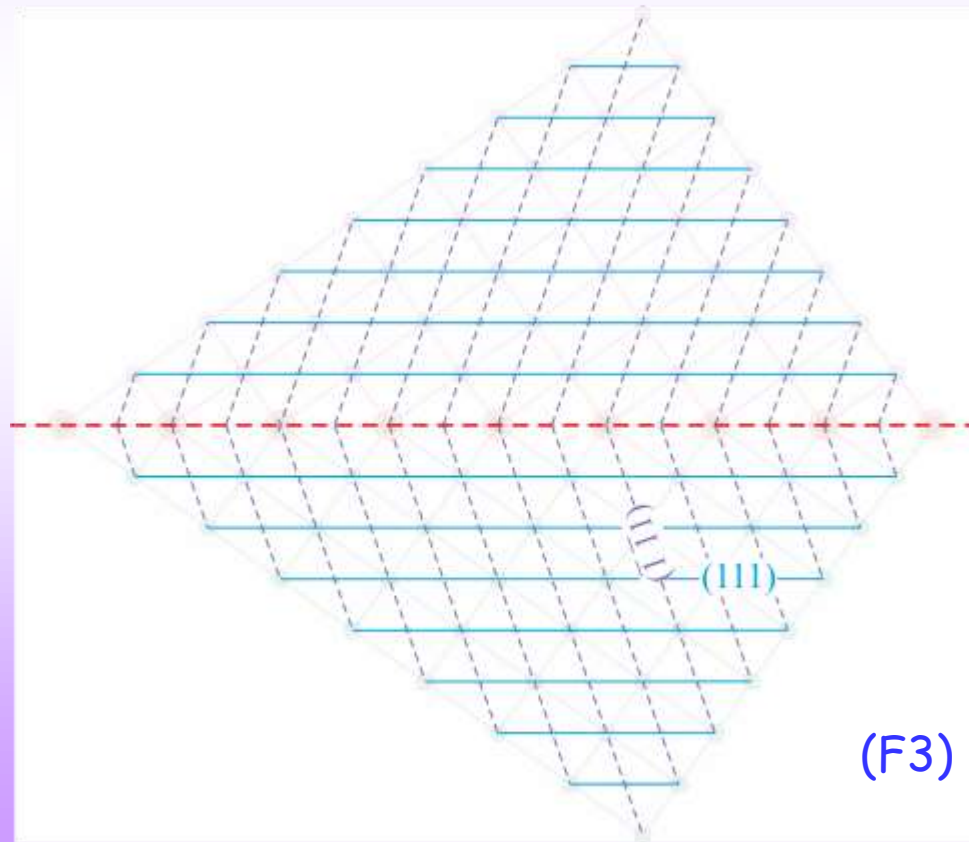
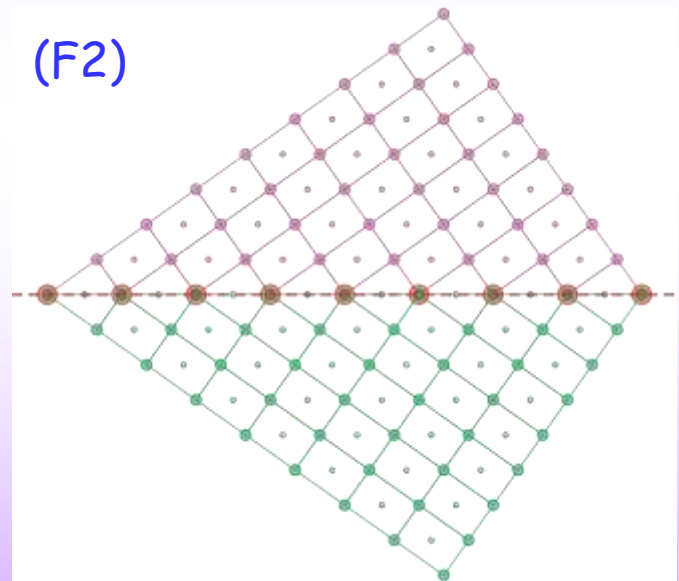
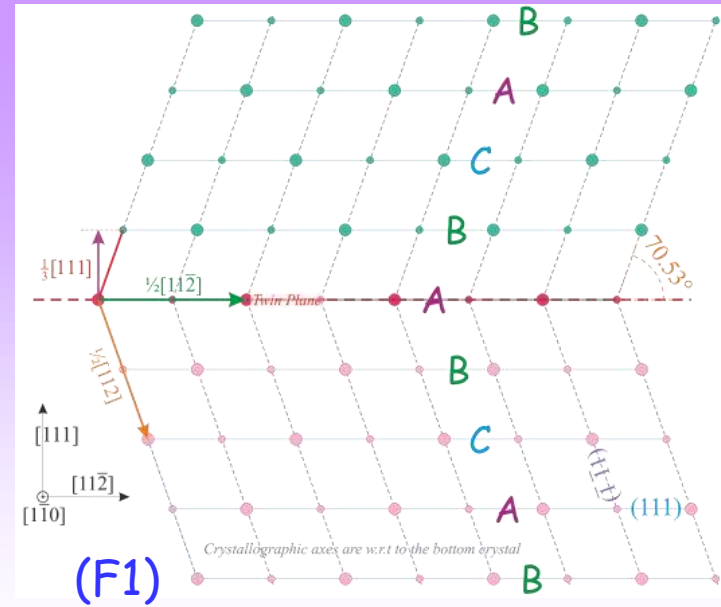
Note the nearly perfect interface

Fig. α

Note that the interface plane is also a mirror plane

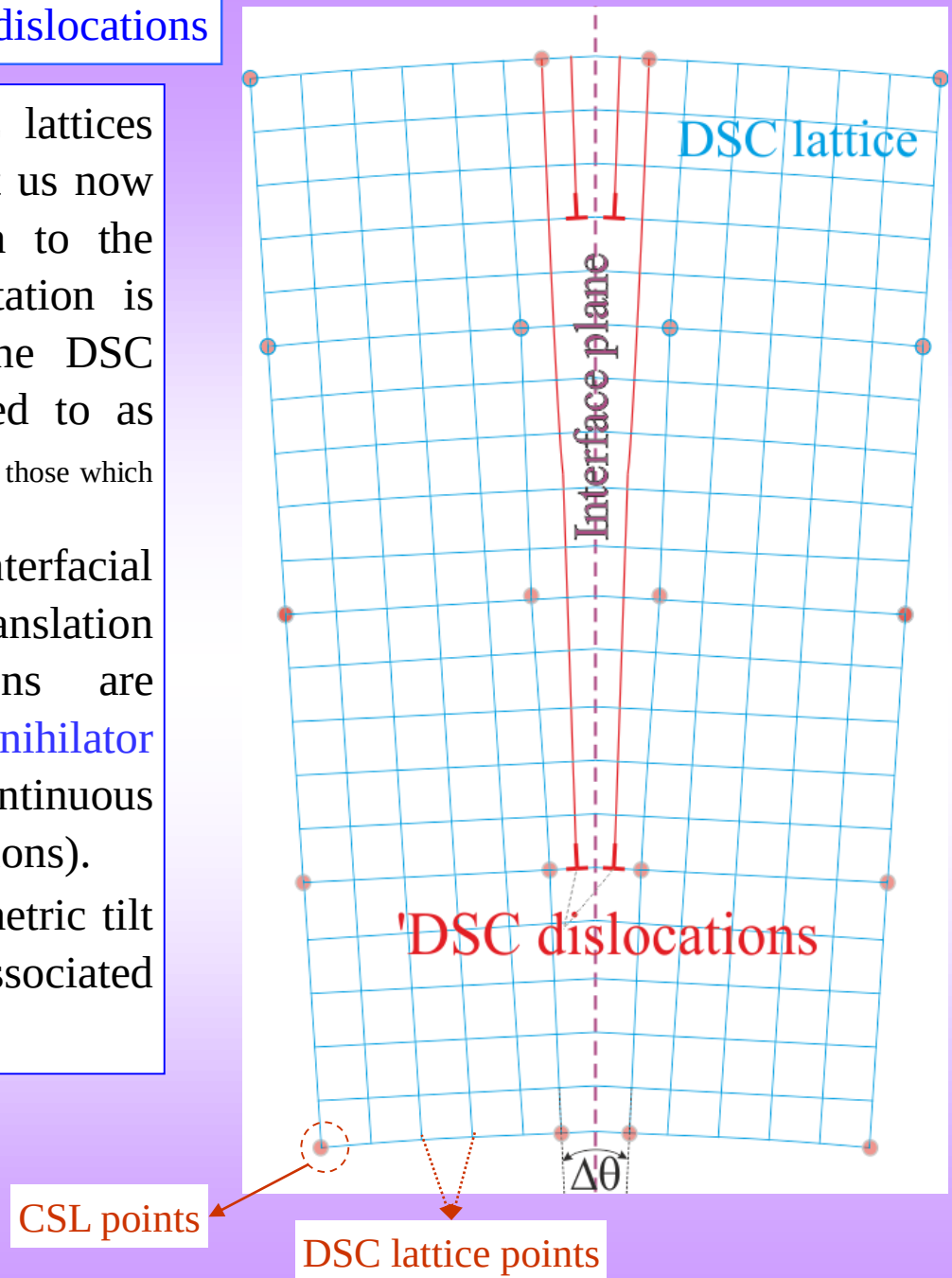
$[1\bar{1}0]_{\text{projection}}$

- We had seen figure on the right (F1) before in the context of the mirror twin. The figure below (F2) is the one we saw in the context of the $\Sigma = 3$ boundary.
- These essentially represent the same ‘picture’. However, in the context of the Σ boundary, we had drawn $\{100\}$ & $\{110\}$ type planes; while in the case of twinning, we had drawn the $\{111\}$ planes (two sets).
- Now in (F2), we draw the $\{111\}$ planes to reconcile the figures (F3).

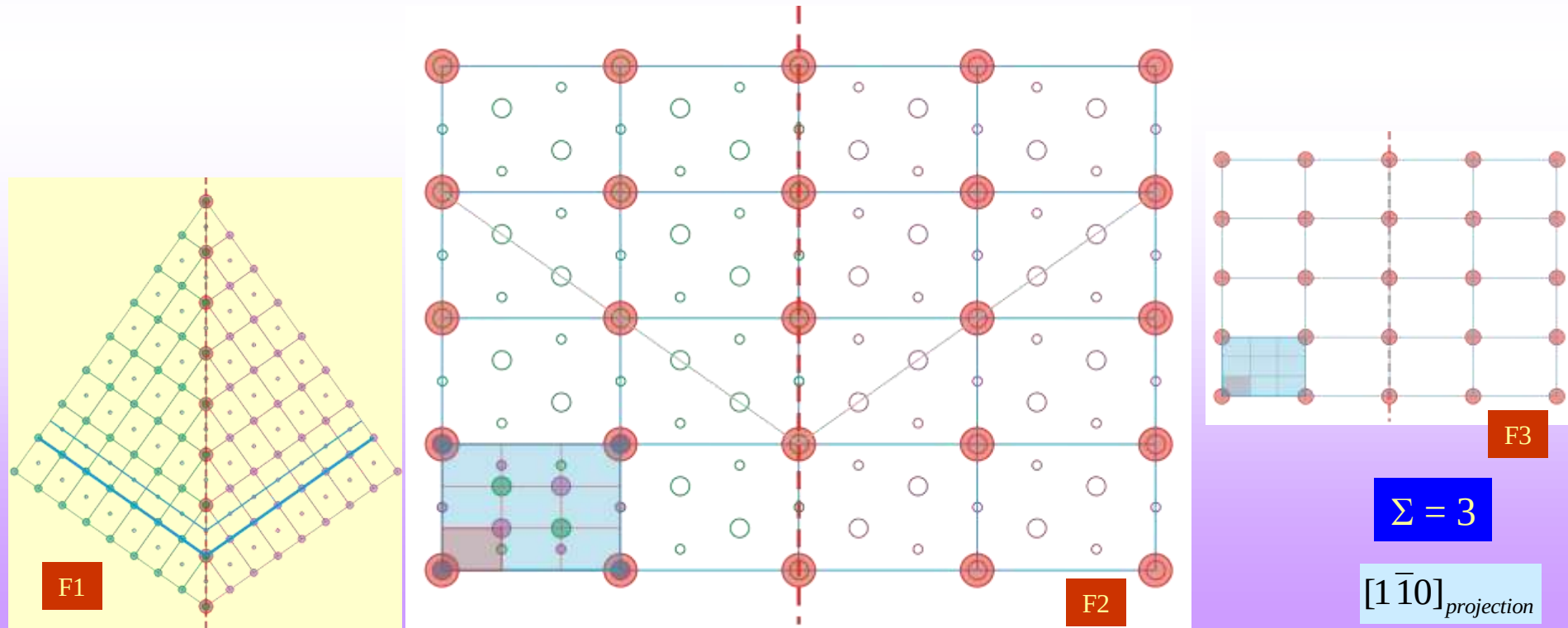


Misorientations in the DSC lattice & DSC dislocations

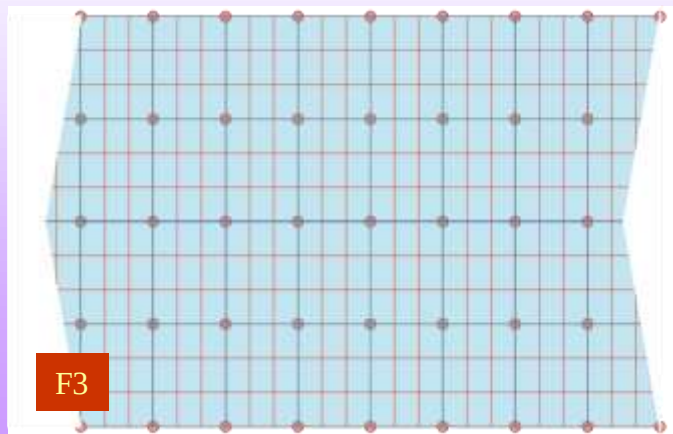
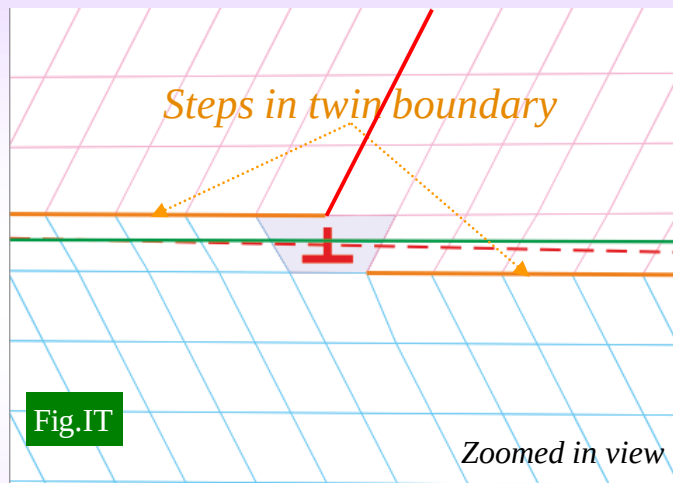
- ❑ The CSL and its derivative the DSC lattices were produced by misorientations. Let us now consider an additional misorientation to the DSC lattice (of $\Delta\theta$). This misorientation is accommodated by dislocations in the DSC lattice. These dislocations are referred to as **secondary** dislocations. The primary ones are those which are associated with the main lattice.
- ❑ The Burgers vector of these interfacial dislocations (**'b'**) is a DSC lattice translation vector. These discrete dislocations are sometimes referred to as **stress annihilator dislocations** (in contrast to a continuous distribution of stress generator dislocations).
- ❑ The figure shows a $\Sigma = 5$ (210) symmetric tilt boundary, with DSC dislocations associated with the boundary.



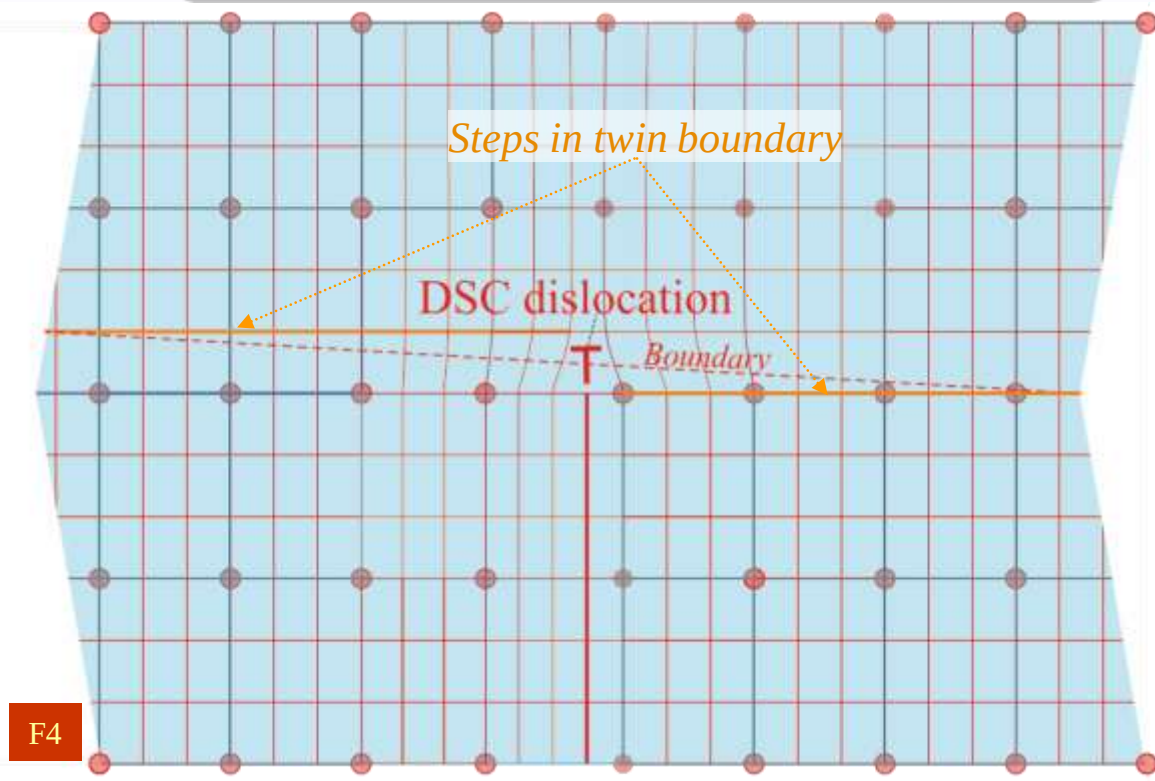
- ❑ We have seen the $\Sigma = 3$ twin boundary (F1). In the picture we have shown the two bounding crystals and not the CSL (or the DSC lattice). Needless to point out, if only the CSL is drawn the misorientation vanishes (F2). *Once again: we should keep in mind that F1 & F2 are mere two representations of the same boundary.* The blue UC and the red circles are constructs of the CSL and do not exist. As before the red lines correspond to the DSC lattice and the red shaded rectangle is the DSC UC.
- ❑ If only the CSL points are drawn (F3), then ‘structure with the interface’ looks perfect.
- ❑ To reiterate, even though all symmetrical tilt boundaries are twin boundaries, the one considered here is the commonly observed twin boundary in CCP materials with low SFE.



- ❑ We had seen an incoherent twin before (Fig.IT also reproduced below), in which the mirror plane did not coincide with the boundary plane.
- ❑ Now, let us start with the figure F3 and introduce further misorientation to the interface– as we had done earlier to get a incoherent twin boundary. Here we consider a 90° rotated, truncated on sides version (so as to give a feel of a twin!) of the figure without the atoms (only with the CSL & DSC entities)– figure F3.
- ❑ This tilt in the boundary (i.e. now the mirror plane does not coincide with the interface plane) is accommodated by a dislocation in the DSC lattice. As before the interface is stepped.



These are two independent descriptions of the same incoherent twin– one in terms of the crystal lattice and one in terms of the DSC lattice.



- The Burgers vector of this dislocation is $(\frac{1}{3})^{\text{rd}}$ of the lattice translation vector along the interface ($\vec{a}_{\text{DSC}} = \frac{1}{2}[11\bar{2}]$); which corresponds to the insertion of one vertical half-plane (red colour in F4).

$$\vec{a}_{\text{DSC}} = \frac{1}{2}[11\bar{2}]$$

$$\vec{b}_{\text{DSC}} = \frac{1}{6}[11\bar{2}]$$

$$(\text{Burgers vector})_{\text{DSC lattice}} = \vec{b}_{\text{DSC}} = \frac{1}{3}\left(\frac{1}{2}[11\bar{2}]\right) = \frac{1}{6}[11\bar{2}]$$

F3

$$\vec{b}_{\text{DSC}} = \frac{1}{3}\left(\frac{1}{2}[11\bar{2}]\right) = \frac{1}{6}[11\bar{2}]$$

F4

DSC dislocation
Boundary

Table 3.2 Measured Grain Boundary Free Energies

Values selected from compilation given in *Interfacial Phenomena in Metals and Alloys*, by L.E. Murr, Addison-Wesley, London, 1975.

Crystal	$\gamma_b/\text{mJ m}^{-2}$	T/°C	γ_b/γ_{sv}
Sn	164	223	0.24
Al	324	450	0.30
Ag	375	950	0.33
Au	378	1000	0.27
Cu	625	925	0.36
γ -Fe	756	1350	0.40
δ -Fe	468	1450	0.23
Pt	660	1300	0.29
W	1080	2000	0.41

Twin-boundary and stacking-fault energies in Al and Pd

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(Received 27 August 1990)

The (111) twin-boundary and stacking-fault energies of aluminum and palladium were investigated with use of the all-electron total-energy linear muffin-tin orbitals method based on the local-density approximation. Fault energies are determined by comparing the total energies obtained for the same-size supercells for each of two cases (with and without fault). The calculated values of the twin, intrinsic, and extrinsic fault energies, 130 ± 15 (97 ± 5), 270 ± 50 , and ~ 330 ergs/cm² for aluminum (palladium) are generally 30% larger than experiment. This discrepancy between the calculated and observed values may be attributed to the neglect of relaxation, the effect of the (finite) size of the supercell (at most 28 atoms/cell), and/or the use of the local-density approximation. Our calculated results appear to verify the correctness of the empirical relationship $2E_{tw} \approx E_{SF}$, where E_{tw} and E_{SF} are the twin and stacking-fault energies, respectively.

TABLE I. Twin and stacking-fault energies for Al and Pd (in ergs/cm²).

		Al			Pd	
		Twin	ISF	ESF	Twin	ISF
Calc.	This work	130 ± 15	280 ± 40	~ 260	97 ± 5	
	Ref. 5	118				
	Ref. 13	57	145	130		
	Ref. 4	61	160	133		
	Ref. 14	55	142			
Expt.	Ref. 1	75	166			180
	Ref. 15	120	200			180
	Ref. 16	75–100	135	180		
0 K (extrapolated)		Ref. 2	100–125	160	200	